Animal Form & Function II

Module 91

Animal Form & Function II

Vertebrate Excretory Systems

Vertebrate Excretory Systems

- Vertebrates face the same problems as invertebrates in controlling water and ion balance.
- Generally, water losses are balanced precisely by water gains.

Vertebrate Excretory Systems

Water Gain:

- Vertebrates gain water from liquids and solid foods through:
  - the small,
  - large intestines, and
  - by metabolic reactions that yield water as an end product.

Vertebrate Excretory Systems

Water Loss:

- They lose water by evaporation:
  - from respiratory surfaces,
  - from the integument,
  - sweating or panting,
  - elimination in feces and
  - excretion by the urinary system.

Vertebrate Excretory Systems

Solute Absorption:

- Vertebrates take in solutes by absorption of minerals from:
  - the small and large intestines,
  - through the integument
  - gills,
  - from secretions of various glands and
  - by metabolism (e.g., the waste products of catabolic reactions).

Vertebrate Excretory Systems

Solute Elimination:

- Vertebrates lose solutes through:
  - sweat,
  - feces,
  - urine, and
  - gill secretions as metabolic wastes.
  - The major metabolic wastes that are eliminated as ammonia, urea, or uric acid.

Vertebrate Excretory Systems

- Different vertebrates face different water and solute problems and solve them in different ways.
- How vertebrates maintain a homeostatic solvent/solute concentrations in their body fluids will be discussed next.

Vertebrate Excretory Systems

- The excretion of certain metabolic waste products is also coupled with osmotic balance and will be discussed with the urinary system.
- Solute losses also must be balanced by solute gains.

Vertebrate Excretory Systems

AVERAGE WATER GAIN AND LOSS IN HUMAN AND KANGROO RAT

VERTEBRATE	WATER GAIN (ml)		WATER LOSS (ml)	
Human	Ingested in solid food	1200	Feces	100
	Ingested in liquids	1000	Urine	1500
	Metabolically produced	350	Skin and lungs	950
	Total	2550		2550
Kangroo rat	Ingested in solid food	6	Feces	3
	Ingested in liquids	0	Urine	13
	Metabolically produced	54	Skin and lungs	44
	Total	60		60

Animal Form & Function II

Module 92

Animal Form & Function II

Osmoregulation in Vertebrates

Osmoregulation in Vertebrates

- A variety of mechanisms have evolved in vertebrates to cope with their osmoregulatory problems, and most of them are adaptations of the urinary system.
- Vertebrates have a closed circulatory system containing blood that is under pressure.

Osmoregulation in Vertebrates

- This pressure forces blood through a membrane filter in a kidney, where the following three key functions take place:
- 1. Filtration
- 2. Reabsorption
- 3. Secretion

Osmoregulation in Vertebrates

1. Filtration,

- In this process blood passes through a filter that retains blood cells, proteins, and other large solutes but lets small molecules, ions, and urea pass through.

Osmoregulation in Vertebrates

2. **Reabsorption,**

- **In reabsorption selective ions and molecules are taken back into the bloodstream from the filtrate.**

Osmoregulation in Vertebrates

3. Secretion,

- In this process select ions and end products of metabolism (e.g., K, H, NH_3) that are in the blood are added to the filtrate for removal from the body.

END

Animal Form & Function II

Module 93

Animal Form & Function II

Vertebrate Kidney Variations

(Pronephros)

Vertebrate Kidney Variations (Pronehros)

Vertebrates

- Vertebrates have two kidneys that are in the back of the abdominal cavity, on either side of the aorta.
- Each kidney has a coat of connective tissue called the renal capsule.
- The inner portion of the kidney is called the medulla; the region between the capsule and the medulla is the cortex.

Vertebrate Kidney Variations (Pronehros)

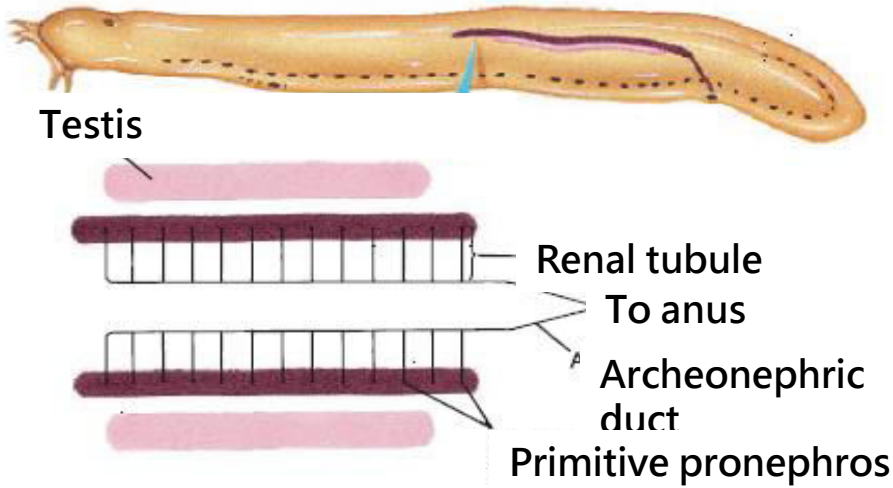
- The structure and function of vertebrate kidneys differ, depending on the vertebrate groups and the developmental stage.
- Overall, there are three kinds of vertebrate kidneys:
- Pronephros,
- Mesonephros, and
- Metanephros.

Vertebrate Kidney Variations

The Pronephros:

- Appears only briefly in many vertebrate embryos, and not at all in mammalian embryos.
- In some vertebrates, the pronephros is the first osmoregulatory and excretory organ of the embryo as in tadpoles and other amphibian larvae.

Vertebrate Kidney Variations (Pronehros)



- **In vertebrates such as in hagfishes, pronephros remains as the functioning kidney.**

  The primitive pronephric kidney is found in adult hagfishes and in the embryos of fishes and amphibians.

Vertebrate Kidney Variations (Pronehros)

- In the anterior half of the body the renal tubules occur in segments that lead from the body of the pronephros to the archinephric duct.
- Notice that the testes are separate from the kidneys.

end

Animal Form & Function II

Module 94

Animal Form & Function II

Vertebrate Kidney Variations

(Mesonephros)

Vertebrate Kidney Variations (Mesonephros)

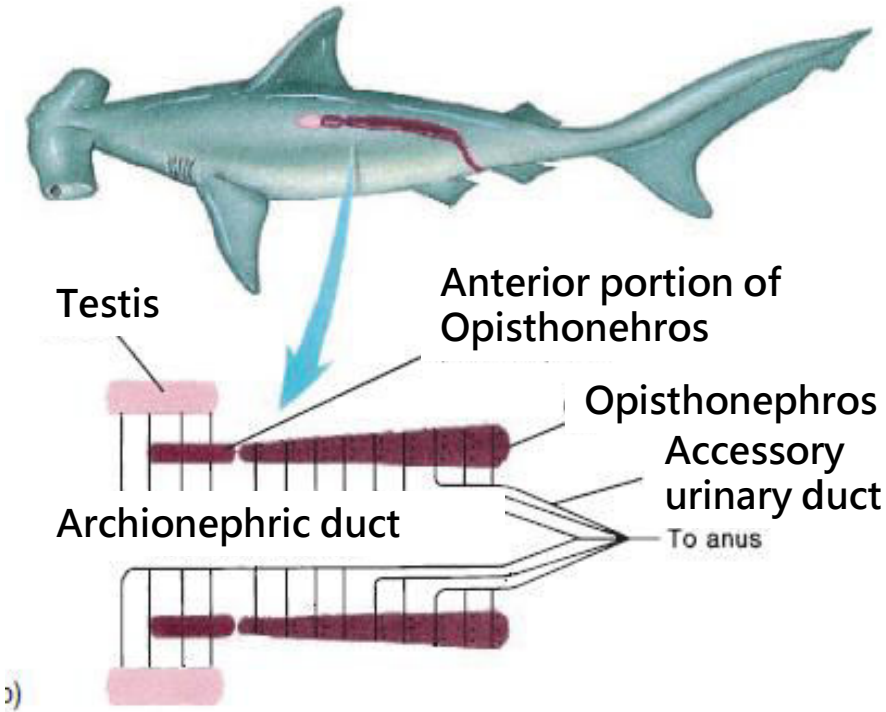
The Mesonephros

- It is one of three excretory organs that develop in vertebrates.
- It is a paired organ consisting of a set of nephrons having capsules that filter blood from the glomerulus and tubules whose cells reabsorb water and nutrients and secrete nitrogenous wastes.

Vertebrate Kidney Variations (Mesonephros)

- It serves as the main excretory organ of aquatic vertebrates and as a temporary kidney in reptiles, birds, and mammals.
- During embryonic development of amniotes, or during metamorphosis in amphibians, the mesonephros replaces the pronephros.

Vertebrate Kidney Variations (Mesonephros)



- **The mesonephros is the functional embryonic kidney of many vertebrates and also adult fishes and amphibians.**
- **It is structurally similar to the nonsegmented opisthonephric (advanced mesonephric) kidney of most nonamniote vertebrates, such as sharks.**

Vertebrate Kidney Variations (Mesonephros)

- The anterior portion of the opisthonephros functions in blood cell formation and secretion of sex hormones.
- Notice that the testes occupy the position of the anterior opisthonephros, and the archinephric duct carries both sperm and urine.

Vertebrate Kidney Variations (Mesonephros)

- Glomeruli are absent in some marine fish; urine then forms solely in the tubules.
- The tubules empty into a long tube, the Wolffian duct, a remnant of the pronephros.
- In more advanced vertebrates the mesonephros develops in the embryo but is replaced after the 10th week by the metanephros as in humans.

end

Animal Form & Function II

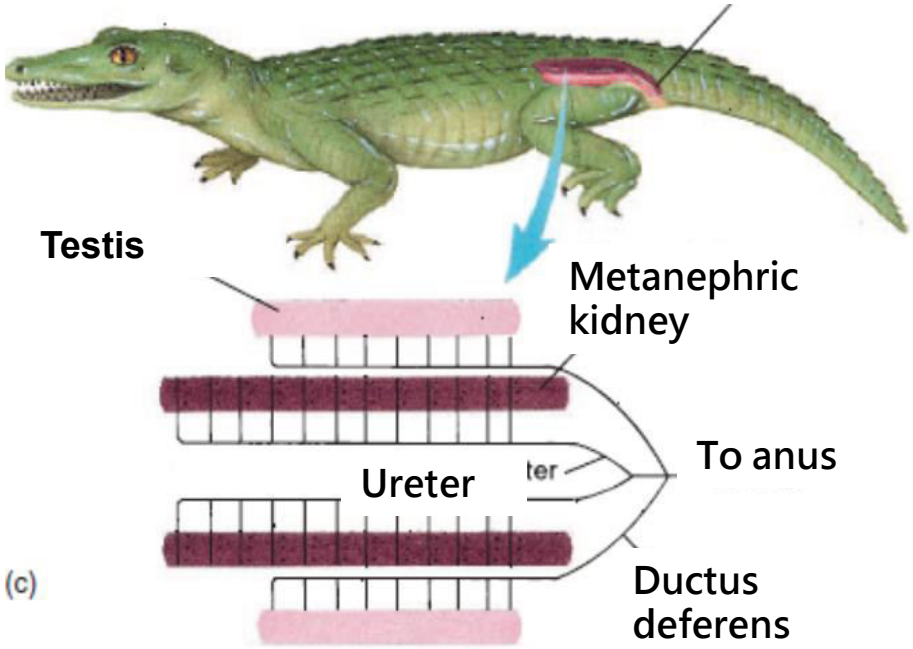
Module 95

Animal Form & Function II

Vertebrate Kidney Variations

(Metanephros)

Vertebrate Kidney Variations (Metanephros)



Metanephros

- Member of the final and most caudal pair of vertebrate renal organs that functions as a permanent adult kidney is in reptiles, birds, and mammals.
- Not present at all in lower forms.

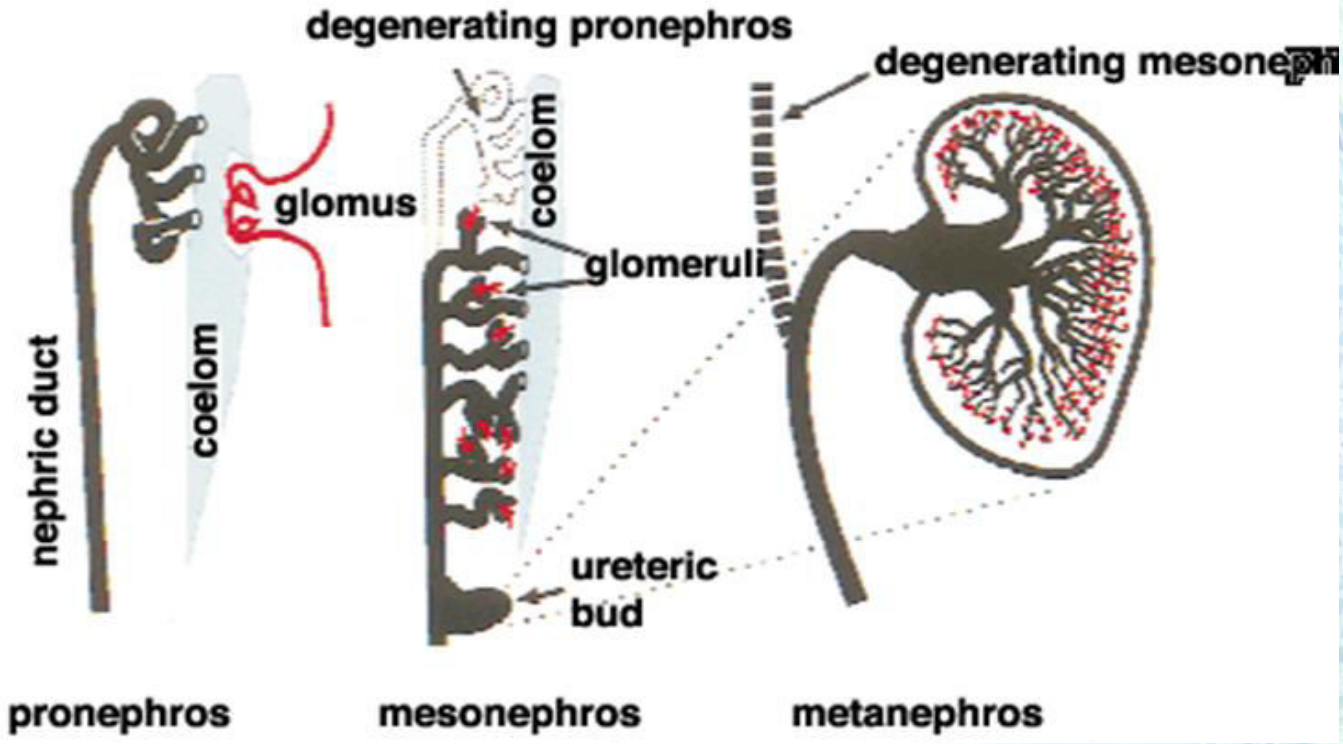
Vertebrate Kidney Variations (Metanephros)

- The physiological differences between these kidney types are primarily related to the number of blood-filtering units.
- The pronephric kidney is formed in the anterior portion of the body cavity and contains fewer blood-filtering units than either the mesonephric or metanephric kidneys.

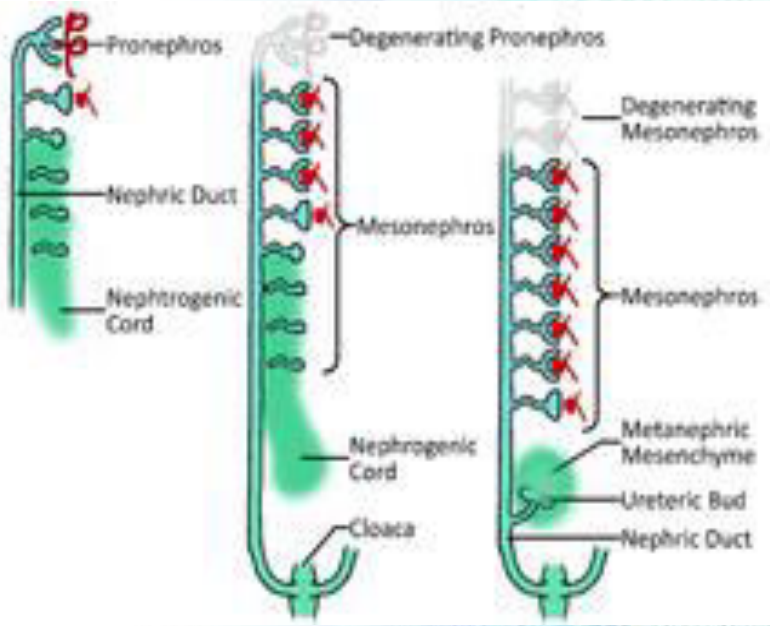
Vertebrate Kidney Variations (Metanephros)

- The larger number of filtering units in the latter has allowed vertebrates to face the rigorous osmoregulatory and excretory demands of freshwater and terrestrial environments.
- What follows is a presentation of how a few vertebrates maintain their water and solute concentrations in different habitats—in the seas, in freshwater, and on land.

Vertebrate Kidney Variations (Metanephros)



Vertebrate Kidney Variations (Metanephros)



end

Animal Form & Function II

Module 96

Animal Form & Function II

Excretion in Sharks

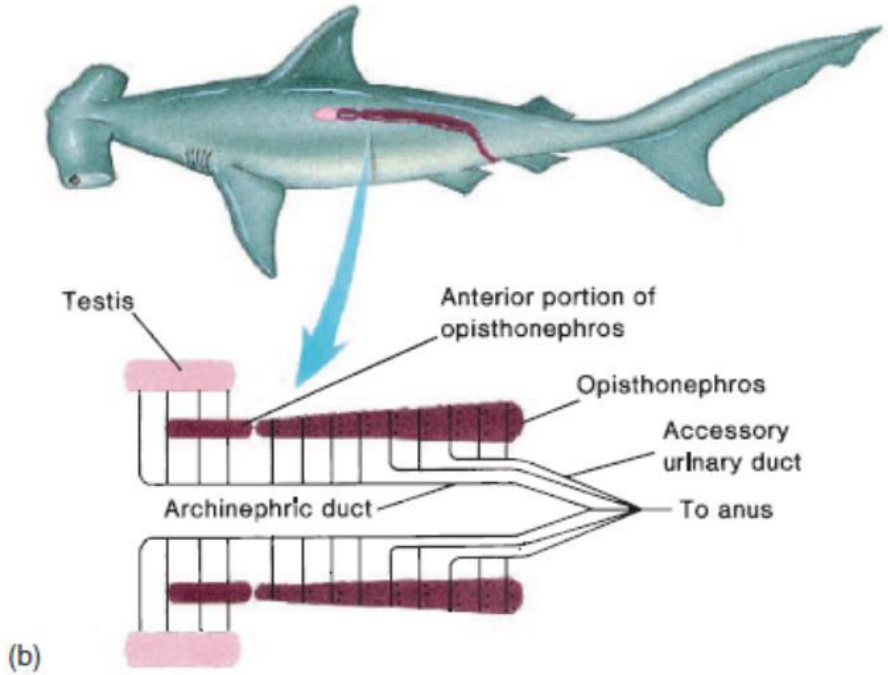
Excretion in Sharks

- **Sharks** and their relatives (skates and rays) have mesonephric kidneys and have solved their osmotic problem in ways different from the bony fishes.
- Instead of actively pumping ions out of their bodies through the kidneys, they have a rectal gland that secretes a highly concentrated salt (NaCl) solution.

Excretion in Sharks

- To reduce water loss, they use two organic molecules—urea and trimethylamine oxide (TMO)—in their body fluids to raise the osmotic pressure to a level equal to or higher than that of the seawater.
- Urea denatures proteins and inhibits enzymes, where asTMO stabilizes proteins and activates enzymes.

Excretion in Sharks



Together in the proper ratio, they counteract each other, raise the osmotic pressure, and do not interfere with enzymes or proteins.

This reciprocity is termed the counteracting osmolyte strategy.

Excretion in Sharks

Animal Form & Function II

Module 97

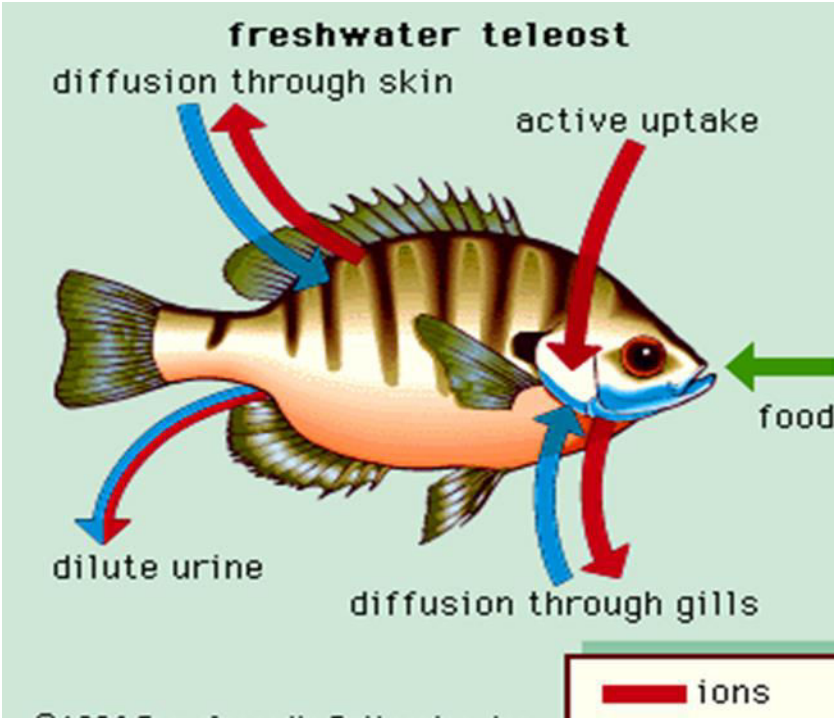
Animal Form & Function II

Excretion in Teleost Fishes

Excretion in Teleost Fishes

- **Most teleost fishes** have mesonephric kidneys.
- Since the body fluids of freshwater fishes are hyper-osmotic relative to freshwater, water tends to enter the fishes, causing excessive hydration or bloating.

Excretion in Teleost Fishes

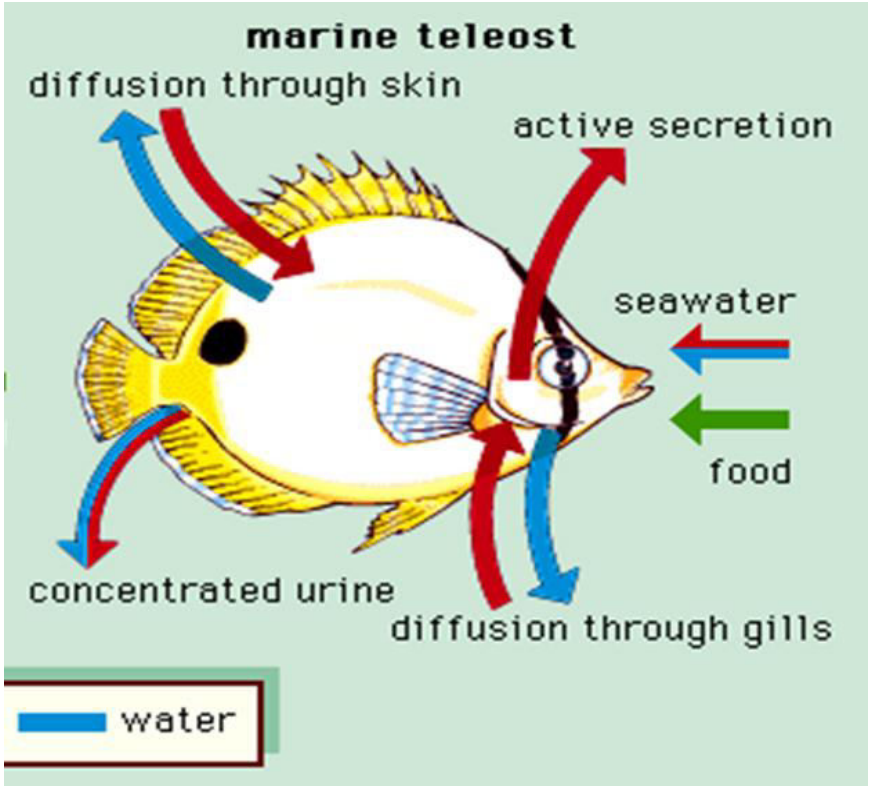


- They absorb salts and ions by active transport across their gills.
- They also excrete a large volume of water as dilute urine.
- Freshwater fishes have enlarged glomeruli and short tubule systems.

Excretion in Teleost Fishes

- They filter large quantities of water from the blood and tubules reabsorb some ions from the filtrate.
- Freshwater fishes produce a hypo-osmotic urine.

Excretion in Teleost Fishes



- Marine fishes face a different problem of water balance—their body fluids are hypoosmotic with respect to seawater and water tends to leave their bodies, resulting in dehydration.
- Marine fishes conserve water by reducing the size of the glomerulus of the nephron, and thus reducing the quantity of water and ions filtered from the blood.

Excretion in Teleost Fishes

- Ions can be secreted from the blood into the kidney tubules.
- Marine fishes can produce urine that is iso-osmotic with the blood.

Excretion in Teleost Fishes

- To compensate, marine fishes drink large quantities of water, and they secrete Na, Cl, and K ions through secretory cells in their gills.
- Channels in plasma membranes of their kidneys actively transport the multivalent ions that are abundant in seawater

Excretion in Teleost Fishes

- (e.g., Ca2, Mg2, SO4 and PO4
- out of the extracellular fluid and into the nephron tubes.
- The ions are then excreted in a concentrated urine.

end

Animal Form & Function II

Module 98

Animal Form & Function II

Excretion in Amphibians

Excretion in Amphibians

- **The amphibian kidney is identical to that of freshwater fishes.**
- **This is not surprising, because amphibians spend a large portion of their time in freshwater, and when on land, they tend to seek out moist places.**

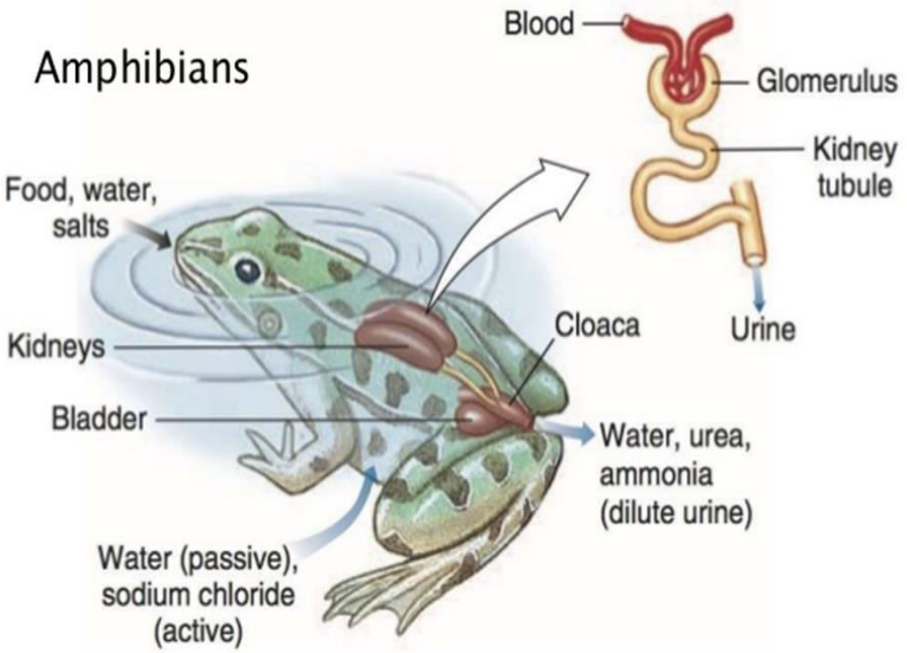
Excretion in Amphibians

- Amphibians take up water and ions in their food and drink, through the skin that is in contact with moist substrates, and through the urinary bladder.
- This uptake counteracts what is lost through evaporation and prevents osmotic imbalance.

Excretion in Amphibians

- The urinary bladder of a frog, toad, or salamander is an important water and ion reservoir.
- For example, when the environment becomes dry, the bladder enlarges for storing more urine.

Excretion in Amphibians



- Water can enter this frog via food, through its highly permeable skin, or from the urinary bladder.
- The skin also actively transports ions such as Na and Cl from the environment.
- The kidney forms a dilute urine by reabsorbing Na and Cl ions.

Excretion in Amphibians

- **If an amphibian becomes dehydrated, a brain hormone causes water to leave the bladder and enter the body fluid.**
- **When water leaves the urinary bladder, the remaining ions are reabsorbed.**

end

Excretion in Amphibians

Animal Form & Function II

Module 99

Animal Form & Function II

Excretion in Reptiles Birds and Mammals

Excretion in Reptiles, Birds and Mammals

- **Reptiles,** birds, and mammals all possess metanephric kidneys.
- Their kidneys are by far the most complex animal kidneys, well suited for these animals' high rates of metabolism.

Excretion in Reptiles, Birds and Mammals

- In most reptiles, birds, and mammals, the kidneys can remove far more water than can those in amphibians, and the kidneys are the primary regulatory organs for controlling the osmotic balance of the body fluids.

Excretion in Reptiles, Birds and Mammals

- **Some desert and marine reptiles and birds build up high salt (NaCl) concentrations in their bodies because they consume salty foods or seawater and they lose water through evaporation, urine and feces.**

Excretion in Reptiles Birds and Mammals

- To rid get of excess salt, the reptiles also have salt glands near the eye or in the tongue that remove excess salt from the blood and secrete it as tear-like droplets.

Excretion in Reptiles, Birds and Mammals

- A major site of water loss in mammals is the lungs.
- To reduce this evaporative loss, many mammals have nasal cavities that act as countercurrent exchange systems.

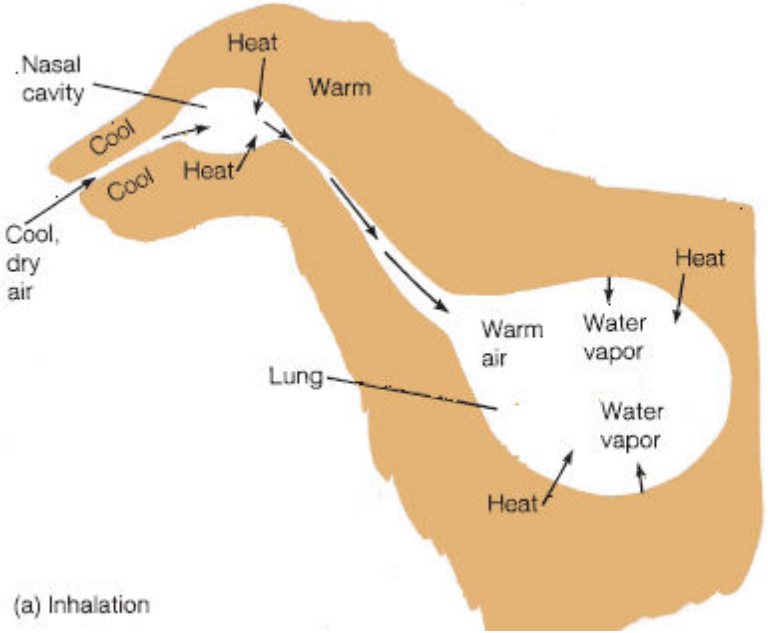
Excretion in Reptiles, Birds and Mammals

- When the animal inhales, air passes through the nasal cavities and is warmed by the surrounding tissues.
- In the process, the temperature of this tissue drops. When the air gets deep into the lungs, it is further warmed and humidified.

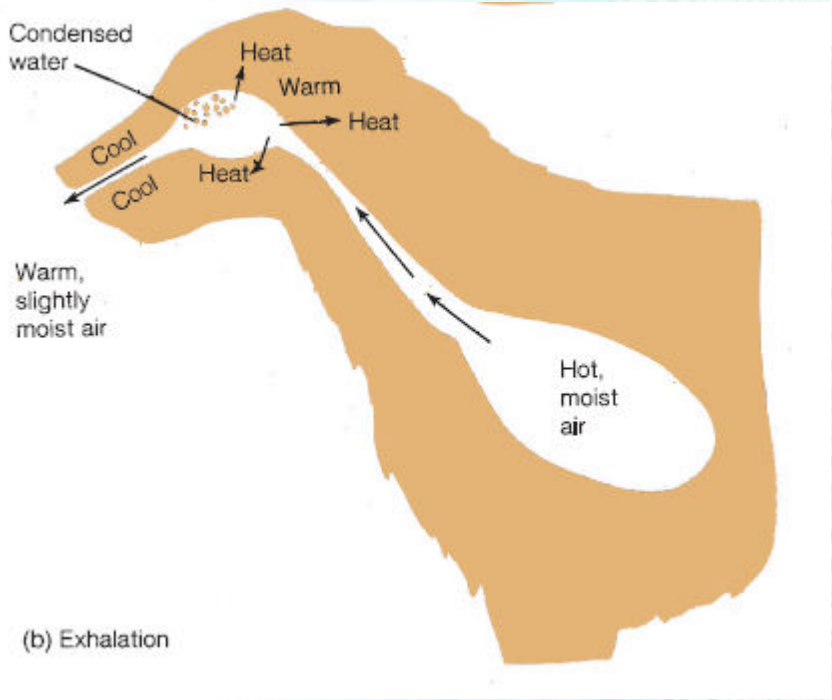
Excretion in Reptiles, Birds and Mammals

- During exhalation, as the warm moist air passes up the respiratory tree, it gives up its heat to the nasal cavity.
- As the air cools, much of the water condenses on the nasal surfaces .
- This mechanism explains why a dog's nose is usually cold and moist.

Excretion in Reptiles, Birds and Mammals



Inhalation



Exhalation

Animal Form & Function II

Module 100

Excretory System of Man

Excretory System of Man

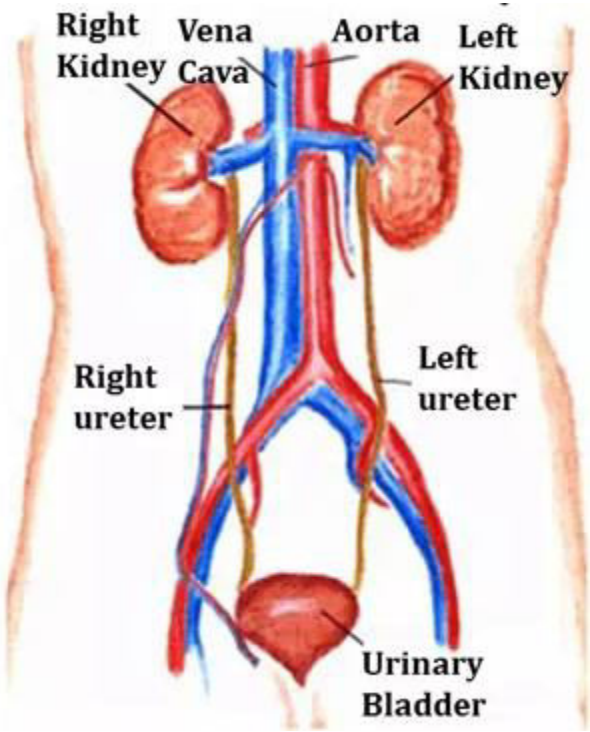
- **The excretory system** is a biological system that removes excess, unnecessary materials from the body fluids of an organism, so as to help maintain internal chemical homeostasis and prevent damage to the body.
- The system is composed of :
- A pair of kidneys
- Ureters
- Urinary bladder and urethra

Excretory System of Man

Kidneys

- There are a pair of kidneys situated towards the back of the lower part of the abdominal cavity on either side of the vertebral column.
- The left kidney is slightly above the right.

Excretory System of Man



- The right kidney is located just below the diaphragm and posterior to the liver.
- The left kidney is situated below the diaphragm and posterior to the spleen.
- Urine from the kidneys pass by a pair of ureters to the bladder where it is stored until it is released via urethra.

Excretory System of Man

- The kidney is a bean-shaped structure with a convex and a concave border.
- A recessed area on the concave border is the renal hilum, where the renal artery enters the kidney and the renal vein and ureter leave.

Excretory System of Man

- The urinary bladder is the organ that collects waste excreted by the kidneys prior to disposal by urination.
- It is a hollow muscular, and distensible (or elastic) organ, present on the pelvic floor.
- Urine enters the bladder via the ureters and streamed out via the urethra.

Excretory System of Man

- Two muscle sphincters surround the urethra where it leaves the bladder.
- One of which is under voluntary control.
- These control the process of urination.

END

Animal Form & Function II

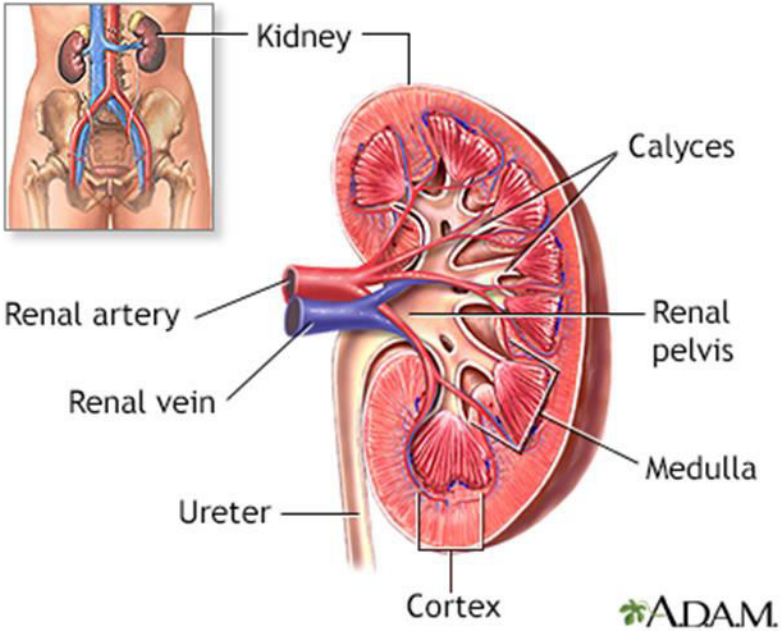
Module 101

Anatomy of Human Kidney

Anatomy of Human Kidney

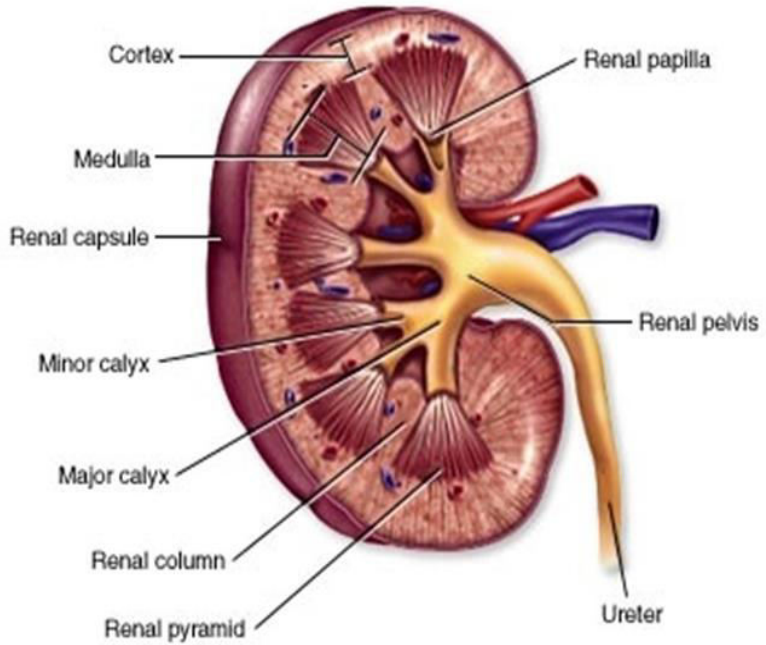
- **The kidney** is divided into two major structures:
- The outer **renal cortex** and the inner **renal medulla.**
- Grossly, these structures take the shape of 85 to 18 cone-shaped renal lobes, each containing renal cortex surrounding a portion of medulla called a renal pyramid.

Anatomy of Human Kidney



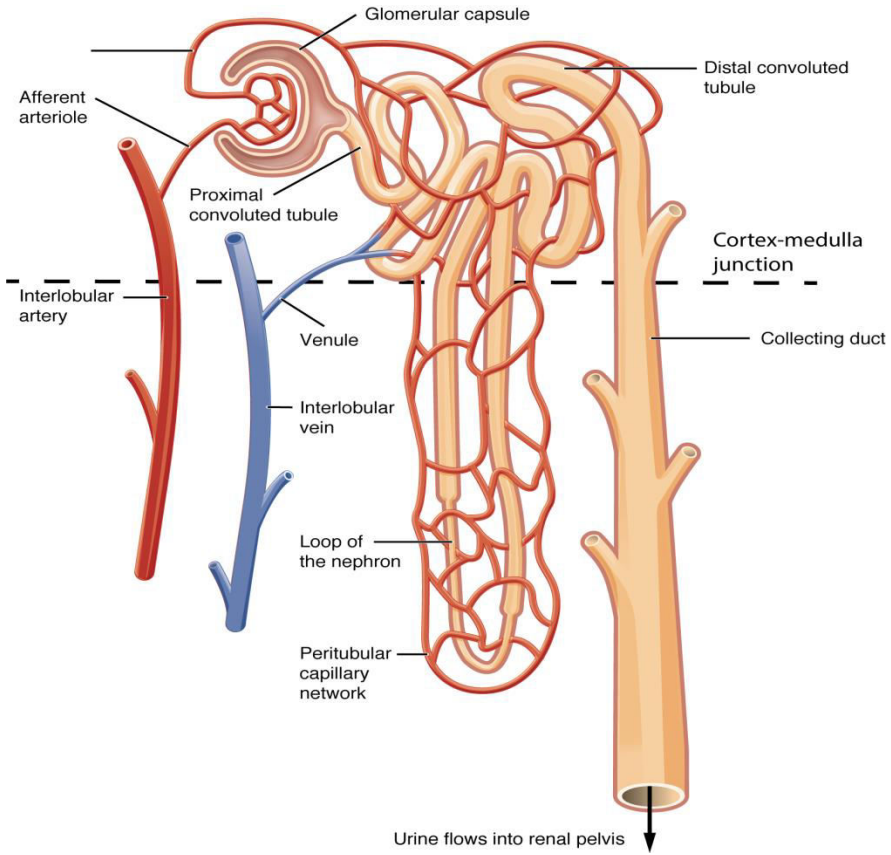
- A frontal section through the kidney reveals an outer region called the renal cortex and an inner region called the medulla.
- The renal columns are connective tissue extensions that converge inward from the cortex through the medulla

Anatomy of Human Kidney



- **The most characteristic features of the medulla are the renal pyramids and renal papillae**
- **The papillae are bundles of collecting ducts that transport urine made by nephrons to the calyces of the kidney for excretion.**

Anatomy of Human Kidney



- **Nephrons**
- **The nephron is the structural and functional unit of the kidney, having 1 M per kidney.**
- **A nephron has two main parts –corpuscle and tubule.**
- **The cup-shaped corpuscle containing the glomerulus is called the Bowman's capsule.**
- **The glomerulus and Bowman's capsule together form the renal corpuscle.**

Anatomy of Human Kidney

- The tubules are small tubes traveling through the inner part of the kidney that regulate the passage of various chemicals to and from the blood.
- The tubule has three parts:

i) the proximal convoluted tubule.

ii) U-shaped curve, the loop of Henle and

iii) the distal convoluted tubule.

Anatomy of Human Kidney

- The cortex is the outer renal tissue composed of:
- The renal corpuscles, as well as the proximal and distal convoluted tubules.
- The medulla is the inner renal tissue, splitting into triangular parts called renal pyramids.

Anatomy of Human Kidney

- **The renal pyramids contain the loops of Henle of each renal tubule, as well as the collecting ducts.**

END

Animal Form & Function II

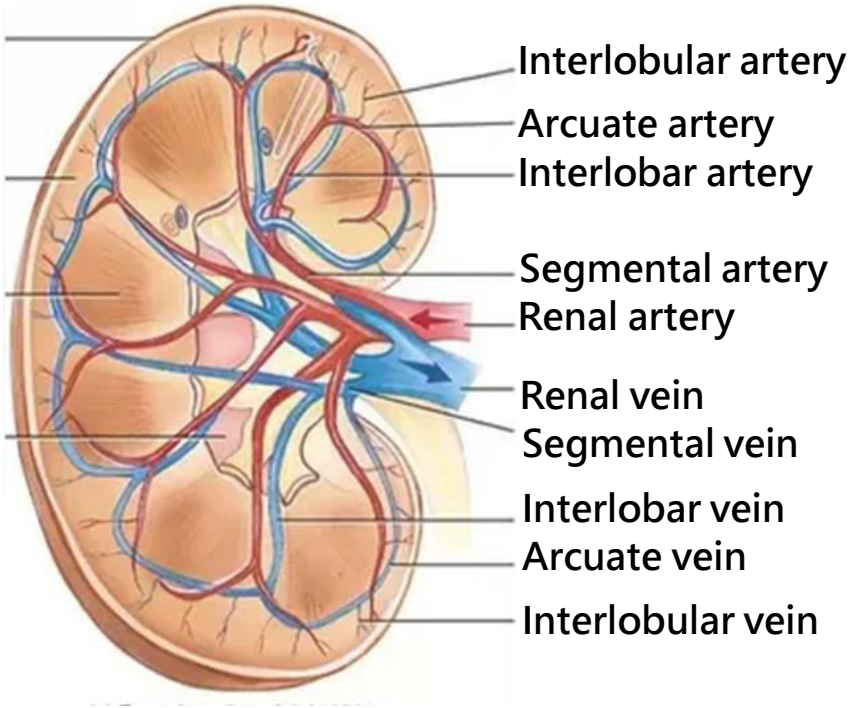
Module 102

Blood Supply to the Kidney

Blood Supply to the Kidney

- **The kidneys** receive blood from the renal arteries, left and right, which branch directly from the abdominal aorta.
- Despite their relatively small size, the kidneys receive approximately 20% of the cardiac output.

Blood Supply to the Kidney

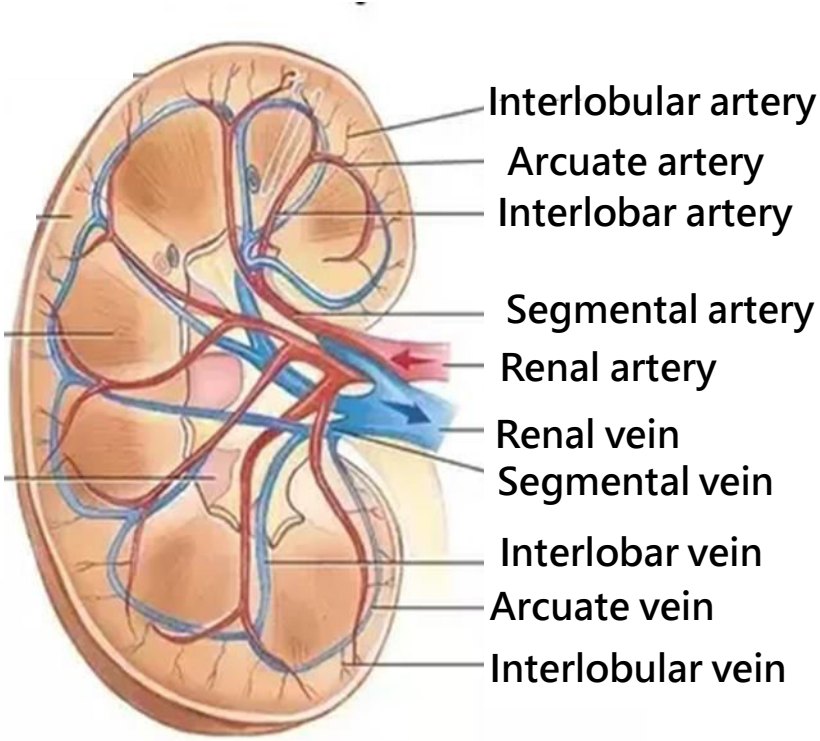


- **Each renal artery branches into segmental arteries, dividing further into interlobar arteries, which penetrate the renal capsule and extend through the renal columns between the renal pyramids.**
- **The interlobar arteries then supply blood to the arcuate arteries that run through the boundary of the cortex and the medulla.**

Blood Supply to the Kidney

- Each arcuate artery supplies several interlobular arteries that feed into the afferent arterioles that supply the glomeruli.

Blood Supply to the Kidney



Frontal section of right kidney

- **Blood drains from the kidneys, ultimately into the inferior vena cava.**
- **After filtration occurs, the blood moves through a small network of small veins (venules) that converge into interlobular veins.**

Blood Supply to the Kidney

- **As with the arteriole distribution, the veins follow the same pattern: the interlobular provide blood to the arcuate veins then back to the interlobar veins, which come to form the renal veins which exiting the kidney .**

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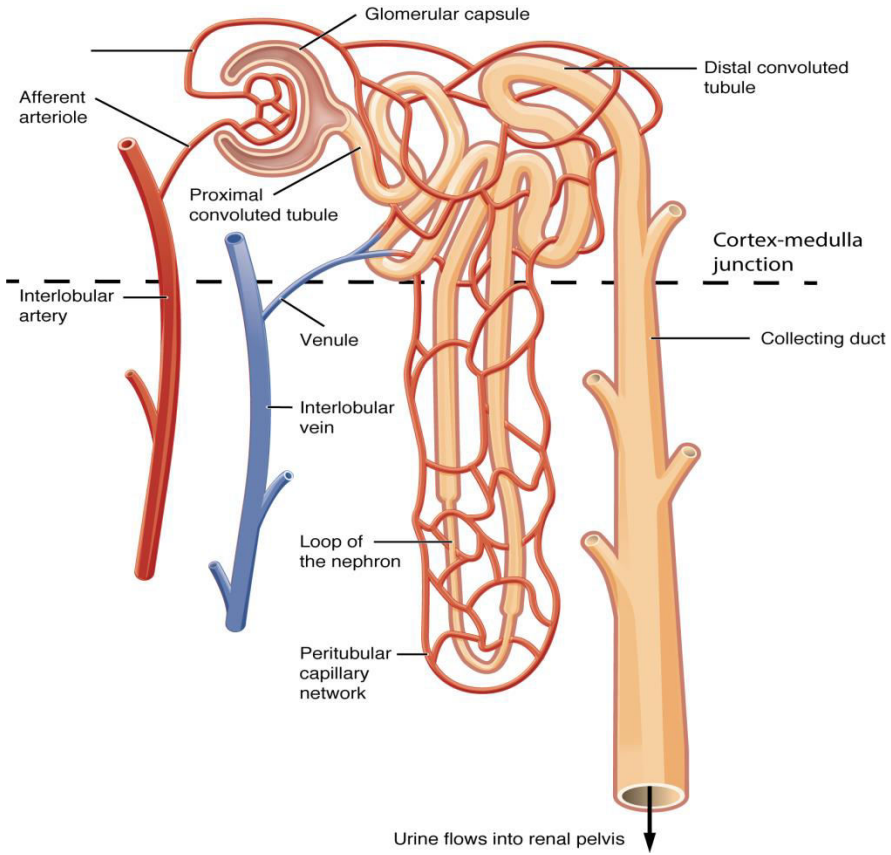
Animal Form & Function II

Module 103

Animal Form & Function II

Structure of Nephron

Structure of Nephron

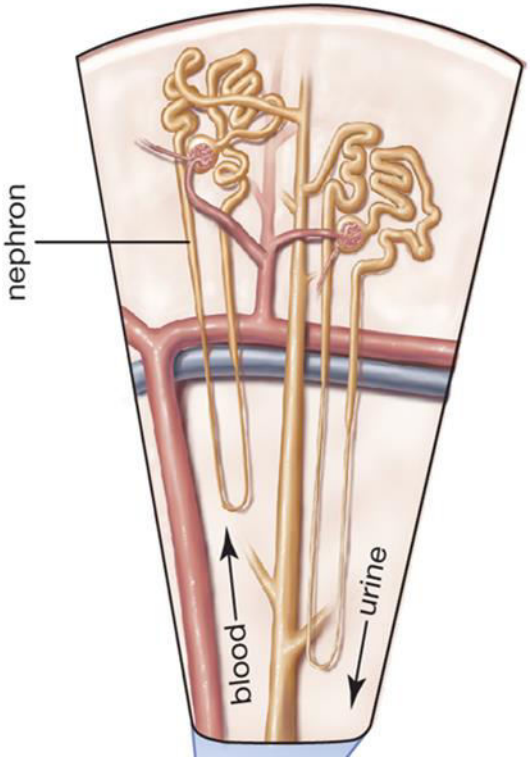


- **The nephron** is the structural and functional unit of the kidney, having 1 M per kidney.
- A nephron has two main parts –corpuscle and tubule.
- The cup-shaped corpuscle containing the **glomerulus** is called the **Bowman's capsule.**
- The glomerulus and Bowman's capsule together form **the renal corpuscle.**

Structure of Nephron

- Nephron is a long tube-like structure, having varying from 35–55 mm long.
- At one end, the tube is expanded, closed, and, folds into a double-walled, a cuplike structure called the Bowman's capsule or renal capsule, which encloses a cluster of microscopic blood vessels called the glomerulus.

Structure of Nephron



- The first part is called the proximal convoluted tubule (PCT) it stays in the renal cortex.
- The second part is called the loop of Henle, (with descending and ascending limbs) that goes through the renal medulla.
- The third part of the renal tubule the distal convoluted tubule (DCT) and this part is also restricted to the renal cortex.

Structure of Nephron

- The cortex is the outer renal tissue which is composed of:
- The renal corpuscles, as well as the proximal and distal convoluted tubules.
- The medulla is the inner renal tissue, splitting into triangular parts called renal pyramids.

Structure of Nephron

- **The renal pyramids contain the loops of Henle of each renal tubule, as well as the collecting ducts.**

END

Animal Form & Function II

Module 104

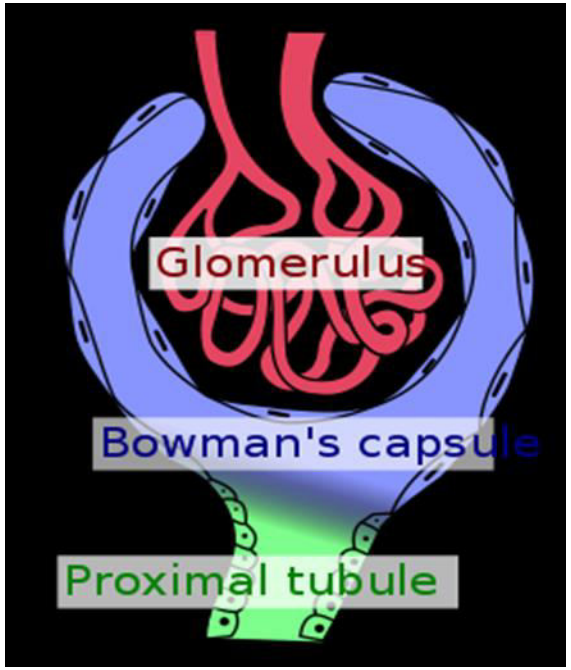
Functioning of the Nephron

Part I

Functioning of the Nephron Part I

- The structure and function of the epithelial cells lining the lumen of the nephron change during the course of the nephron, and have segments named by their location which reflects three different functions:
- 1. Filtration
- 2. Reabsorption
- 3. Secretion

Functioning of the Nephron Part I



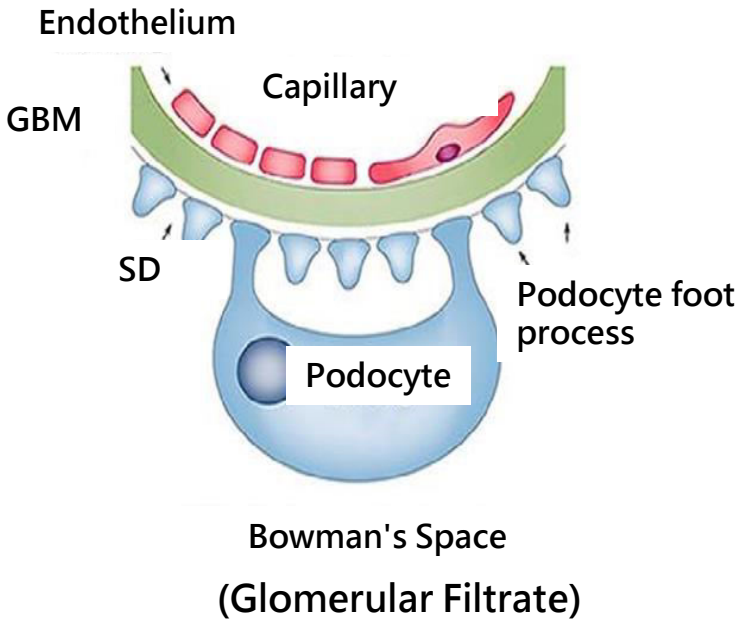
Filtration:

- The glomerulus is the tuft of filtering capillaries located in Bowman's capsule.
- Each glomerulus receives blood supply from an afferent arteriole of the renal circulation.
- The glomerular blood pressure provides the driving force for water and solutes to be filtered out of the blood plasma.

Functioning of the Nephron Part I

- The wall through which the blood filters has a unique structure:
- It is composed of three layers:
- 1. Capillary endothelium,
- 2. Basement membrane, and
- 3. Epithelium (podocytes) of Bowman’s capsule
- The filtrate comes into the space called Bowman's space.

Functioning of the Nephron Part I



- After passing through the endothelial cells of the capillaries, the glomerular basement membrane GBM, and between podocyte foot processes, the filtrate enters the capsule as ultrafiltrate or glomerular filtrate.
- The filtrate contains:
- Glucose, amino acids, ions(Ca^{++}, PO_4^{3-}) and the primary nitrogenous wastes i.e. urea or uric acid.

Animal Form & Function II

Module 106

Animal Form & Function II

Loop of Henle as Counter Current Multiplier

Loop of Henle as Countercurrent Multiplier

Countercurrent Exchange

- The Loop of Henle increases the efficiency of reabsorption by countercurrent flow similar to that in the gills of fishes.
- Generally, the longer the loop of Henle, the more water and ions that can be reabsorbed.

Loop of Henle as Counter Current Multiplier

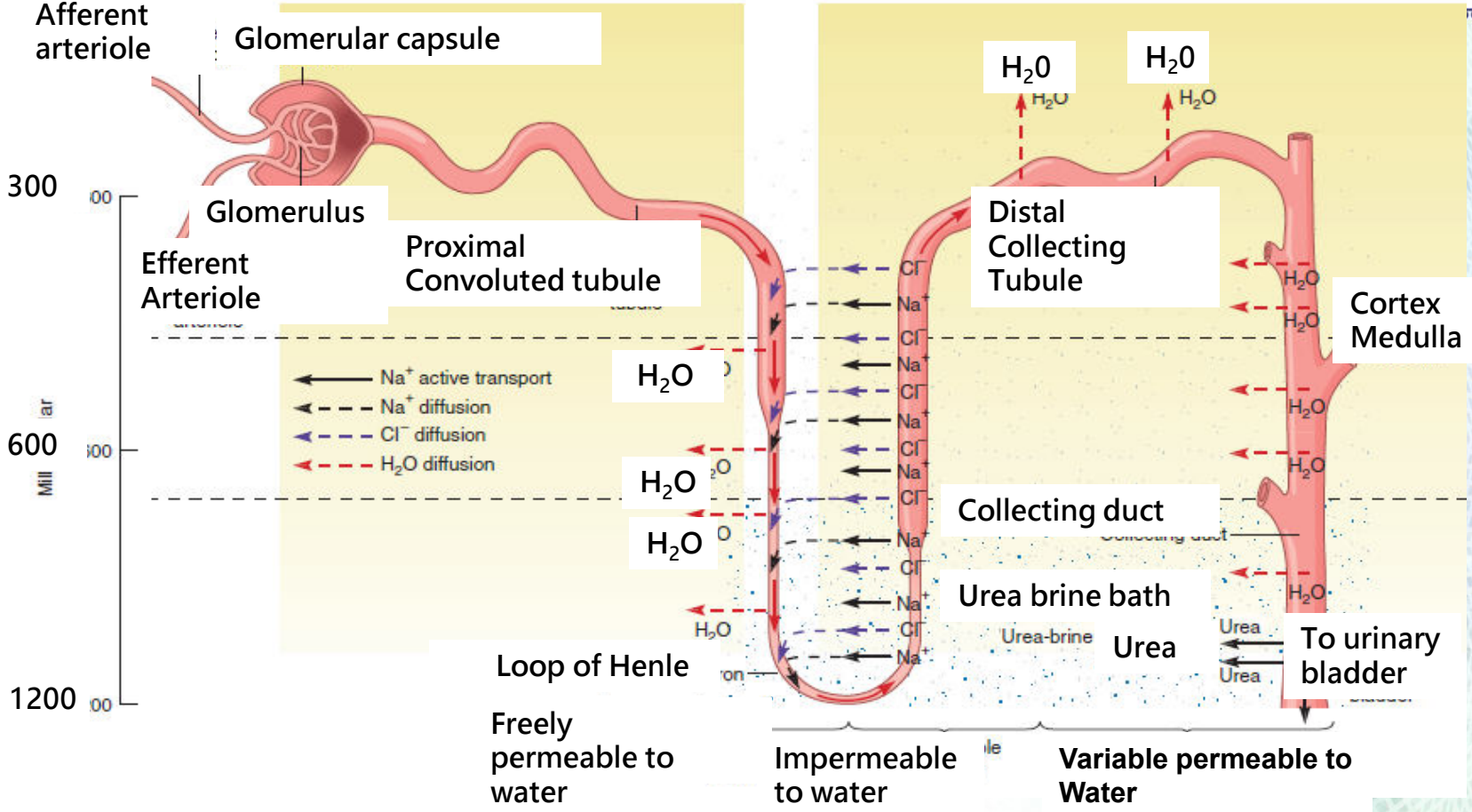
It follows that:

- **Amphibians lack the loop because that are closely associated with aquatic habitats .**
- **Desert rodents (e.g., the kangaroo rat) that live in dry conditions have very long nephron loops and form highly concentrated urine.**

Loop of Henle as Counter Current Multiplier

- The diagram next, shows the countercurrent flow mechanism for concentrating urine.
- The process of reabsorption in the proximal convoluted tubule removes water and some salt (NaCl) from the glomerular filtrate.
- This reduces its volume by approximately 25%.

Loop of Henle as Counter Current Multiplier



Loop of Henle as Counter Current Multiplier

- However, the concentrations of salt and urea are still isosmotic with the extracellular fluid.
- As the filtrate moves to the descending limb of the loop it becomes further reduced in volume and more concentrated.

Loop of Henle as Counter Current Multiplier

- Water moves out of the tubule by osmosis due to the high salt concentration (the “brine-bath”) in the extracellular fluid.
- The highest urea-brine bath concentration is around the lower portion of the loop of the nephron.

Loop of Henle as Counter Current Multiplier

- As the filtrate passes into the ascending limb, sodium (Na) ions are actively transported out of the filtrate into the extracellular fluid, with chloride (Cl) ions following passively.
- Water cannot flow out of the ascending limb because the cells of the ascending limb are impermeable to water.

Loop of Henle as Counter Current Multiplier

- Thus, the salt concentration of the extracellular fluid becomes very high.
- The salt flows passively into the descending loop, only to move out again in the ascending loop, creating a recycling of salt through the loop and the extracellular fluid.

Loop of Henle as Counter Current Multiplier

- **Because the flows in the descending and ascending limbs are in opposite directions, a countercurrent gradient in salt is set up.**
- **The osmotic pressure of the extracellular brine bath is made even higher because of the abundance of urea that moves out of the collecting ducts.**

END

Animal Form & Function II

Module 107

Animal Form & Function II

Overall Functioning of the Metanephric Kidney

Overall Functioning of the Metanephric Kidney

- **Metanephric Kidneys** corresponds to the mature and functional kidney in higher vertebrates.
- The metanephros plays an important role in the urogenital system.

Overall Functioning of the Metanephric Kidney

- Following are the chief functions :
- 1. electrolyte balancing,
- 2. blood filtering,
- 3. BP regulation
- 4. formation of urine

Overall Functioning of the Metanephric Kidney

- To make urine, three physiological mechanisms are involved:
  - Filtration of plasma
  - Reabsorption of needed materials
  - Secretion of a few substances, such as potassium and hydrogen ions, into the renal tubule.

Overall Functioning of the Metanephric Kidney

- About 99% of the filtrate is reabsorbed from the renal tubules into the blood.
- Reabsorption is a highly selective process that returns usable materials to the blood but leaves wastes and excesses of other substances to be excreted in the urine.

Overall Functioning of the Metanephric Kidney

- In secretion, hydrogen ions, certain other ions, and some drugs are actively transported into the renal tubule to become part of the urine.
- Production of concentrated urine depends on a high salt and urea concentration in the interstitial fluid of the kidney medulla.

Overall Functioning of the Metanephric Kidney

- A counter flow of fluid through the two limbs of the loop of Henle concentrates filtrate as it moves down the descending loop and dilutes it as it moves up the ascending loop.
- Water is drawn by osmosis from the filtrate as it passes through the collecting ducts.
- This process concentrates urine in the collecting ducts.

Overall Functioning of the Metanephric Kidney

- **Urine is a watery solution of nitrogenous wastes, excess salts, and other substances not needed by the body.**

END

Animal Form & Function II

Module 108

Animal Form & Function II

How Various Vertebrates Maintain Water and Salt Balance

How Various Vertebrates Maintain Water & Salt Balance

Freshwater Fishes

- Environmental Concentration relative to body fluids is hypoosmotic.
- Urine concentration relative to blood is hypoosmotic.
- Major nitrogenous waste is ammonia.
- Key adaptation: absorb ions through gills.

How Various Vertebrates Maintain Water & Salt Balance

Seawater Fishes

- Environmental Concentration relative to body fluids is hyperosmotic.
- Urine concentration relative to blood is isoosmotic.
- Major nitrogenous waste is ammonia.
- Key adaptation: secrete ions through gills.

How Various Vertebrates Maintain Water & Salt Balance

Sharks

- Environmental Concentration relative to body fluids is isoosmotic.
- Urine concentration relative to blood is isoosmotic.
- Major nitrogenous waste is ammonia.
- Key adaptation: secrete ions through rectal glands..

How Various Vertebrates Maintain Water & Salt Balance

Amphibians

- Environmental Concentration relative to body fluids is hypoosmotic.
- Urine concentration relative to blood is very hypoosmotic.
- Major nitrogenous waste is ammonia and urea.
- Key adaptation: absorb ions through skin.

How Various Vertebrates Maintain Water & Salt Balance

Marine reptiles

- Environmental Concentration relative to body fluids is hyperosmotic.
- Urine concentration relative to blood is isoosmotic.
- Major nitrogenous waste is ammonia and urea.
- Key adaptation: secrete ions through salt glands..

How Various Vertebrates Maintain Water & Salt Balance

Marine Mammals

- **Environmental Concentration relative to body fluids is hyperosmotic.**
- **Urine concentration relative to blood is very hyperosmotic.**
- **Major nitrogenous waste is urea.**
- **Key adaptation: drink some water**

How Various Vertebrates Maintain Water & Salt Balance

Desert Mammals

- **Environmental Concentration relative to body fluids is non comparison.**
- **Urine concentration relative to blood is very hyperosmotic..**
- **Major nitrogenous waste is ammonia and urea.**
- **Key adaptation: produce metabolic water.**

How Various Vertebrates Maintain Water & Salt Balance

Marine reptiles

- **Environmental Concentration relative to body fluids is no comparisn.**
- **Urine concentration relative to blood is weakly hyperosmotic.**
- **Major nitrogenous waste is uric acid.**
- **Key adaptation: drink sea water and use salt glands.**

How Various Vertebrates Maintain Water & Salt Balance

Terrestrial Birds

- **Environmental Concentration relative to body fluids is no comparison.**
- **Urine concentration relative to blood is weakly hyperosmotic.**
- **Major nitrogenous waste is uric acid.**
- **Key adaptation: drink fresh water.**

END

Animal Form & Function II

Module 109

Animal Form & Function II

Hormonal Control of Kidney Function

Hormonal Control of Kidney Function

The Kidneys

- Maintain fluid volume and electrolyte concentration.
- They help maintain the concentration of Na^+ and K^+ in the blood within narrow limits and also regulate blood pH.
- There are many hormones that regulate the homeostatic functions of the kidney.

Hormonal Control of Kidney Function

- These are:
- Renin–Angiotensin–Aldosterone, Endothelin · Natriuretic Hormones, Parathyroid Hormone, Antidiuretic Hormone (ADH)
- Here we discuss only ADH.

Hormonal Control of Kidney Function

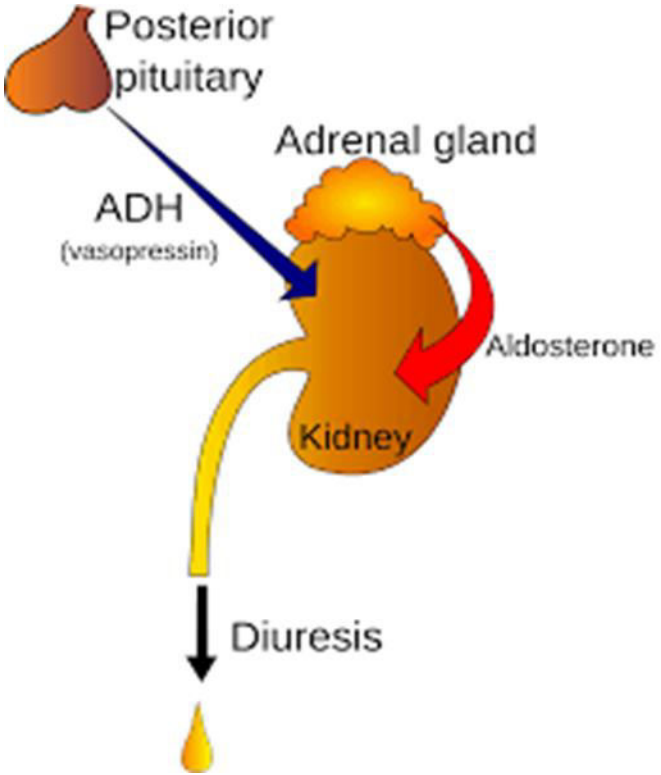
ADH increases water reabsorption

- The amount of urine produced depends on the body's need of water.
- When fluid intake is low, the body begins to dehydrate, and the blood volume decreases.

Hormonal Control of Kidney Function

- As blood volume decreases, the concentration of salts in the blood becomes greater, causing an increase in osmotic pressure.
- Certain receptors in the hypothalamus are sensitive to this osmotic change.
- They signal the posterior lobe of the pituitary gland to release antidiuretic hormone (ADH).

Hormonal Control of Kidney Function



- **ADH makes the collecting ducts more permeable to water so that more water is reabsorbed.**
- **As a result, a small volume of concentrated urine is produced.**

Hormonal Control of Kidney Function

- When we drink a large volume of water, our blood becomes diluted and its osmotic pressure falls.
- This condition inhibits the release of ADH, and decrease in reabsorption of water from the collecting ducts.
- The kidneys then produce a large volume of dilute urine.

Hormonal Control of Kidney Function

- In diabetes insipidus, the pituitary gland malfunctions and does not release enough ADH.
- A person with severe diabetes insipidus may excrete up to 25 quarts of urine each day, a serious water loss.
- The affected individual becomes dehydrated and must drink almost continually to counterbalance fluid loss.

Hormonal Control of Kidney Function

- **Injections of ADH or use of an ADH nasal spray can often control diabetes insipidus.**

END

Animal Form & Function II

Module 110

Animal Form & Function II

Common Disorders of Human Excretory System

Common Disorders of Human Excretory System

Excretory System consists of:

- Kidneys,
- Ureters,
- Urinary bladder and
- Urethra.

Common Disorders of Human Excretory System

Examples of urinary disorders:

- Kidney failure
- Kidney stones
- Urinary tract infections
- Hydronephrosis
- Polycystic Kidney disease
- Uremia
- Inability to control urine flow
- Hypertension due to Renin Secretion
- Diabetes Insipidus

Common Disorders of Human Excretory System

Kidney failure

- Kidney failure means when they stop working well enough for survival without dialysis or a kidney transplant.
- Kidney failure the last stage of chronic kidney disease.

Common Disorders of Human Excretory System

END

- **Complete and irreversible kidney failure is sometimes called end-stage renal disease, or ESRD.**
- **When kidneys stop working completely, the body fills with extra water and waste products.**
- **This condition is called uremia.**
- **Untreated uremia may lead to death.**

Animal Form & Function II

Module 111

Animal Form & Function II

Kidney Stones

Kidney Stones

Kidney stones

- are hard deposits made of minerals and salts that form inside the kidneys. Also called renal calculi.
- Main causes of kidney stone formation are:
- Diet, obesity, certain supplements , and sometimes medications.

Kidney Stones

- Kidney stones are formed when urine contains more crystal-forming substances — such as calcium, oxalate and uric acid.
- At the same time, urine may lack substances that prevent crystals from sticking together, creating an ideal environment for kidney stones to form.

Kidney Stones

Types of Kidney Stones

- Calcium Oxalate Stones. (The most common type of kidney stone.)
- Calcium Phosphate Stones.
- Uric Acid Stones.
- Cystine Stones.

Kidney Stones

Lithotripsy

- It is a procedure that uses shock waves or LASER *(Light Amplification by Stimulated Emission of Radiation)* to break up stones into small particles in the kidney and parts of the ureter.
- The particles of small stone then exit the body when a person urinates.

Kidney Stones

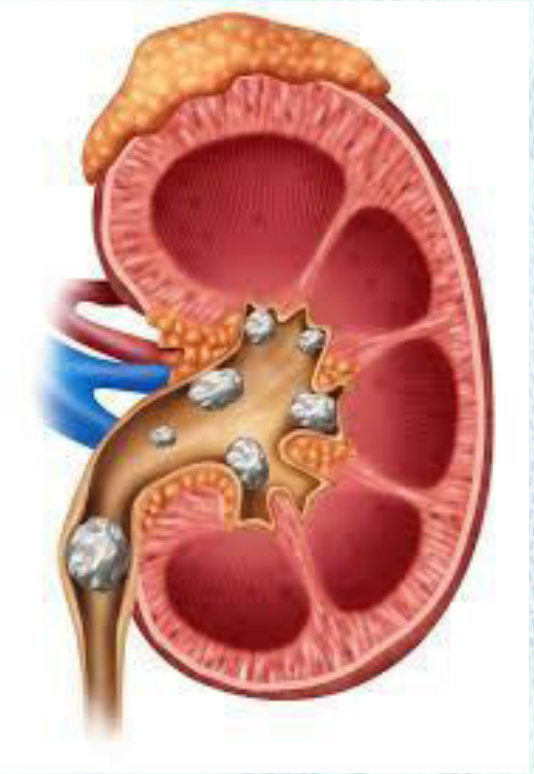
Solutions for Kidney Pain at-Home

- Stay Hydrated.
- Drink Cranberry Juice.
- Take Probiotics.
- Use Non-Aspirin Pain Killers.
- Take a Warm Epsom Salt Bath ($Mg\ SO_4$)

Kidney Stones











Animal Form & Function II

Module 112

Animal Form & Function II

Urinary Tract Infection

Urinary Tract Infection

- **A urinary tract infection (UTI)**
- **is an infection in any part of the urinary system —kidneys, ureters, bladder and urethra.**
- **Most infections involve the lower urinary tract — the bladder and the urethra.**
- **Women are at greater risk of developing UTI than men.**

Urinary Tract Infection

- UTI is caused by microorganisms — usually bacteria — that enter the urethra and bladder, causing inflammation and infection.
- Though UTI most commonly happens in the urethra and bladder, bacteria can also travel up the ureters and infect the kidneys.

Urinary Tract Infection

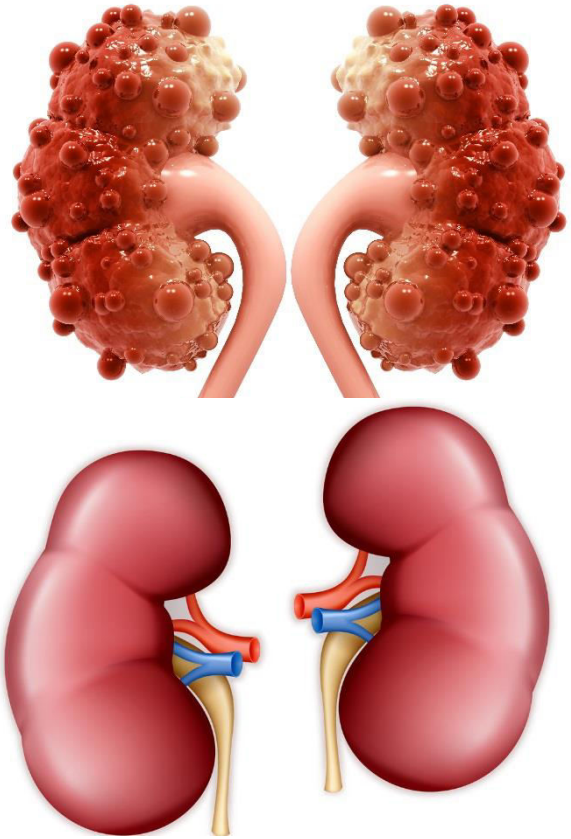
- Some signs of UTI include:
- cloudy urine
- painful urination
- burning with urination
- a weak urine stream
- back pain and fever

Urinary Tract Infection

Natural Remedy of UTI

- Drinking lot of water
- Emptying bladder when needed.
- Eat Vitamin C that help destroy bad bacteria due to its acidity.
- Vitamin C also help protect from future infections

Polycystic Kidney Disease



Polycystic kidney disease (PKD)

- It is an inherited disorder in which clusters of cysts develop within the kidneys, causing kidneys to enlarge and lose function over time.
- Cysts are noncancerous round sacs containing fluid.
- The growth of cysts causes the kidneys to become enlarged and can lead to kidney failure.

Urinary Tract Infection

- **UTIs are one of the most common complications of hydronephrosis.**

END

Animal Form & Function II

Module 113

Hydronephrosis

Hydronephrosis

- **Hydronephrosis** is a condition that typically occurs when a kidney swells due to urine failing to properly drain from the kidney to the bladder.
- This swelling most commonly affects only one kidney, but it can involve both kidneys.

Hydronephrosis

- Hydronephrosis is not a primary disease.
- It is a secondary condition that results from some other underlying disease.
- It is the result of a blockage or obstruction in the urinary tract.
- Hydronephrosis is said to affect about 1 in every 100 babies.

Hydronephrosis

- Normally, urine flows through the urinary tract with minimal pressure.
- Pressure can build up if there is an obstruction in the urinary tract.
- If this pressure remains build up for an extended period, kidney can enlarge.

Hydronephrosis

- **Kidney may become so enlarged with urine that it starts pressing nearby organs.**
- **If it is left untreated for too long, this pressure can cause kidneys to lose function permanently.**

END

Animal Form & Function II

Module 114

Animal Form & Function II

Summary of Chapter- Temperature and Body Fluid Regulation

Summary of the Chapter Part I

- 1. Thermoregulation is a complex and important physiological process for maintaining heat homeostasis despite environmental changes.
- 2. Ectotherms generally obtain heat from the environment, whereas endotherms generate their own body heat from metabolic processes.

Summary of the Chapter Part I

- 3. Homeotherms generally have a relatively constant core body temperature, while heterotherms have a variable body temperature.
- 4. The high, constant body temperature of birds and mammals also depends on:
- insulation,
- panting,
- sweating

Summary of the Chapter Part I

- specific behaviors,
- vasoconstriction
- vasodilation of peripheral blood vessels and in some species
- a rete mirabile system
- Thermogenesis involves mainly shivering, enzymatic activity, brown fat, and high cellular metabolism.

Summary of the Chapter Part I

- The hypothalamus is the temperature regulating center that functions as a thermostat with a fixed set point.
- This set point can either rise or fall during hibernation or torpor.
- Some invertebrates have:
  - contractile vacuoles,
  - flame-cell systems,
  - antennal (green) glands,

Summary of the Chapter Part I

- maxillary glands, coxal glands, nephridia, or Malpighian tubules for osmoregulation.
- The osmoregulatory system of vertebrates governs the concentration of water and ions; the excretory system eliminates metabolic wastes, water, and ions from the body.

Summary of the Chapter Part I

- **Freshwater animals tend to lose ions and take in water.**
- **To avoid hydration, freshwater fishes rarely drink much water, have impermeable body surfaces covered with mucus, excrete a dilute urine, and take up ions through their gills.**

END

Animal Form & Function II

Module 115

Animal Form & Function II

Summary of Chapter- Temperature and Body Fluid Regulation Part II

Summary of the Chapter Part II

- Marine animals tend to take in ions from the seawater and to lose water.
- To avoid dehydration, they frequently drink water, have relatively permeable body surfaces, excrete a small volume of concentrated urine, and secrete ions from their gills.

Summary of the Chapter Part II

- Amphibians can absorb water across the skin and urinary bladder wall.
- Desert and marine reptiles and birds have salt glands to remove and secrete excess salt (NaCl).
- In reptiles, birds, and mammals, the kidneys are important osmoregulatory structures.

Summary of the Chapter Part II

- The functional unit of the kidney is the nephron, composed of the glomerular capsule, proximal convoluted tubule, loop of the nephron, distal convoluted tubule, and collecting duct.
- The loop of the nephron and the collecting duct are in the kidney's medulla; the other nephron parts lie in the kidney's cortex.

Summary of the Chapter Part II

- Urine passes from the pelvis of the kidney to the urinary bladder.
- To make urine, kidneys produce a filtrate of the blood and reabsorb most of the water, glucose, and needed ions, while allowing wastes to pass from the body.

Summary of the Chapter Part II

- Three physiological mechanisms are involved:
- Filtration of the blood through the glomerulus,
- Reabsorption of the useful substances, and
- Secretion of toxic substances.

Summary of the Chapter Part II

- **In those animals with a loop of the Henle, salt (NaCl) and urea are concentrated in the extracellular fluid around the loop, allowing water to move by osmosis out of the loop and into the peritubular capillaries.**

END

Animal Form & Function II

Module 116

Animal Form & Function II

Reproduction and its Types

Reproduction and its Types

- **Reproduction** is giving rise of new individuals like their parents.
- Reproduction is the basic characteristic of living things.
- Reproduction is vital for the survival and the continuity of each species.
- The members of a species produce new individuals to replace those that die.

Reproduction and its Types

- There are two types of reproduction:
- 1. Asexual reproduction and
- 2. Sexual reproduction

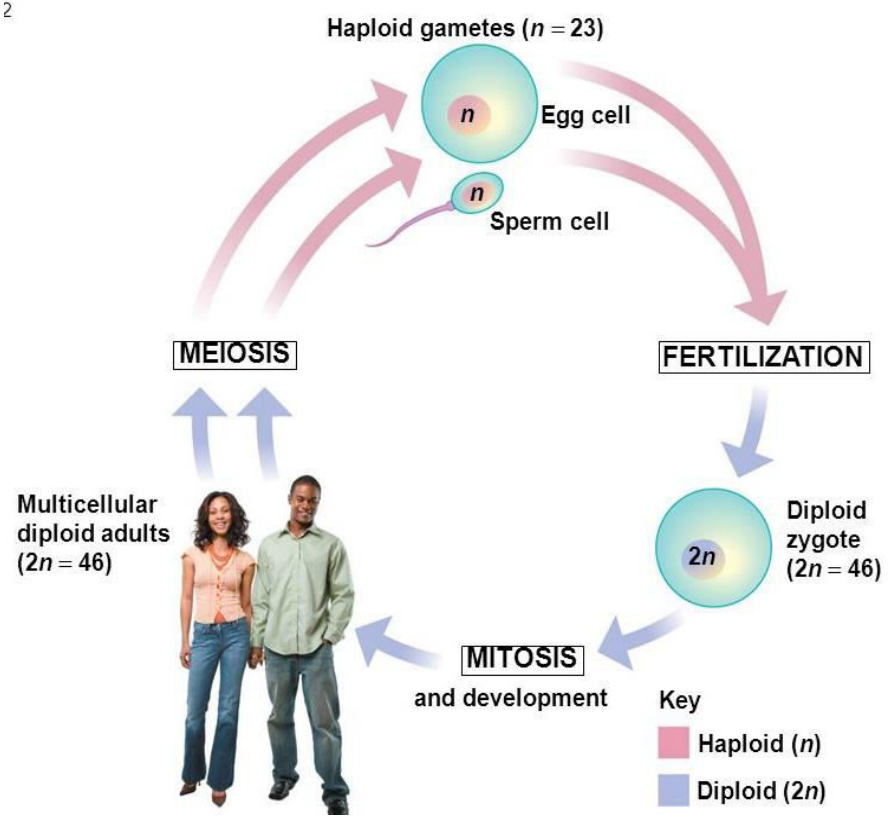
In **Asexual reproduction**

- the organisms reproduce by mitotic divisions producing offspring that are identical to their parents.
- It does not involve the fusion of gametes.

Reproduction and its Types

- Organisms that reproduce asexually are usually found in stable environment in which they are best suited.
- Examples: Amoeba, Plasmodium etc.

Reproduction and its Types



Sexual reproduction

- Involves the fusion of male and female gametes i.e sperms and eggs to form zygote.
- This zygote develops to form a unique new individual having two kinds of cells:
- somatic cells and
- germ cells

Reproduction and its Types

- **Meiotic division of germ cells give rise to haploid gametes in both male and female parents.**
- **The ovum and sperm fuse to form zygote.**
- **The zygote then divides by mitotic divisions to produce large number of cells that differentiates to form different organs.**

END

Animal Form & Function II

Module 117

Animal Form & Function II

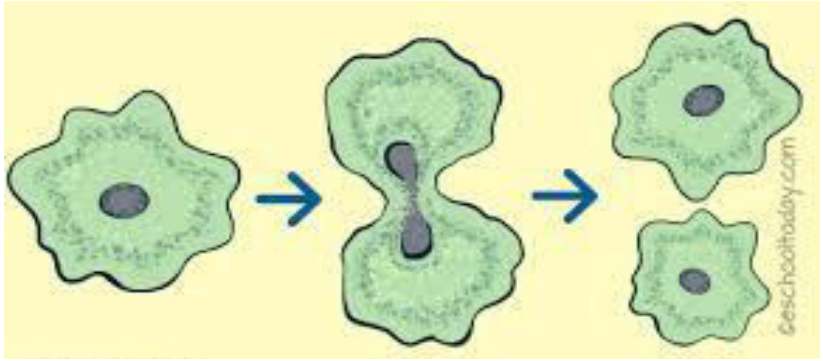
Asexual Reproduction in Invertebrates

Asexual Reproduction in Invertebrates

Asexual Reproduction:

- Following are the forms:
  - Fission
    a. binary fission
    b. multiple fission
  - Budding (both internal and external)
  - Fragmentation
  - Parthenogenesis

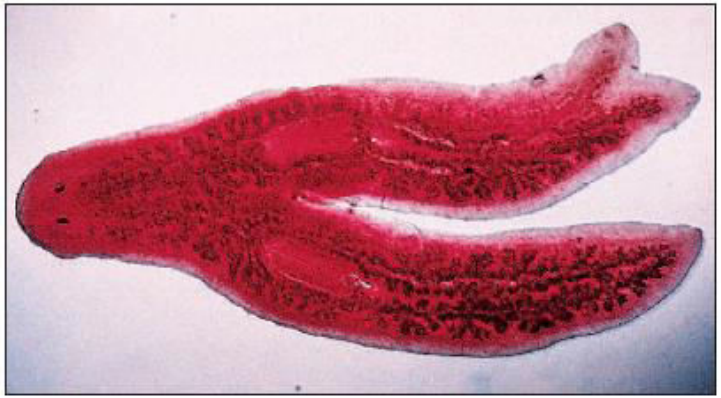
Asexual Reproduction in Invertebrates



Fission :

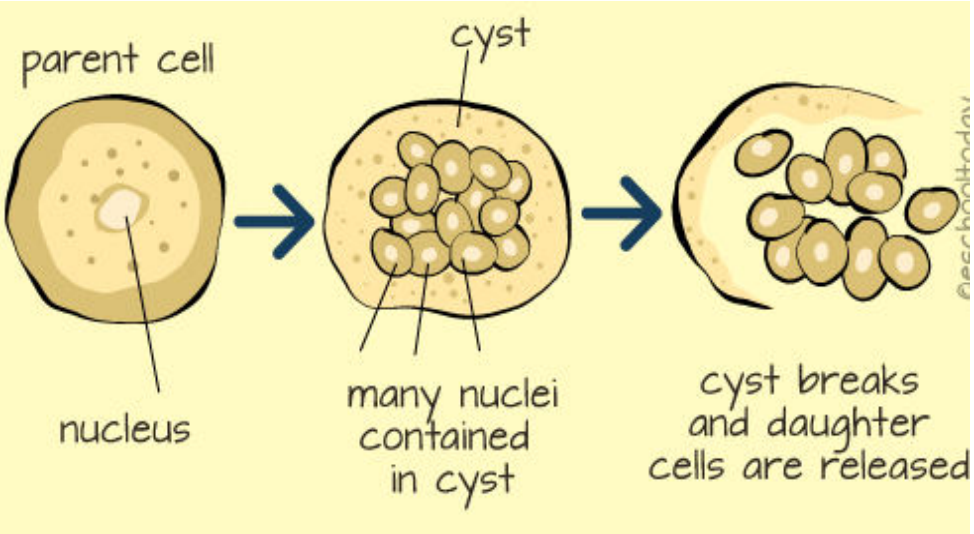
- **Binary Fission is the division of one cell, body, or body part into two.**
- **In this process, the *Amoeba* pinches in two by an inward furrowing of the plasma membrane.**
- **Binary fission is common in protozoa (*Amoeba, Plasmodium, Paramecium* etc)**

Asexual Reproduction in Invertebrates



- In fission, the plane of division may be asymmetrical, transverse, or longitudinal, depending on the species.
- For example*, Planarian* reproduce by longitudinal fission

Asexual Reproduction in Invertebrates



Multiple fission

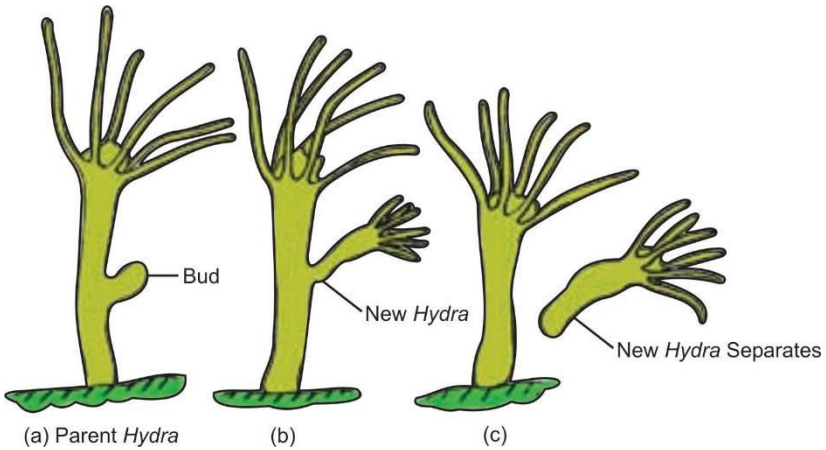
- In this process, the nucleus divides mitotically to produce a large number of nuclei before the cell divides.
- Each nucleus, with the surrounding cytoplasm, forms a daughter cell.
- These daughter cells then separate.

Asexual Reproduction in Invertebrates



Some flatworms and annelids reproduce by forming numerous constrictions along the length of the body; a chain of daughter individuals results This type of asexual reproduction is an other example of multiple fission

Asexual Reproduction in Invertebrates



- **Budding** is a form of asexual reproduction that results from the outgrowth of body region leading to a separation from the original organism into two individuals.
- It may be External or Internal.
- External Budding occurs commonly in some invertebrate animals such as corals and hydras.

Asexual Reproduction in Invertebrates

- The bud cells proliferate to form a cylindrical structure, which develops into a new animal, usually breaking away from the parent.
- If the buds remain attached to the parent, they form a colony.

Asexual Reproduction in Invertebrates

- **Internal budding** is by the formation of gemmules which are collections of many cells surrounded by a body wall.
- When the body of the parent dies and degenerates, each gemmule gives rise to a new individual.
- Example is freshwater sponges.

Asexual Reproduction in Invertebrates



- **Fragmentation** is a type of asexual reproduction that occurs in some cnidarians, platyhelminths, and echinoderms.

  For example, in sea anemones, as the organism moves, small pieces break off from the adult which develop into new individuals

Animal Form & Function II

Module 118

Asexual Reproduction in Invertebrates Part II

Asexual Reproduction in Invertebrates Part II

Asexual Reproduction:

- Following are the forms:
  - Fission,
    a. binary fission
    b. multiple fission
  - Budding (both internal and external),
  - Fragmentation
  - Parthenogenesis

Asexual Reproduction in Invertebrates Part II

- **Parthenogenesis**
- **Parthenogenesis is a form of asexual reproduction in which development of egg occur without being fertilized.**

Asexual Reproduction in Invertebrates Part II

Types of Parthenogenesis:

- Diploid Parthenogenesis
- Haploid Parthenogenesis

Diploid Parthenogenesis:

- In this type, no meiosis occurs and the eggs are formed by mitosis.
- The eggs develop and the offspring are the copies or clones of the parents.

Asexual Reproduction in Invertebrates Part II

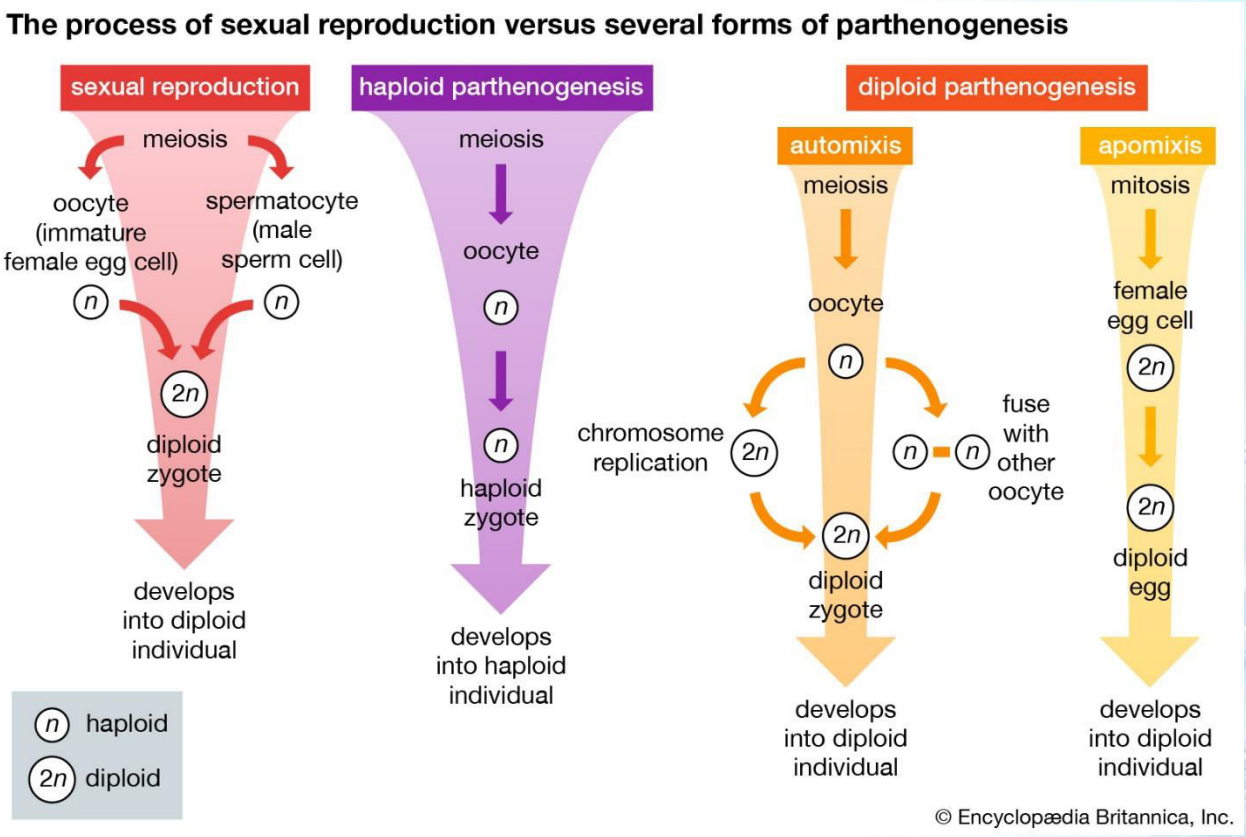
- In some cases egg are formed by modified form of meiosis involving total non-disjunction of the chromosomes thus retaining the diploid number.
- Such eggs develop into young females.
- Examples: some species of flat worm, rotifers and insects.

Asexual Reproduction in Invertebrates Part II

Haploid Parthenogenesis:

- In this type of Parthenogenesis haploid ova are formed by meiosis and they may or may not be fertilized by sperm.
- If fertilized, the zygotes develop into females (Queen or sterile female workers in bee).
- If unfertilized the eggs develop to become haploid males (drones). On maturity they produce sperms by mitosis.

Asexual Reproduction in Invertebrates Part II



Asexual Reproduction in Invertebrates Part II

- Overall, animals that reproduce parthenogenetically have substantially less genetic variability than do animals with chromosome sets from two parents.
- Parthenogenesis also plays an important role in social organization in colonies of certain bees, wasps, and ants.

Asexual Reproduction in Invertebrates Part II

- **Parthenogenesis has the advantage of accelerating the rate of reproduction.**

Asexual Reproduction in Invertebrates Part II

- **In fact, mature eggs of species that do not undergo parthenogenesis can sometimes be activated to develop without fertilization by pricking them with a needle, by exposing them to high concentrations of calcium, or by altering their temperature.**

END

Animal Form & Function II

Module 119

Advantages and Disadvantages of Asexual Reproduction

Advantages & Disadvantages of Asexual Reproduction

Advantages:

1. Large number of offspring are produced very quickly from only one parent when conditions are favorable.
2. Large colonies can form that can out compete other organisms for nutrients and water.

Advantages & Disadvantages of Asexual Reproduction

3. Large number of organisms mean that species may survive when conditions or number of predators change.
4. Energy is not required to find mate

Advantages & Disadvantages of Asexual Reproduction

Disadvantage

1. Offspring are genetic clones.
2. A negative mutation can make asexually produced organisms susceptible to disease and can destroy large number of offspring.
3. Some methods of asexual reproduction produce offspring that are close together and compete for food and space.

Advantages & Disadvantages of Asexual Reproduction

4. **Usnfavorable conditions such as extreme temperatures can wipe out entire colonies.**

END

Animal Form & Function II

Module 120

Animal Form & Function II

Fertilization and its Types

Fertilization and Its Types

Fertilization

- Fertilization is the fusion of haploid gametes, egg and sperm, to form the diploid zygote.
- There are delicate differences in the fertilization process which occurs naturally within the body or through reproductive technologies outside the body.
- In both cases diploid zygote is formed.

Fertilization and Its Types

Stages of Fertilization Process:

- 1) sperm and egg preparation,
- 2) sperm-egg recognition
- 3) fusion of sperm and egg membranes
- 4) Penetration of sperm nucleus into the egg.
- 5) Fusion of pronuclei and activation of the zygote.

Fertilization and Its Types

Types of fertilization

- **External fertilization-**
- **Fertilization out of the body i.e. in water.**
- **Internal fertilization- In this type of fertilization the eggs are fertilized within the female reproductive tract.**

END

Animal Form & Function II

Module 121

Animal Form & Function II

External Fertilization

External Fertilization

External fertilization

- **It is a mode of reproduction in which a male organism's sperm fertilizes a female organism's egg outside of the female's body.**
- **The release of eggs and sperm into the water is known as spawning.**

External Fertilization

- Many invertebrates (e.g., sponges and corals) simply release their gametes into the water in which they live allowing external fertilization to occur.
- In these invertebrates, the transient structures, such as coelomic ducts, metanephridia, sperm ducts, or oviducts are used to release gametes from the body.

External Fertilization

- **Invertebrates utilizing external fertilization are mostly benthic, sessile, or both, including animals such as coral, sea anemones, and tube-dwelling polychaetes.**
- **Among vertebrates, external fertilization is most common in amphibians and fish.**

END

Animal Form & Function II

Module 122

Animal Form & Function II

Internal Fertilization

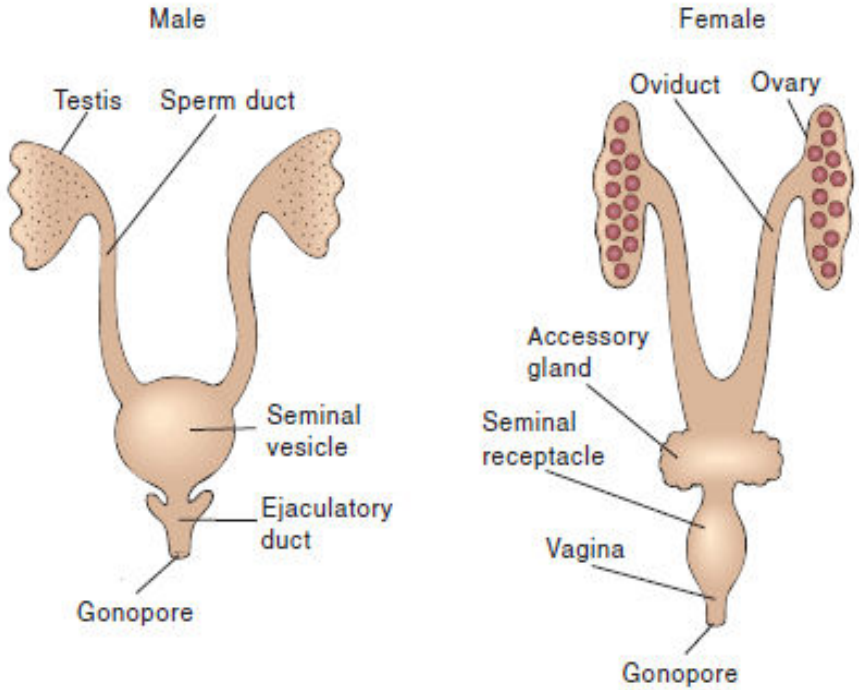
Internal Fertilization

- **Internal fertilization** is the union of an egg cell with a sperm during sexual reproduction inside the female body.
- For this to happen there needs to be a method for the male to introduce the sperm into the female's reproductive tract.

Internal Fertilization

- In some invertebrates (e.g., arrow worms, leeches, some insects) the sperms are produced in the testes and transported via a sperm duct to a storage packet area called spermatophores.
- Spermatophores provide a protective casing for sperms and facilitate the transfer of large numbers of sperm with minimal loss.

Internal Fertilization



- Some spermatophores are even motile and act as independent sperm carriers.
- Sperm or the spermatophores are then passed into an ejaculatory duct to a copulatory organ (e.g. cirrus, gonopore).
- The copulatory organ is used as an intromittent structure to introduce sperm into the female's system.

Internal Fertilization

- Various accessory glands (e.g., seminal vesicle) may be present in males that produce seminal fluid.
- In the female, ova (eggs) are produced in the ovaries and transported to the oviduct.

Internal Fertilization

- Sperm move up the oviduct, where they encounter the ova and fertilize them.
- Accessory glands (e.g.,those that produce egg capsules or shells) may also be present in females.
- In some sexually reproducing animals occasionally depart from this basic reproductive mode and exhibit variant forms of sexual reproduction.

Internal Fertilization

- **The various forms of sexual reproduction are:**
- **1. Hermaphroditism**
- **2. Sequential hermaphroditism**

END

Animal Form & Function II

Module 123

Animal Form & Function II

Hermaphroditism

Hermaphroditism

- **Hermaphroditism, is the condition of having both male and female reproductive organs.**
- **Hermaphroditic animals are mostly invertebrates such as worms, bryozoans, trematodes (flukes), snails, slugs, and barnacles.**

Hermaphroditism



- They are usually parasitic, slow-moving, or permanently attached to another animal or plant.
- Hermaphroditism is especially beneficial to sessile animals that occasionally encounter the opposite sex.

Hermaphroditism

- Another variation is sequential hermaphroditism.
- This occurs when an animal is one sex during one phase of its life cycle and the opposite sex during another phase.
- Hermaphrodites are either Protogynous or Protandrous

Hermaphroditism

- **Protogynous** animals are born female and at some point of their life span change sex to males.
- Examples: ***Bluehead wrasses, Rana temporaria.***
- **Protandrous** animals are male during their early life history and female later in the life history.
- Examples: ***Many fish, mollusks, and crustaceans.***

Hermaphroditism

- **A change in the sex ratio of a population is one factor that can induce sequential hermaphroditism, which is common in oysters.**

END

Animal Form & Function II

Module 124

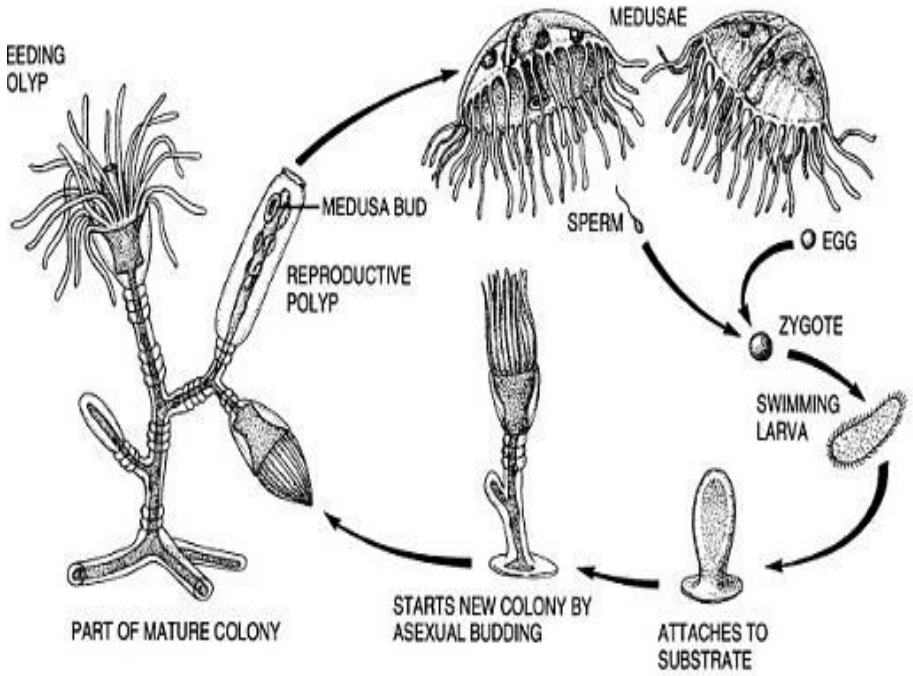
Sexual Reproduction in Invertebrates

Sexual Reproduction in Invertebrates

Sexual Reproduction:

- Most invertebrates reproduce partly through sexual reproduction.
- They produce specialized reproductive cells that undergo meiosis to produce smaller, motile spermatozoa or larger, non-motile ova.
- These fuse to form zygotes, which develop into new individuals.

Sexual Reproduction in Invertebrates



- **Cnidarians usually reproduce sexually, using eggs and sperms (Medusae).**
- **Depending on the species, cnidarians can be monoecious (also called hermaphroditic), with individuals capable of producing both eggs and sperm, or they can be dioecious, with individuals of separate sexes for gamete production.**

Sexual Reproduction in Invertebrates

- **Flatworms** are hermaphroditic (having both male and female sex organs) and they typically reproduce both sexually and asexually.
- The majority of sexual reproduction is through cross-fertilization (where both individuals fertilize each other).

Sexual Reproduction in Invertebrates

- **Polychaete** species normally have separate sexes, while most oligochaetes are hermaphrodites.
- In hermaphroditic species, each individual has both male and female sex organs.

Sexual Reproduction in Invertebrates

- **Arthropods** have a great diversity in sexual reproduction.
- A few species of insects such as honey bee reproduce by parthenogenesis.
- Aquatic arthropods may breed by external fertilization, as frogs do, or by internal fertilization.

Sexual Reproduction in Invertebrates

- All known terrestrial arthropods are dioceous and use internal fertilization.
- Once the eggs are fertilized, the female usually lays the egg, and continues developing outside of the mother's body.
- Most male terrestrial arthropods produce waterproof sperm packets, spermatophores,, which the females take into their bodies.

Sexual Reproduction in Invertebrates

- **Most arthropods lay eggs, but scorpions are ovoviparous: they produce live young after the eggs have hatched inside the mother, and are noted for prolonged maternal care.**

END

Animal Form & Function II

Module 125

Animal Form & Function II

Advantages & Disadvantages of Sexual Reproduction

Advantages & Disadvantages of Sexual Reproduction

Advantages of Sexual Reproduction

- Offspring are genetically unique because of crossing over, Independent assortment and random fusion of gametes.
- Slower rate of reproduction but faster adaptation/evolution.
- Favorable when the environment is not stable.
- Lower extinction rate.

Advantages & Disadvantages of Sexual Reproduction

- Fast removal of bad mutations.
- Better adaptation to host-parasite race.
- Offspring are dispersed widely to end up in different places from their parents.

Advantages & Disadvantages of Sexual Reproduction

END

Disadvantages

- **Slower reproductive rate, fewer offspring are produced over longer time span.**
- **Potential for spread of sexually transmitted diseases through population.**
- **Energetically costly, i.e. requires a lot of ongoing energy input from the parent.**

Animal Form & Function II

Module 126

Animal Form & Function II

Sexual Reproduction in Vertebrates

Sexual Reproduction in Vertebrates

Sexual reproduction

- in animals involves the production and fusion of two types of gametes—sperm and eggs.
- Typically, two different individuals are required.
- A male parent contributes sperm, and a female parent contributes an egg, or ovum.

Sexual Reproduction in Vertebrates

- The sperm provides genes coding for some of the male parent's traits, and the egg contributes genes coding for some of the female parent's traits.
- The egg is typically large and non-motile, with a store of nutrients that supports the development of the embryo.

Sexual Reproduction in Vertebrates

- The sperm is usually small and motile and adapted to propel itself by beating its long, whip-like flagellum.
- When sperm and egg unite, a zygote, or fertilized egg, is produced.
- The zygote develops into a new animal, similar to both parents but not identical to either.

Sexual Reproduction in Vertebrates

- Sexual reproduction typically involves remarkably complex structural, functional, and behavioral processes.
- The strong drive to mate or reproduce dominates the lives of many vertebrates.
- Females of most mammal species come into heat or estrus about the same time each year.

Sexual Reproduction in Vertebrates

- **Genetic, hormonal, and nervous system controls the estrus cycle so that the young are born when environmental conditions are favorable.**

END

Animal Form & Function II

Module 127

Animal Form & Function II

Basic Reproductive Strategies in Vertebrate

Basic Reproductive Strategies in Vertebrates

- **Fishes** have various reproductive methods, structures, and physiology to adapt themselves to a great variety of aquatic habitats.
- High fecundity i.e. releasing of eggs in thousands to millions. is well known strategy.
- Cod fish for example releases about five million eggs in a single spawn.

Basic Reproductive Strategies in Vertebrates

- **In Amphibians** the reproductive strategies are much more diverse than those observed in other groups of vertebrates.
- Anurans (toads and frogs) for example have a variety of reproductive adaptations towards terrestriality.

Basic Reproductive Strategies in Vertebrates





- Ovoviviparity, and Viviparity are the adaptations for successful invasion of mountainous environments by amphibians.
- A few frogs and toads, the fire salamander and several species of caecilians, are viviparous:
- These adaptations are evolutionary conquest of terrestrial environments by vertebrates.

Basic Reproductive Strategies in Vertebrates

- **Reptiles, Birds, and early mammals** exhibit reproductive adaptations of advanced mammals including humans.

i. The reptilian system includes: Shelled, desiccation-resistant eggs.

ii. Three basic embryonic membranes.

Basic Reproductive Strategies in Vertebrates

iii. Flat embryo.
iv. Gastrulation atop a huge yolk mass.

- The same process of gastrulation is seen in mammalian embryos even though the massive yolk mass has been lost.

Basic Reproductive Strategies in Vertebrates

- During gestation the embryo is nourished with nutrients and oxygen, and protected from any attack by the mother's immune system.
- After birth, the mammals nurse their young with milk from the mammary glands, just as primates do today.

Basic Reproductive Strategies in Vertebrates

- Females apes and monkeys mate only when in estrus, increasing the probability of fertilization.
- Human females show a less distinctive estrus phase and can reproduce throughout the year.
- They can also engage in sexual activity without reproductive purpose.

END

Animal Form & Function II

Module 128

Animal Form & Function II

Reproduction in Fishes

Reproduction in Fishes



- All fishes reproduce in aquatic environments.
- In bony fishes, fertilization is usually external, and eggs contain only enough yolk to sustain the developing fish for a short time.
- After the yolk is consumed, the growing fish must seek food.

Reproduction in Fishes

- Although many thousands of eggs are produced and fertilized, few survive and grow to maturity.
- Some succumb to fungal and bacterial infections, others to siltation and still others to predation.

Reproduction in Fishes

- **To assure reproductive success, the fertilized egg develops rapidly, and the young achieve maturity within a short time.**

END

Animal Form & Function II

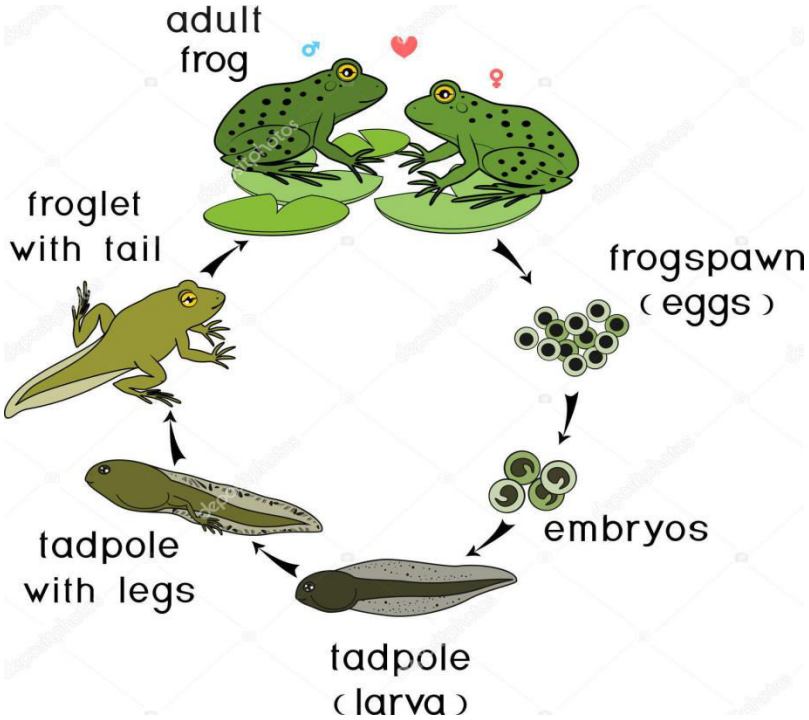
Module 129

Reproduction in Amphibians

Reproduction in Amphibians

- The amphibians were the first vertebrates to invade the land.
- They have not, however, become adapted to a completely terrestrial environment; their life cycle is still inseparably linked to water.

Reproduction in Amphibians



- Among most amphibians, fertilization is still external, just as it is among the fishes.
- Among the frogs and toads, the male grasps the female and discharges fluid containing sperm onto the eggs as she releases them into the water.

Reproduction in Amphibians

- The developmental period is much longer in amphibians than in fishes, although the eggs do not contain appreciably more yolk.
- An evolutionary adaptation present in amphibians is the presence of two periods of development: larval and adult stages.

Reproduction in Amphibians

- The aquatic larval stage develops rapidly, and the animal spends much time eating and growing.
- After reaching a sufficient size, the larval form undergoes a developmental transition called metamorphosis into the adult (often terrestrial) form.

END

Animal Form & Function II

Module 130

Animal Form & Function II

Reproduction in Reptiles

Reproduction in Reptiles

- The reptiles were the first group of vertebrates to completely abandon the aquatic habitat because of adaptations that permitted sexual reproduction on land.
- A crucial adaptation first found in reptiles is internal fertilization.

Reproduction in Reptiles

- Internal fertilization protects the gametes from drying out, freeing the animals from returning to water to breed.
- Many reptiles are oviparous, and the eggs are deposited outside the body of the female.
- Others are ovoviviparous.
- They form eggs that hatch in the body of the female, and the young are born alive.

Reproduction in Reptiles

- The shelled egg and extra mbryonic membranes, also first seen in reptiles, constitute two other important evolutionary adaptations to life on land.
- These adaptations allowed reptiles to lay eggs in dry places without danger of desiccation.

Reproduction in Reptiles

- As the embryo develops, the extra embryonic chorion and amnion help protect it, the latter by creating a fluid-filled sac for the embryo.
- The allantois permits gas exchange and stores excretory products.
- Complete development can occur within the eggshell.

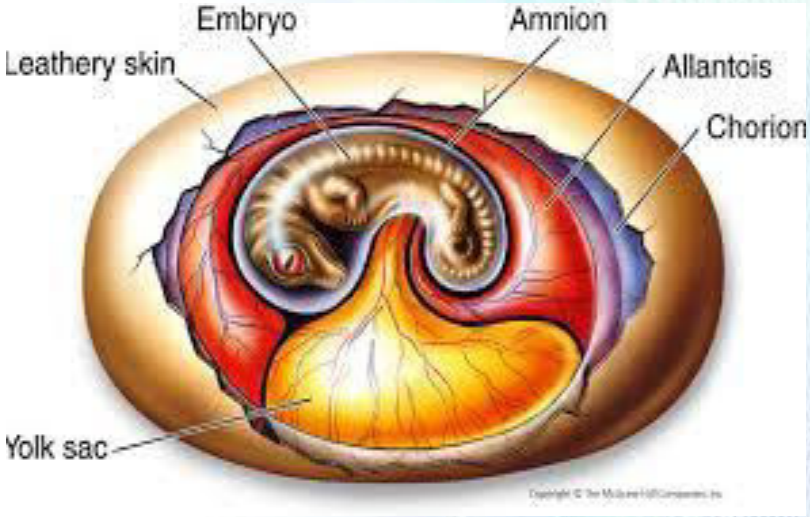
Reproduction in Reptiles

- When the animal hatches, it has developed to the point that it can survive on its own or with some parental care.

Reproduction in Reptiles







Animal Form & Function II

Module 131

Animal Form & Function II

Reproduction in Birds

Reproduction in Birds

- Birds have retained the important adaptations for life on land that evolved in the early reptiles.
- With the exception of most waterfowl, birds lack a penis.
- Males simply deposit semen against the cloaca for internal fertilization.

Reproduction in Birds

- Sperm then migrate up the cloaca and fertilize the eggs before hard shells form.
- All birds are oviparous, and the eggshells are much thicker than those of reptiles.
- Thicker shells permit birds to sit on their eggs and warm them.

Reproduction in Birds

- This brooding, or incubation, hastens embryo development.
- When many young birds hatch, they are incapable of surviving by their own.
- Extensive parental care and feeding of young are more common among birds than fishes, amphibians, or reptiles.

Reproduction in Birds





Animal Form & Function II

Module 132

Animal Form & Function II

Reproduction in Mammals

Reproduction in Mammals





- The most primitive mammals, the monotremes (e.g., the duckbilled platypus and spiny anteater), lay eggs, as do the reptiles from which they evolved.
- All other mammals are viviparous.
- Mammalian viviparity was another major evolutionary adaptation, and it has taken two forms.

Reproduction in Mammals



- **The Marsupials** developed the ability to nourish their young in a pouch after a short gestation inside the female.
- **The Placentals** much larger group, retain the young inside the female, where the mother nourishes them by means of a placenta.
- Even after birth, mammals continue to nourish their young.

Reproduction in Mammals

- **Some mammals nurture their young until adulthood, until they are able to secure for themselves.**

END

Animal Form & Function II

Module 133

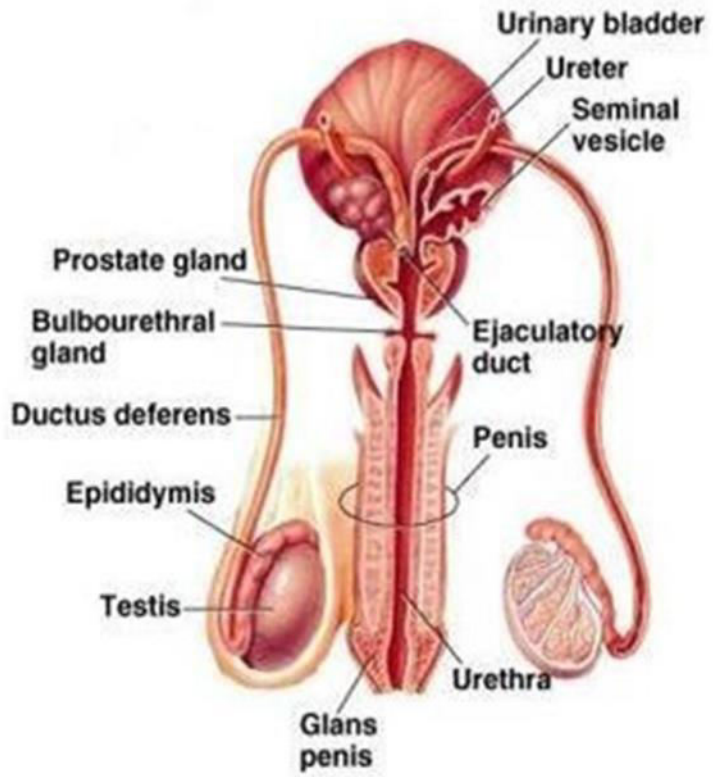
Animal Form & Function II

Human Male Reproductive System

Human Male Reproductive System

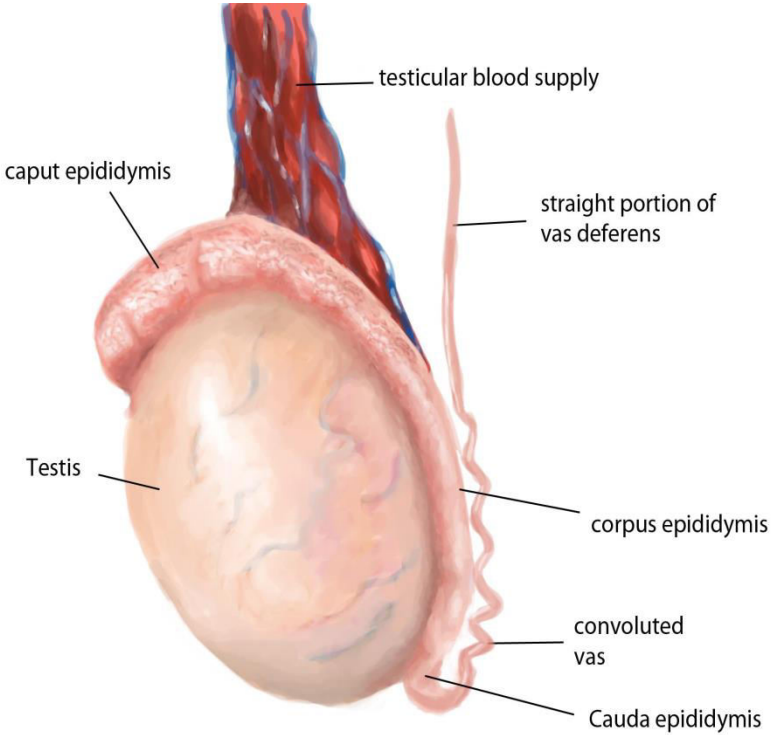
- **The main function of the male reproductive system is to produce sperms and male sex hormone for fertilization of the ovum.**

Human Male Reproductive System



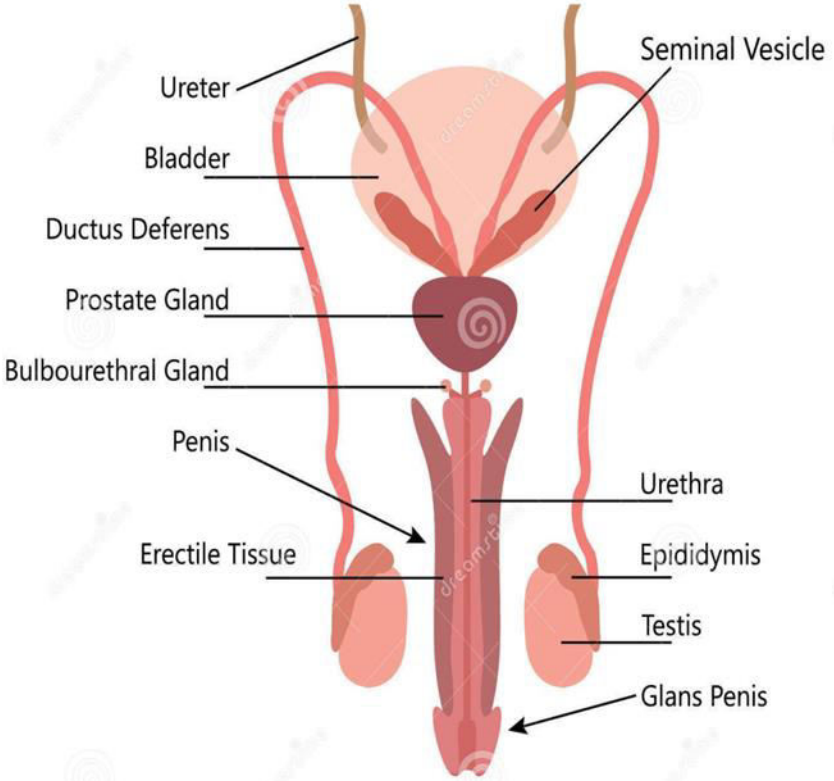
- The male reproductive system can be divided into three categories.
- **The first category consists of organs located outside the body** produces and stores sperms.
- These are produced in the testes, which are housed in the temperature-regulating pocket, scrotum.

Human Male Reproductive System



- **Immature sperms travel to the epididymis for development and storage.**
- **The testes also produce the male sex hormone, the testosterone.**

Human Male Reproductive System



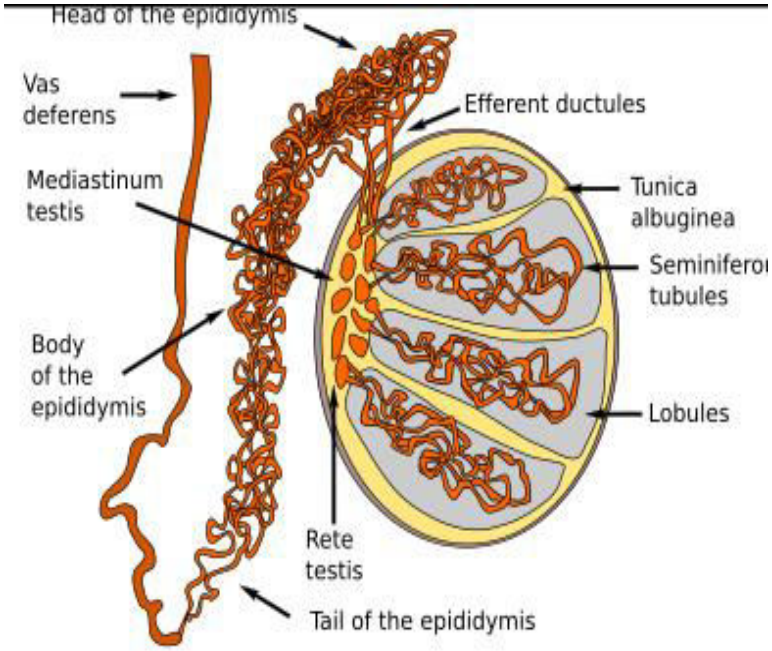
- **The second category** are the structures located around the pelvis region of male.
- These structures include fluid producing glands (bulbo-urethral gland, seminal vesicles, prostate gland) and ductus deferense.
- **The third category** are those used for copulation and deposition of semen in the vagina (the penis and urethra).

Animal Form & Function II

Module 134

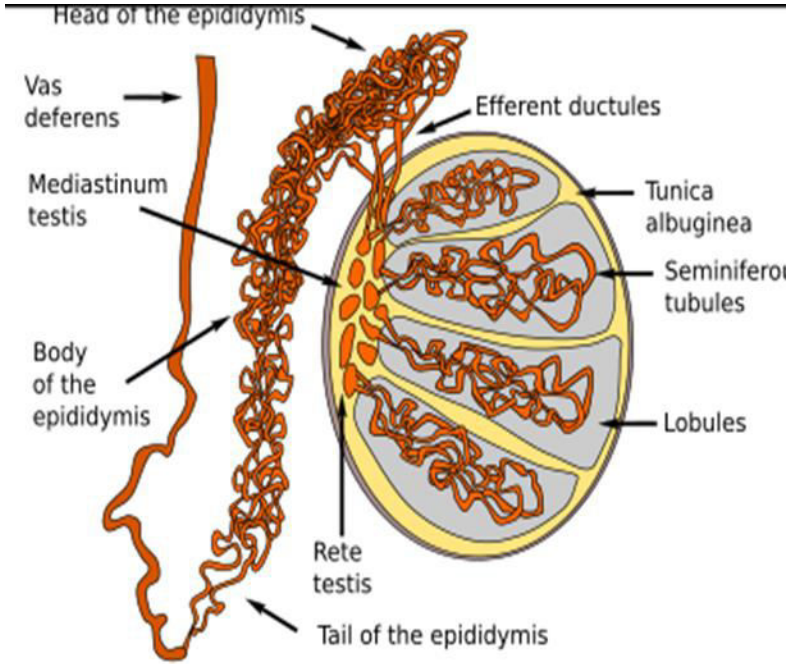
Histology of Human Testis

Histology of Human Testis



- Each testis is covered by a thick capsule called Tunica Albuginea.
- It is composed mainly of collagenous fibers and smooth muscle cells.
- Posteriorly the tunica albuginea is thickened and is called mediastinum testis.

Histology of Human Testis



- Thin fibrous septa, called Septula testis, radiate from the midiastinum to the capsule and divide the interior of the testis into 200 to 300 Pyramidal Lobules.
- The septula are not complete partitions; they show numerous gaps, through which the testicular lobules communicate quite freely.

Histology of Human Testis

- Beneath the tunica albuginea there is a layer of loose areolar connective tissue containing networks of blood vessels.
- This layer is known as tunica vasculosa, also lines the Septula Testis.

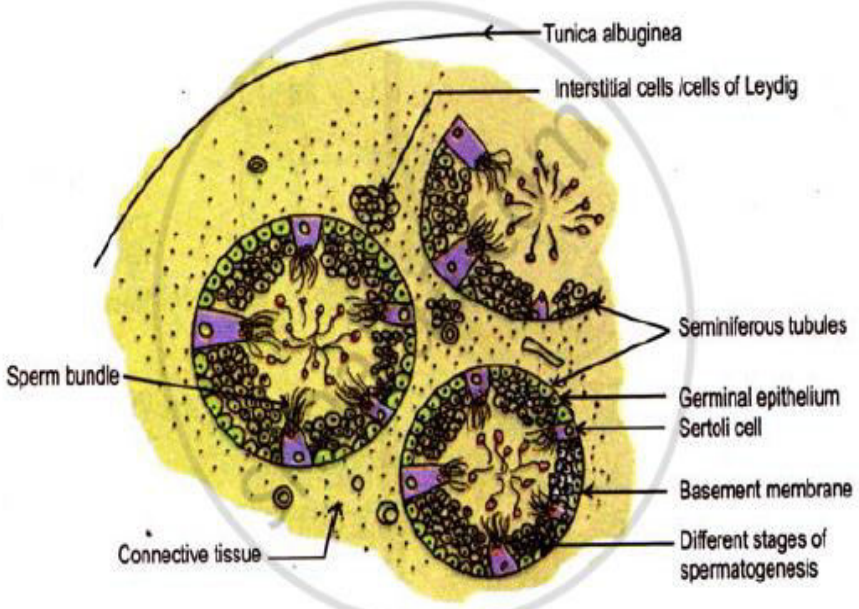
Histology of Human Testis

- Each testicular lobule contains 1-4 highly convoluted Seminiferous Tubules,
- Each seminiferous tubule is intricately coiled, averaging about 0.2 mm diameter and 50 cm in length.
- Individual tubules usually commence as free blind ends but neighboring tubules may form anastomosing loops.

Histology of Human Testis

- At the apex of the testicular lobule seminiferous tubules lose their convolutions and become straight.
- This makes the initial part of the male genital duct system.
- Seminiferous tubules are lined by a complex Germinal Epithelium.

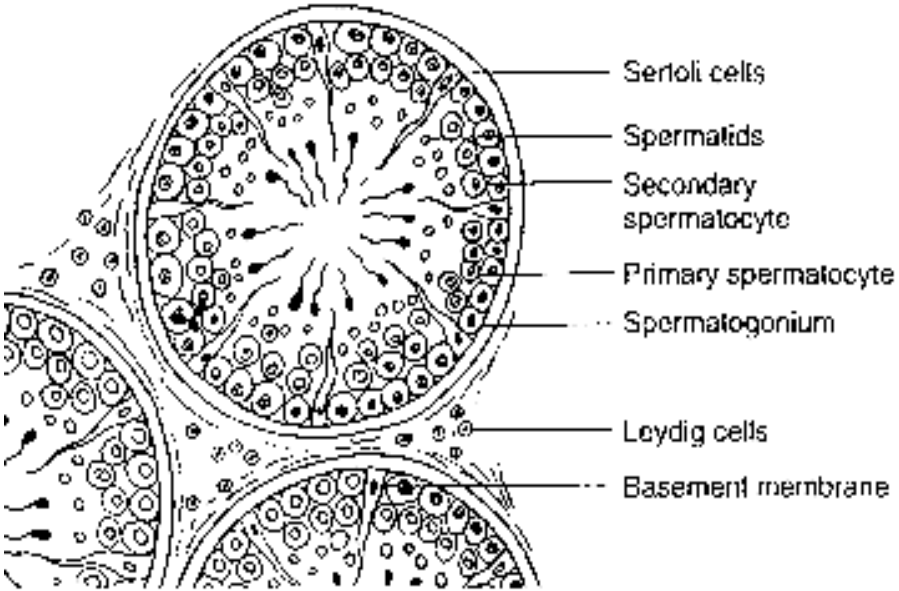
Histology of Human Testis



TS of Human Testis

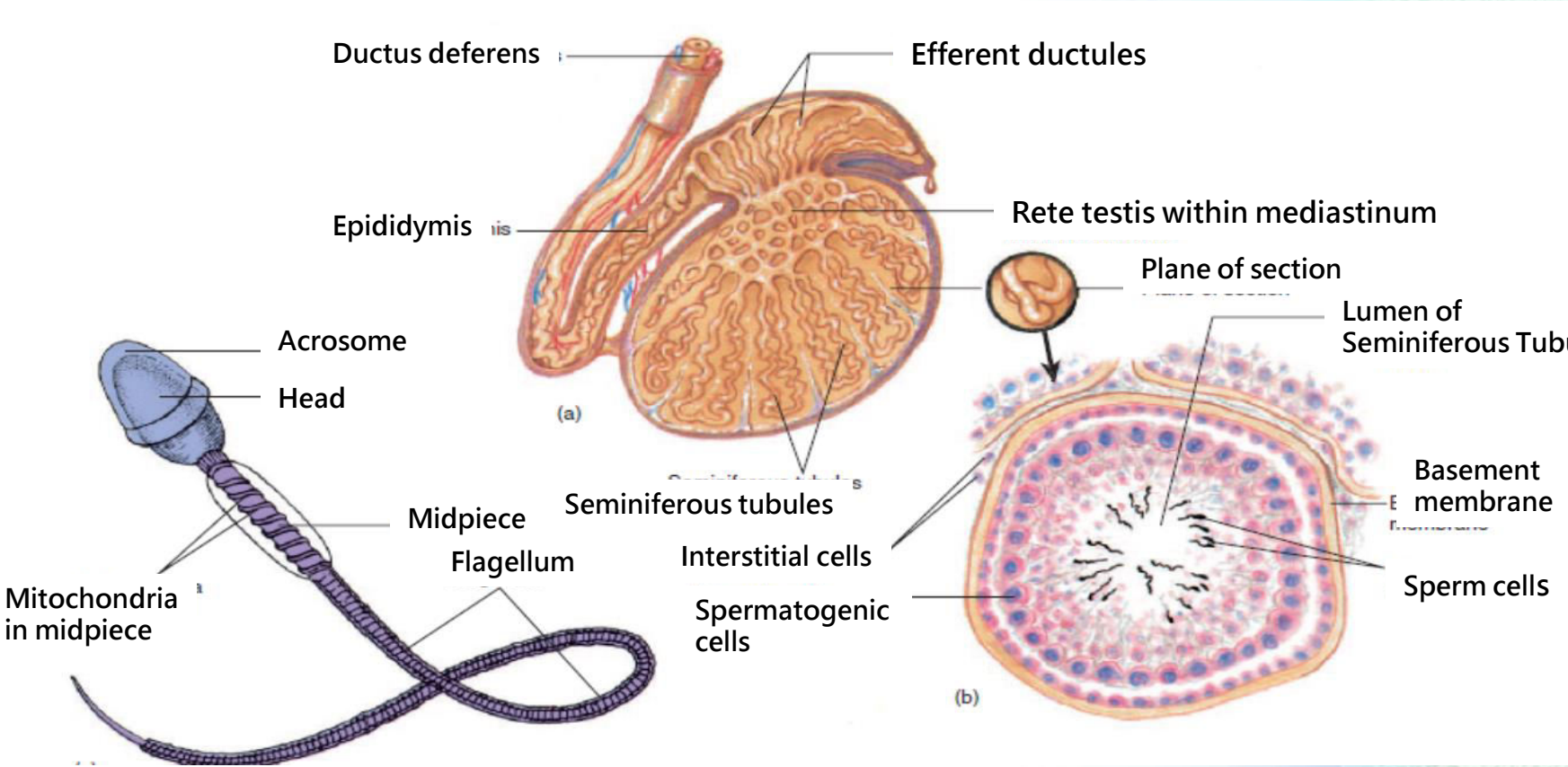
- The germinal cells are present in several layers forming the wall of Seminiferous Tubules.
- Sertoli Cells are supportive in the seminiferous tubules and help in the process of spermatogenesis.
- Spermatogenesis takes place in the seminiferous tubules.
- This includes mitosis and meiosis to form haploid gametes.

Histology of Human Testis



- **Spermatogonium passes through Primary spermatocyte, Secondary Spermatocyes, Spermatids, and finally haploid sperms.**

Histology of Human Testis



Animal Form & Function II

Module 135

Animal Form & Function II

Production of Sperms

Production of Sperms

- The paired testes are the male reproductive organs that produce sperms.
- Shortly after birth, the testes descend from the abdominal cavity into the scrotum which hangs between the thighs.
- Testes hang outside the body, the temperature inside the scrotum is about 34° C compared to a 37° C core temperature.

Production of Sperms

- The lower temperature is necessary for active sperm production and survival.
- Muscles elevate or lower the testes, depending on the outside air temperature.
- Each testis contains over eight hundred tightly coiled seminiferous tubules which produce thousands of sperm each second in healthy young men.

Production of Sperms

- The walls of the seminiferous tubules are lined with two types of cells:
- **Spermatogenic cells**, which give rise to sperm, and **Sustentacular cells,** which nourish the sperm as they form and secrete a fluid and a hormone, inhibin, into the tubules to provide a liquid medium for the sperm.

Production of Sperms

- **Between the seminiferous tubules are clusters of endocrine cells, called Leydig cells, that secrete the male sex hormone, the testosterone.**

END

Animal Form & Function II

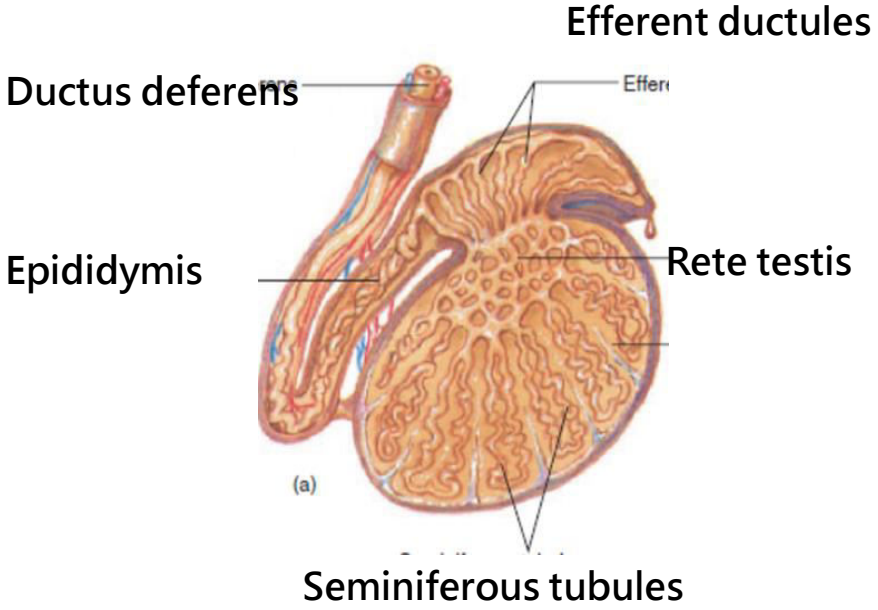
Module 136

Transport of Sperms in Man

Transport of Sperms in Man

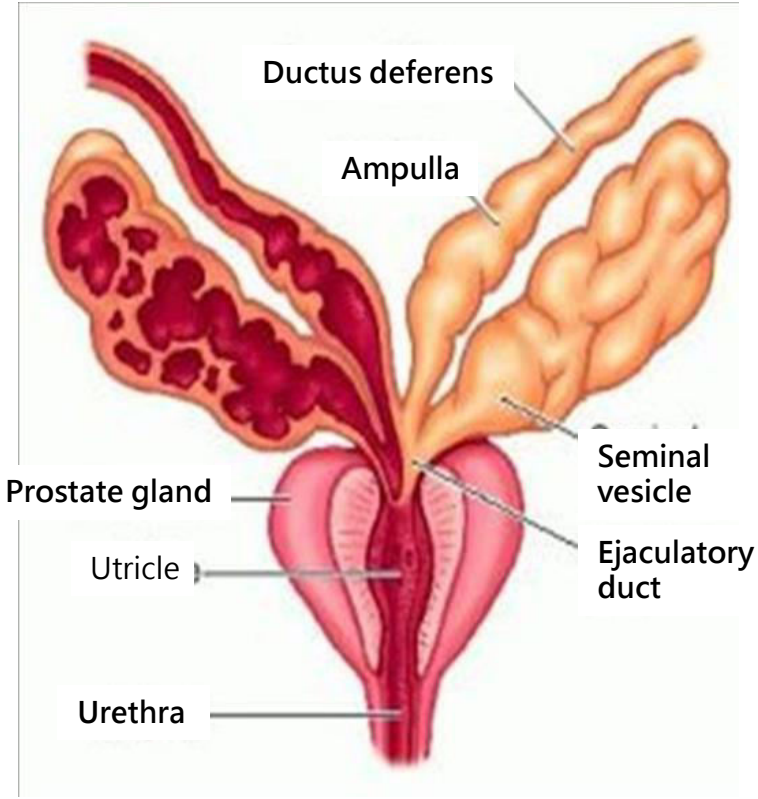
- A system of tubes carries the sperms from the testes to the penis.
- The seminiferous tubules merge into a network of tiny tubules called the rete testis, which merges into a coiled tube called the epididymis.

Transport of Sperms in Man



- The epididymis has three main functions:
- 1. it stores sperms until they mature and ready to be ejaculated,
- 2. it contains smooth muscle that helps propel the sperm toward the penis by peristaltic contractions,
- 3. it serves as a duct system for sperm to pass from the testis to the ductus deferens.

Transport of Sperms in Man



- The ductus deferens or sperm duct after leaving the scrotum, continues upward, passes around the urinary bladder and enlarges to form the ampulla.
- The ampulla stores some sperm until they are ejaculated.
- Distal to the ampulla, the ductus deferens becomes the ejaculatory duct.

Transport of Sperms in Man

- **The urethra is the final section of the reproductive duct system.**

END

Animal Form & Function II

Module 137

Animal Form & Function II

Accessory Glands and their Secretions

Accessory Glands & Their Secretions

- After the ductus deferens passes around the urinary bladder, several accessory glands add their secretions to the sperm as they are propelled through the ducts.

  These accessory glands are:

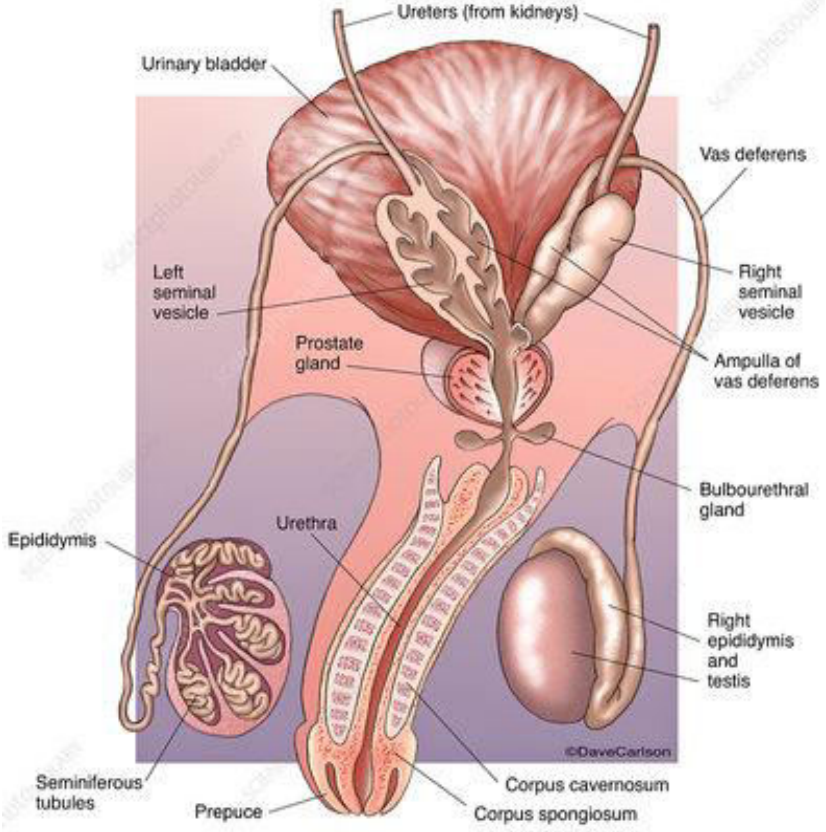
- 1. Seminal vesicles
- 2. Prostate gland, and
- 3. Bulbourethral glands

Accessory Glands & Their Secretions

Seminal Vesicles:

- These vesicles are paired secrete water, fructose, prostaglandins, and vitamin C.
- This secretion provides an energy source for the motility of sperms.
- Also helps to neutralize the natural protective acidity of the vagina. (The pH of the vagina is about 3 to 4).

Accessory Glands & Their Secretions



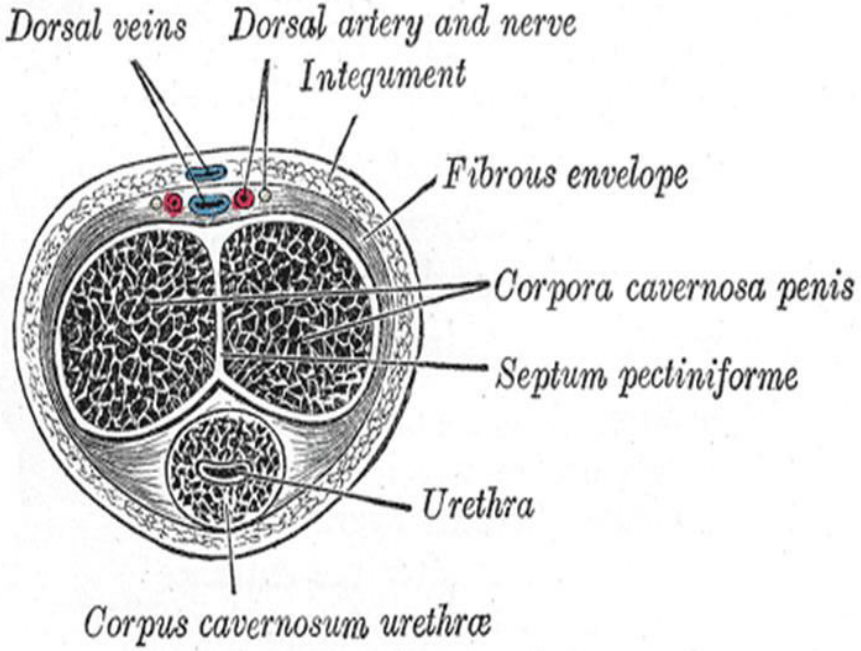
- When vaginal pH increases to about 6 the sperm motility and fertility enhances.
- **The Prostate Gland**
- It is walnut sized gland, located between the bladder and penis.
- It secretes water, enzymes, cholesterol, buffering salts, and phospholipids.

Accessory Glands & Their Secretions

The Bulbourethral Glands:

- The bulbourethral glands (also known as Cowper's glands) are a pair of pea shaped exocrine glands urethra.
- These glands secrete a clear, alkaline fluid that lubricates the urethra to facilitate the ejaculation of semen and lubricates the penis prior to sexual intercourse.

Accessory Glands & Their Secretions



- The penis has two functions. It carries urine through the urethra to the outside during urination, and it transports semen through the urethra during ejaculation.
- In addition to the urethra, the penis contains three cylindrical strands of erectile tissue: two corpora cavernosa and the corpus spongiosum.

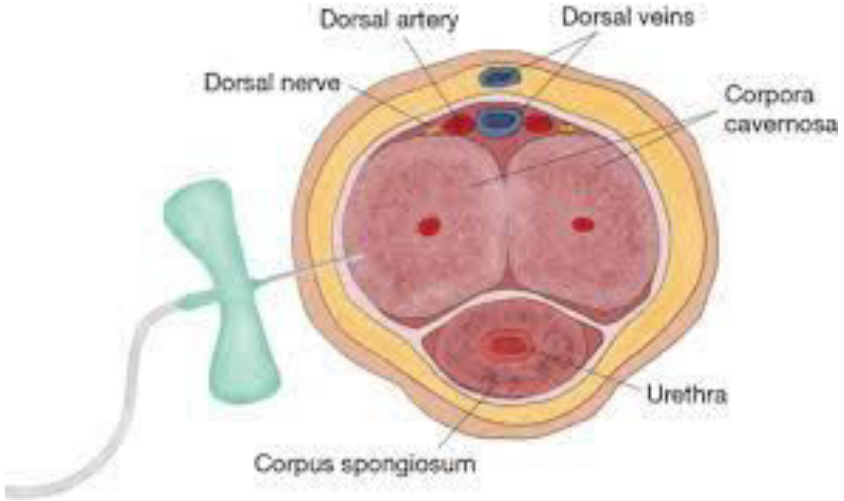
Accessory Glands & Their Secretions

- The corpus spongiosum extends beyond the corpora cavernosa and becomes the expanded tip of the penis called the glans penis.
- The loosely fitting skin of the penis folds forward over the glans to form the prepuce or foreskin.
- Circumcision is the removal of the prepuce for religious or health reasons.

Accessory Glands & Their Secretions

- The fluid that results from the combination of sperm and glandular secretions is semen.
- The average human ejaculation is about 3 to 4 ml of semen and contains 300 to 400 million sperms.

Accessory Glands & Their Secretions



- **Today, many circumcisions are performed in the belief that they lessen the likelihood of cancer of the penis.**

Animal Form & Function II

Module 138

Structure of Human Sperm

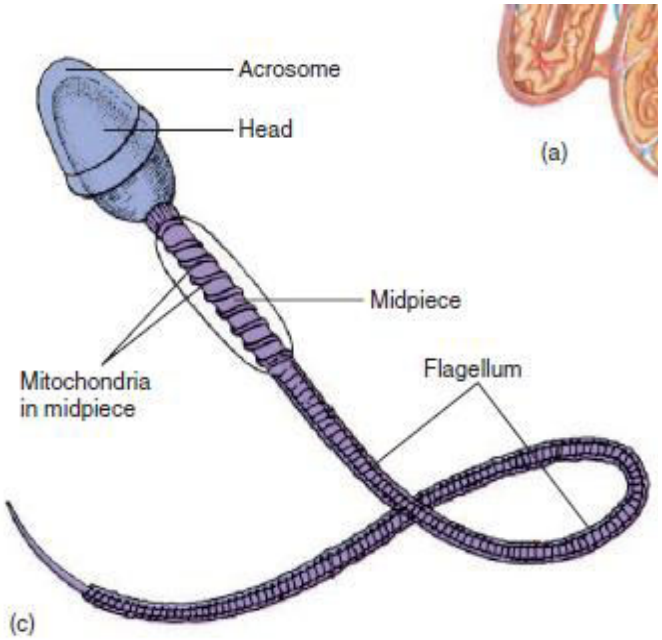
Structure of Human Sperm

- A mature human sperm consists of a head, midpiece, and tail.
- The head is flat, disc shaped 5.1 μm by 3.1 μm and contains the haploid nucleus, which is mostly DNA.

Structure of Human Sperm

- **The acrosome, a cap over most of the head, contains an enzyme called *acrosin* that assists the sperm in penetrating the outer layer surrounding a secondary oocyte.**

Structure of Human Sperm



- The spiral mitochondria in the midpiece supply the ATP necessary for these movements.
- The sperm tail is 50 μm long contains an array of microtubules that bend to produce whip like movements.

Structure of Human Sperm

Animal Form & Function II

Module 139

Hormonal Control of Male Reproductive Function

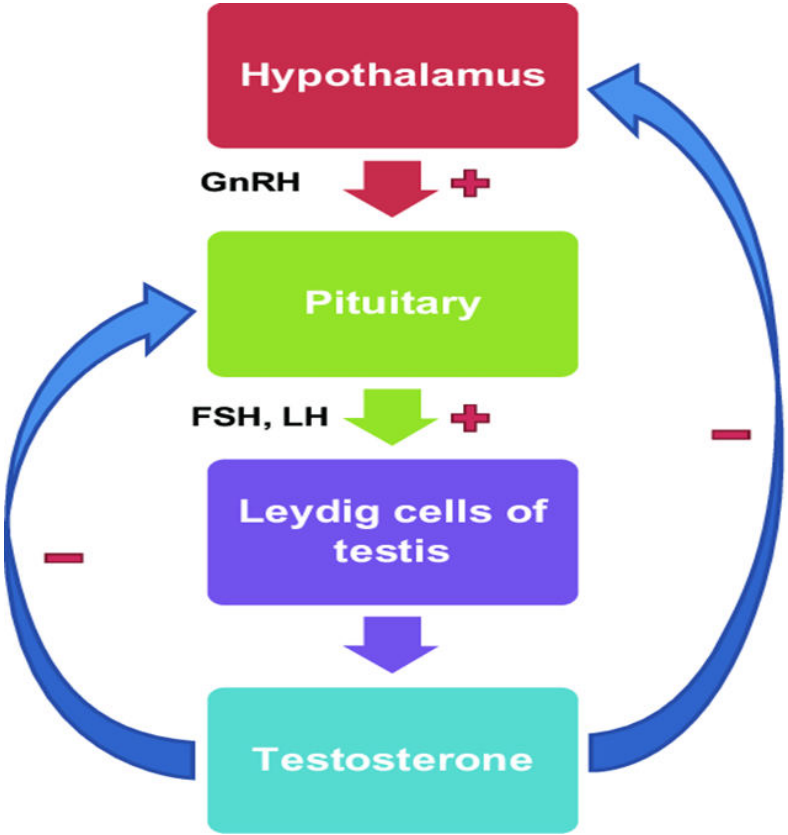
Hormonal Control of Male Reproductive Function I

- Before a male can mature and function sexually, special regulatory hormones must come into play.
- Male sex hormones are collectively called androgens.
- The hormones that travel from the brain and pituitary gland to the testes and ovaries are called gonadotropins.

Hormonal Control of Male Reproductive Function I

- As previously noted, the interstitial cells (leydig cells) produce the male sex hormone testosterone.
- Figure shows the negative feedback mechanisms that regulate the production and secretion of testosterone, as well as its actions.

Hormonal Control of Male Reproductive Function I



- When the level of testosterone in the blood decreases, the hypothalamus is stimulated to secrete GnRH (gonadotropin-releasing hormone).
- GnRH stimulates the secretion of FSH (follicle-stimulating hormone) and LH (luteinizing hormone), also called ICSH (interstitial cell-stimulating hormone), into the bloodstream.

Hormonal Control of Male Reproductive Function I

- (FSH and LH were first named for their functions in females, but their molecular structure is exactly the same in males.)
- FSH causes the spermatogenic cells in the seminiferous tubules to initiate spermatogenesis, and LH stimulates the interstitial cells to secrete testosterone.

Hormonal Control of Male Reproductive Function I

- The cycle is completed when testosterone inhibits the secretion of LH, and another hormone, inhibin, is secreted.
- Inhibin inhibits the secretion of FSH from the anterior pituitary.
- This cycle maintains a constant rate (homeostasis) of spermatogenesis.

Animal Form & Function II

Module 140

Human Female Reproductive System

Human Female Reproductive System

- The main external structures of the female reproductive system include:
- **Labia majora:** enclose and protect the other external reproductive organs.
- **Labia minora**: They lie just inside the labia majora, and surround the openings to the vagina.

Human Female Reproductive System

- **Clitoris:** a small, sensitive protrusion at the point where the two labia minora meet.
- This is comparable to the penis in males.
- The clitoris is covered by a fold of skin, called the Hilum.
- **Vagina:** The vagina is a canal that joins the cervix (the lower part of uterus) to the outside of the body. It also is known as the birth canal.

Human Female Reproductive System

- In most young women, the vaginal opening is partially covered by a thin membrane, the hymen, which may be ruptured during normal strenuous activities or may be stretched or broken during sexual activity.

Human Female Reproductive System

- **Uterus :** The uterus is a hollow, pear-shaped organ that is the home to a developing fetus.
- The uterus is divided into two parts: the cervix, which is the lower part that opens into the vagina, and the main body of the uterus is called corpus.
- The corpus can easily expand to hold a developing baby.

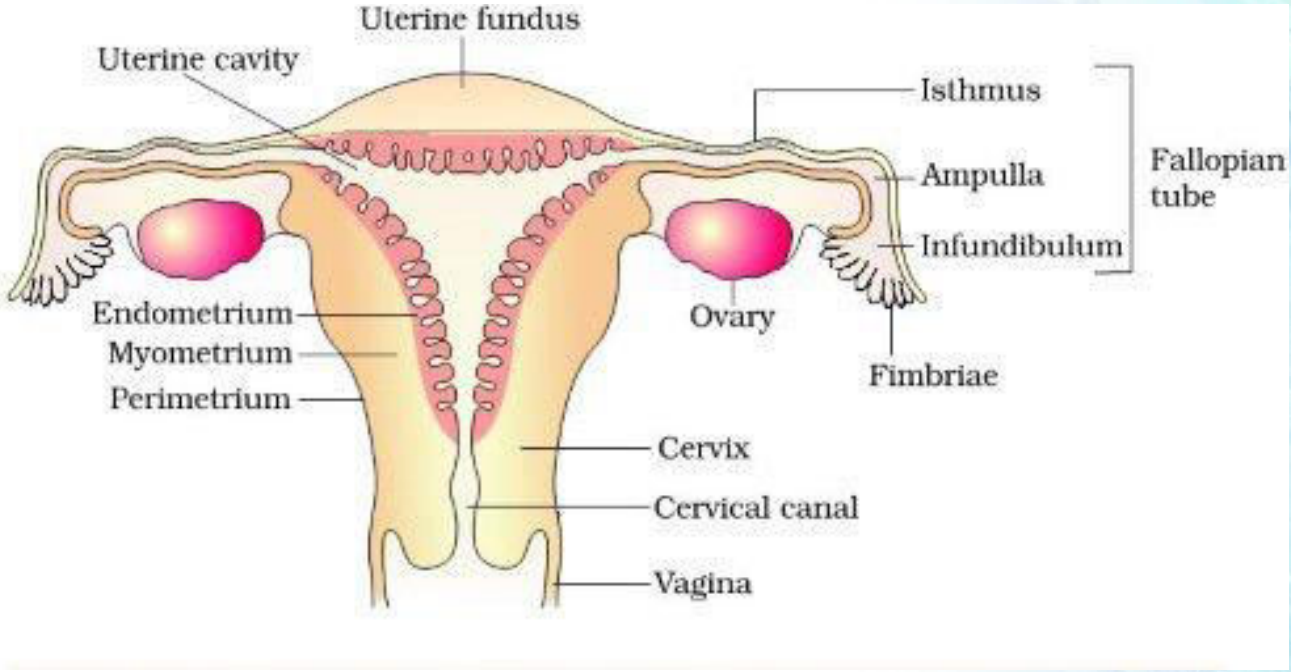
Human Female Reproductive System

- A canal through the cervix allows sperm to enter and menstrual blood to exit.
- **Ovaries:** The ovaries are almond sized, oval-shaped glands that are located on either side of the uterus.
- The ovaries produce eggs and hormones.

Human Female Reproductive System

- **Fallopian tubes:** These are narrow tubes that are attached to the upper part of the uterus and serve as pathways for the ova (egg cells) to travel from the ovaries to the uterus.
- Fertilization of an egg by a sperm normally occurs in the fallopian tubes.
- The fertilized egg then moves to the uterus, where it implants to the uterine lining.

Human Female Reproductive System



Female Reproductive System

Animal Form & Function II

Module 141

Animal Form & Function II

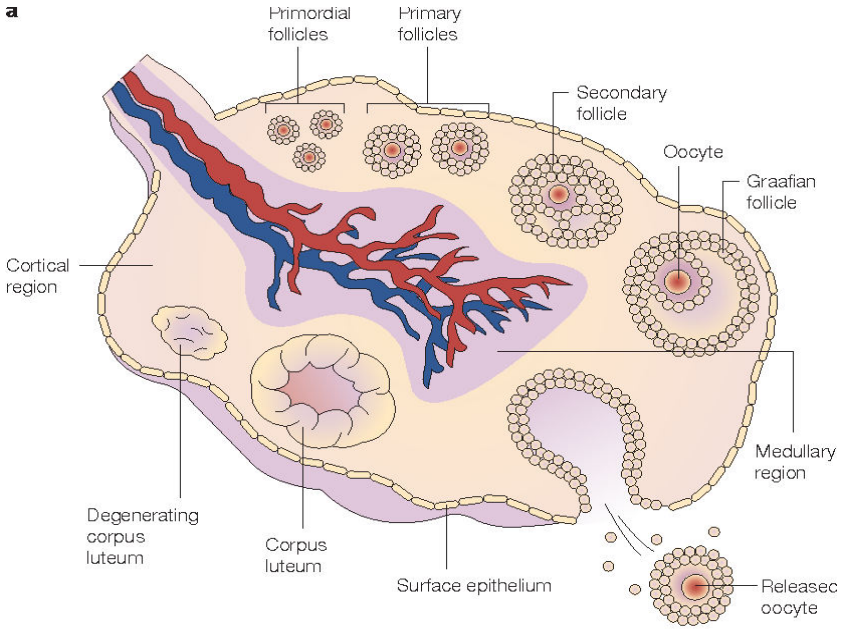
Histology of Human Ovary

Histology of Human Ovary

- **The ovaries** are held in place by ligaments (mesovarium) attached to the uterus through which vessels and nerves pass to the ovary.

- The ovaries have two functions:
- 1. Production of ova
- 2. Secretion of hormones.

Histology of Human Ovary



- The surface of the ovary is covered by a single layer of cuboidal epithelium, called germinal epithelium.
- Fibrous connective tissue forms a thin capsule, the tunica albuginea, immediately beneath the epithelium.
- Ovary is divided into an outer cortex and an inner medulla.

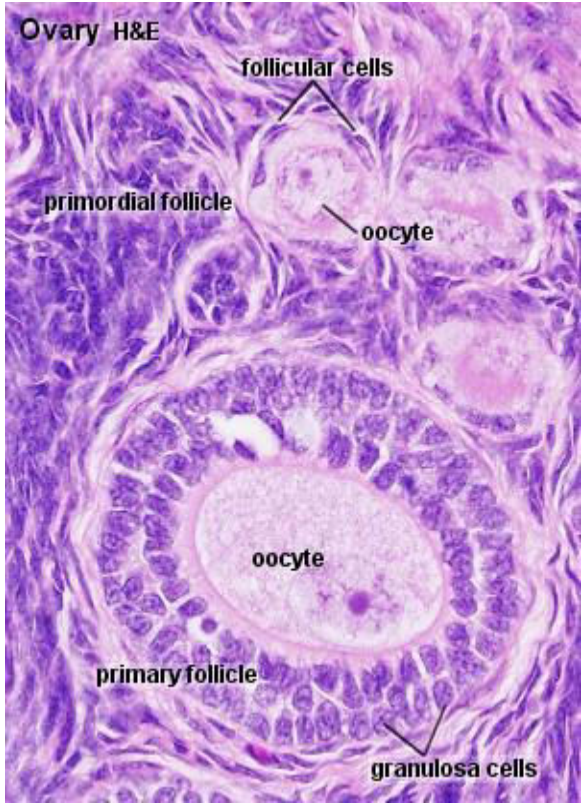
Histology of Human Ovary

- The cortex consists of a very cellular connective tissue stroma in which the ovarian follicles are embedded.
- The medulla is composed of loose connective tissue, which contains blood vessels and nerves.

Histology of Human Ovary

- **Ovarian Follicles**
- **Ovarian follicles consist of one oocyte and surrounding follicular cells.**
- **Follicular development can be divided into a number of stages.**
- **Primordial follicles**
- **are located in the cortex just beneath tunica albuginea. One layer of flattened follicular cells surround the oocyte.**

Histology of Human Ovary

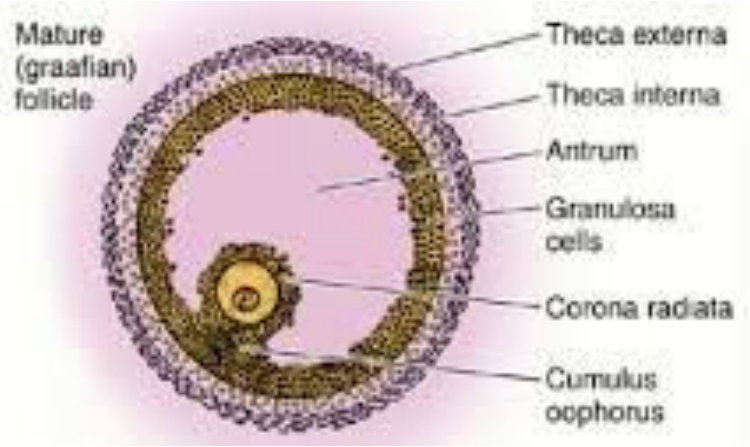


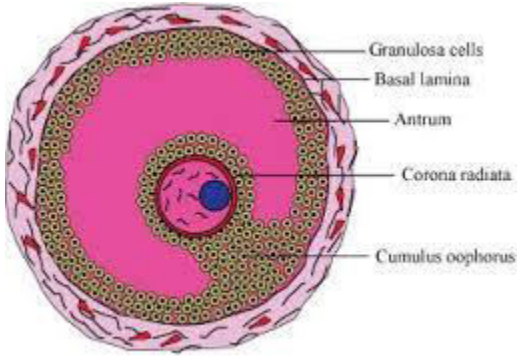
- **The primary follicle**
- **is the first morphological stage that marks the onset of follicular .**
- **The previously flattened cell surrounding the oocyte now form a cuboidal or columnar epithelium surrounding the oocyte.**

Histology of Human Ovary

- **Secondary follicle**
- **Small fluid-filled spaces become visible between the granulosa cells as the follicle reaches a diameter of about 400 µm.**
- **These spaces enlarge and fuse to form the follicular antrum.**

Histology of Human Ovary





Structure of the Graafian follicle

- Graafian follicle or
- mature follicle forms a small "bump" on the surface of the ovary.
- The Graafian follicle finally ruptures and the oocyte is released from the ovary.

Histology of Human Ovary

- After ovulation, the lining of the follicle grows
- inward, forming the corpus luteum (“yellow body”), which serves as a temporary endocrine tissue and continues to secrete the female sex hormones estrogen and progesterone.

Animal Form & Function II

Module 142

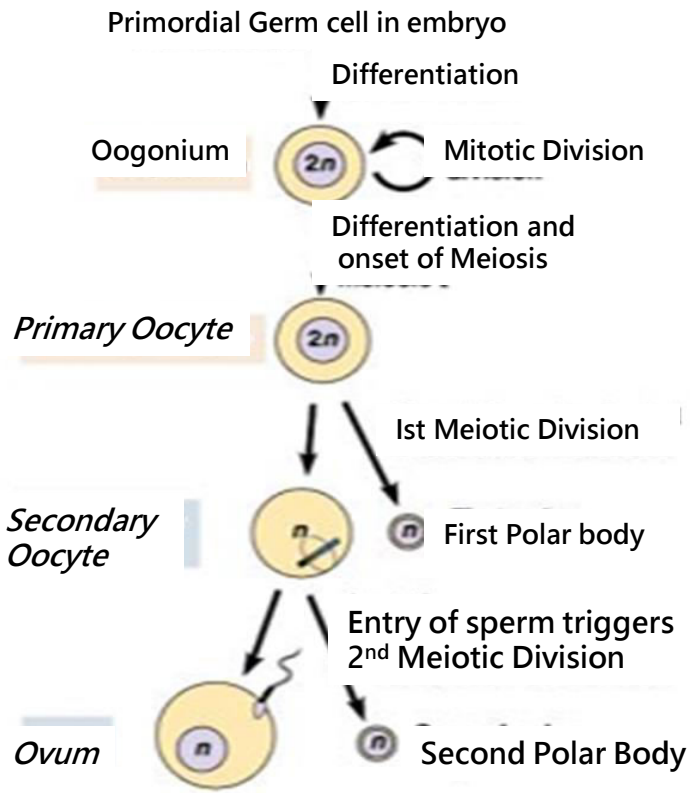
Animal Form & Function II

Production of Eggs
or
Oogenesis

Production of Eggs or Oogenesis

- **WHAT IS OOGENESIS?**
  Oogenesis is the formation of ovum or egg.
- **In human oogenesis starts in the ovaries of the fetus before birth.**
- **There are 5 stages of oogenesis process:**
- **1. The germinal epithelial cells divide repeatedly until many (5-7) million diploid oogonia are formed.**

Production of Eggs or Oogenesis



- 2. The oogonia grow to form primary oocytes that become surrounded by a layer of follicle cells.
- 3. The primary oocytes undergoes the first meiotic division to become a secondary oocyte and a first polar body.(haploid).
- 4. The secondary oocyte surrounded by the follicle cells develop into the secondary follicle.

Production of Eggs or Oogenesis

- The characteristic feature that distinguishes secondary from primary follicles is the appearance of a follicular antrum within the granulosa layer.
- Within the secondary follicle the antrum further enlarges to form Graafian follicle.
- This is the stage before ovulation. The oocyte is now a haploid.

Production of Eggs or Oogenesis

END

- The Graafian follicle is the stage after the first meiotic division has completed but before ovulation.
- When Graafian follicle matures, it moves towards the surface of the ovary wall and rupture to release the secondary oocyte(n).
- 5. The final product of meiosis II is the ovum and the second polar body.

Animal Form & Function II

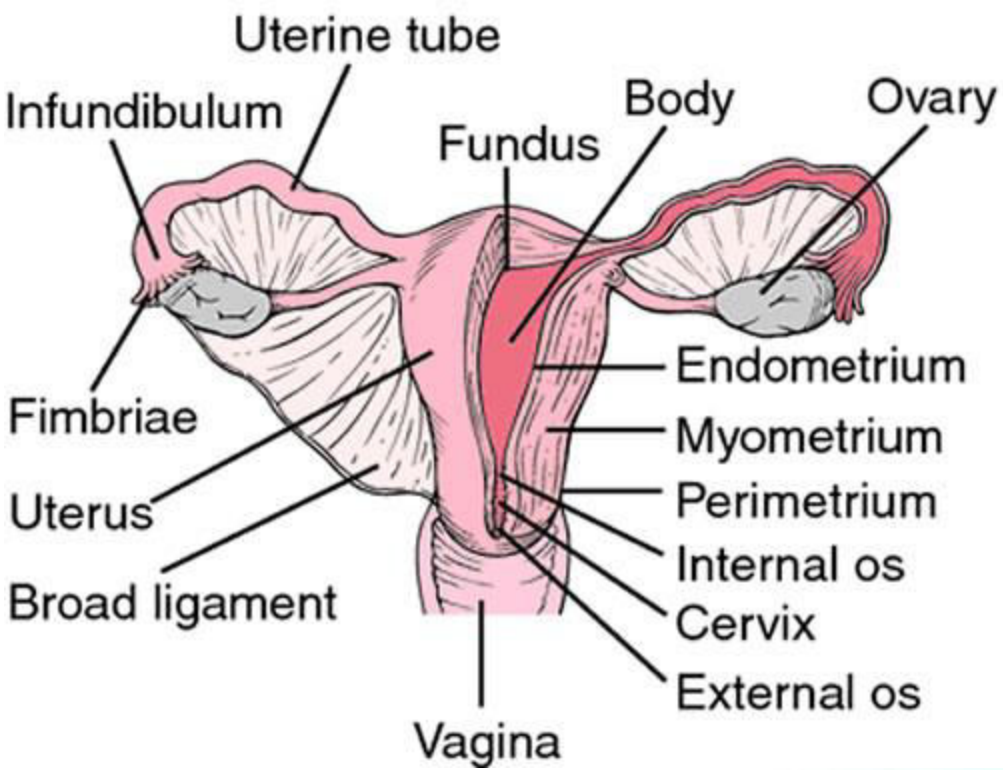
Module 143

Transport of Ovum in Human Female

Transport of Ova in Human Female

- The paired tubes that receive the secondary oocyte from the ovary and convey it to the uterus are called either the uterine tubes or fallopian tubes.
- Feathery fimbriae fringe the part of the uterine tube that encircles the ovary.

Transport of Ova in Human Female



Transport of Ova in Human Female

- Each month, as the secondary oocyte is released, the motion of the fimbriae sweep it across a tiny space between the uterine tube and the ovary into the tube.
- Unlike sperm, the secondary oocyte cannot move on its own.

Transport of Ova in Human Female

- Instead, the peristaltic contractions of the tube and the waving motions of the cilia in the mucous membrane of the tube carry the secondary oocyte along.
- Fertilization usually occurs in the uppermost third of the uterine tube.

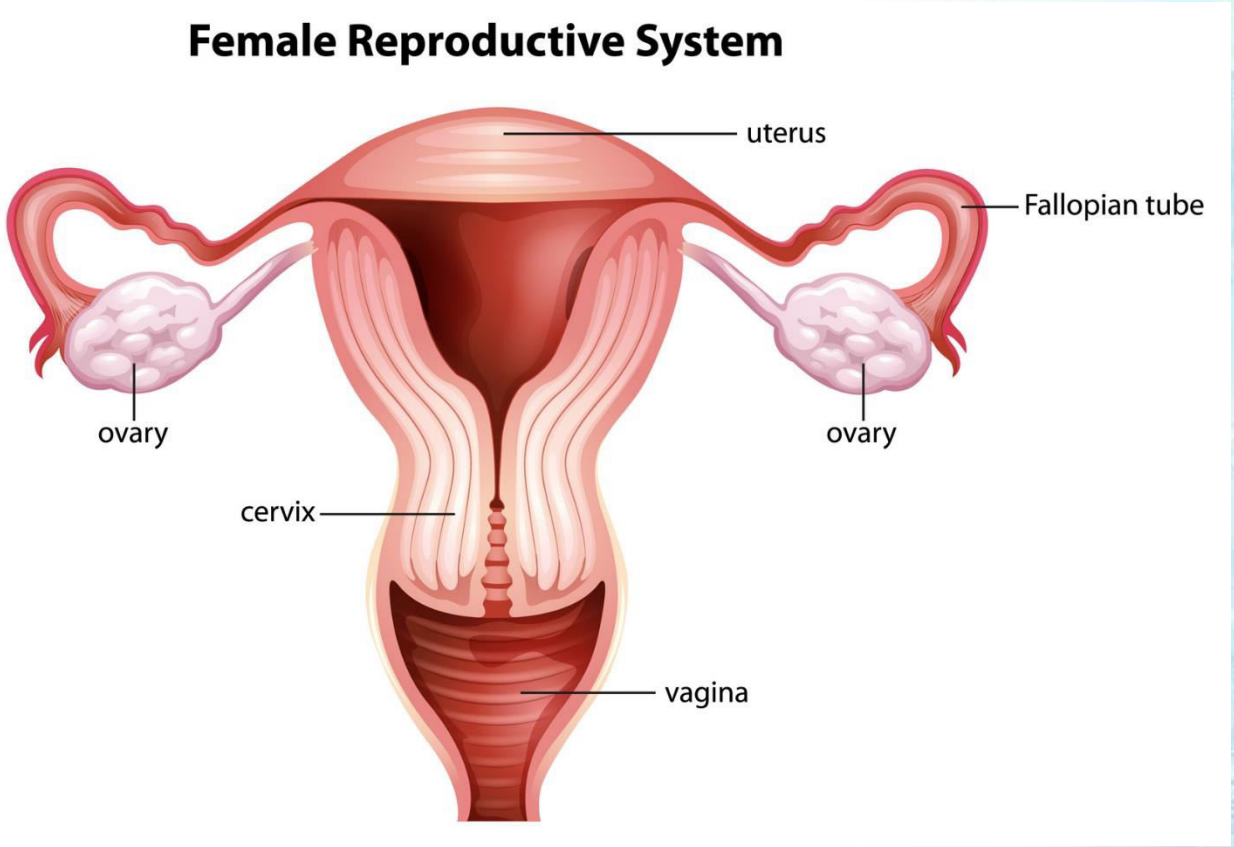
Transport of Ova in Human Female

- A fertilized oocyte (zygote) continues its journey toward the uterus, where it implants.
- The journey takes four to seven days.
- If fertilization does not occur, the secondary oocyte degenerates in the uterine tube.
- The uterine tubes terminate in the uterus,

Transport of Ova in Human Female

- The uterus terminates in a narrow portion called the cervix, which joins the uterus to the vagina.

Transport of Ova in Human Female



Animal Form & Function II

Module 144

Animal Form & Function II

Comparison Between Spermatogenesis and Oogenesis

Comparison b/w Spermatogenesis & Oogenesis

1. Spermatogenesis (smg) is the production of sperms from spermatogonia.

whereas

- Oogenesis (ogs) is the production of eggs from oogonia.

2. Smg. occurs inside the testes in male.

whereas

- Ogs. occurs inside the ovaries in female.

Comparison b/w Spermatogenesis & Oogenesis

3. In spg. all stages occur inside the testis.

whereas

- In og. all stages except last stage occur inside the ovary.

4. Spg. is a continuous process occurring after puberty till death.

whereas

- In Ogs. early stages occur during fetal life and the rest of the process occur after puberty till menopause.

Comparison b/w Spermatogenesis & Oogenesis

5. Sperms develop from germinal epithelial lining of the seminiferous tubules.

whereas

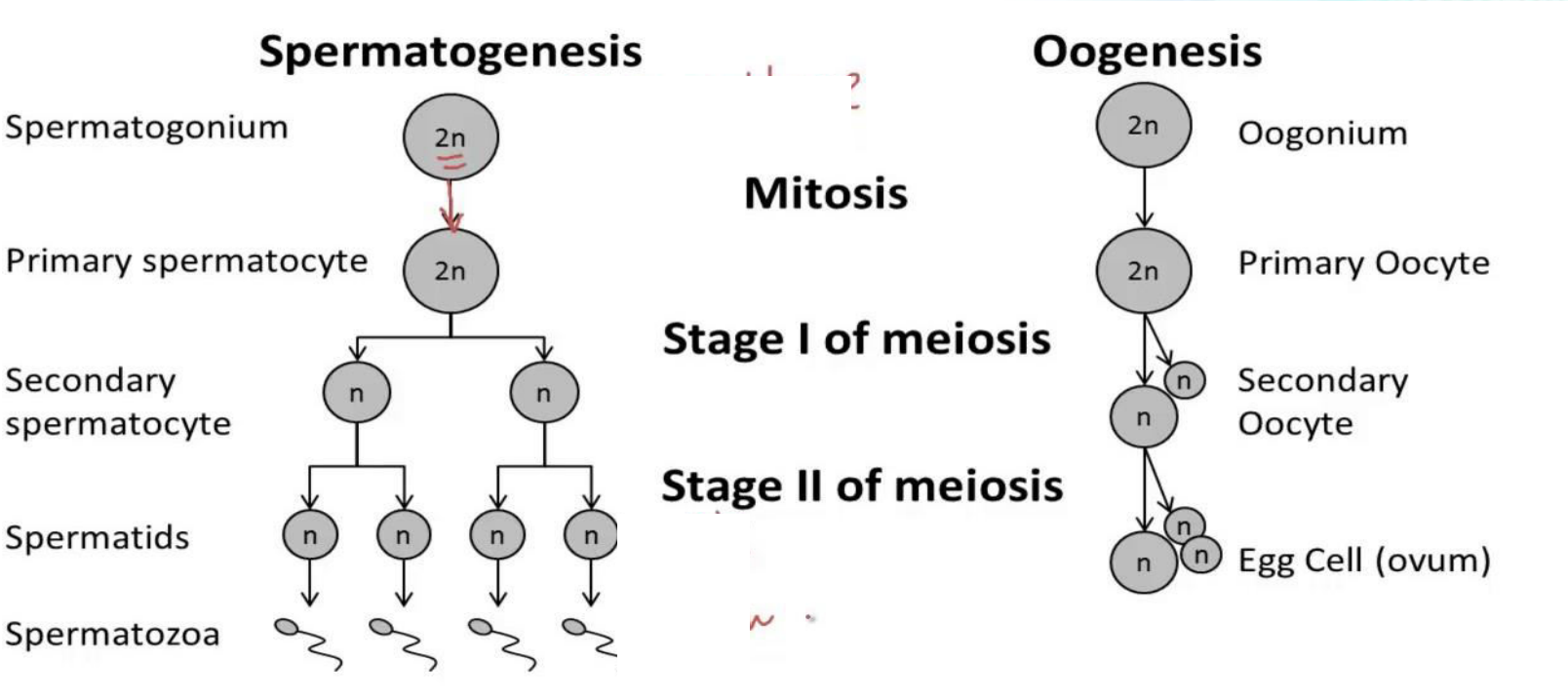
- Eggs develop from germinal epithelium which overlays the ovary.

6. All the spermatogonia divide by meiosis producing 4 sperm cells from each.

whereas

- Some of the oogonia divide by meiosis producing one egg.

Comparison b/w Spermatogenesis & Oogenesis



Comparison b/w Spermatogenesis & Oogenesis

7. In spg. growth phase is too short,

whereas in

- **In ogs. growth phase is very long.**

8. Primary spermatocyte divide by meiosis I forming two secondary spermatocytes.

whereas in

- **Primary oocyte divide by meiosis I forming one secondary oocyte and one polar body.**

Comparison b/w Spermatogenesis & Oogenesis

9. In spg. motile gametes are formed.

whereas in

- **In oogenesis non motile gametes are formed.**

END

Animal Form & Function II

Module 145

Animal Form & Function II

Mammary Glands and Production of Milk

Mammary Glands and Production of Milk

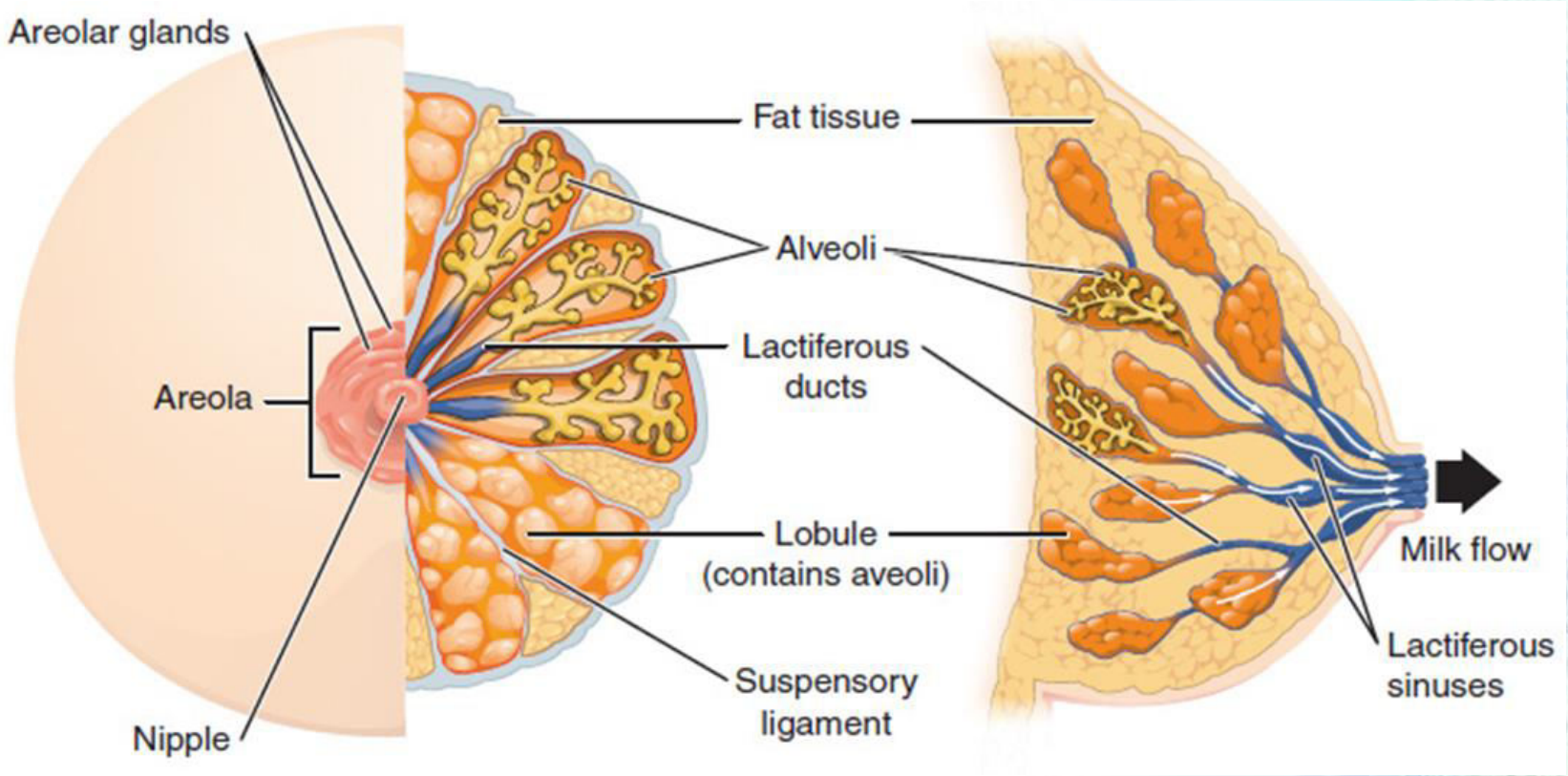
The Mammary Glands

- MG are modified sweat glands.
- MS produce and secrete milk.
- They contain varying amounts of adipose tissue.
- The amount of adipose tissue determines the size of the breasts, but the amount of mammary tissue does not vary widely from one woman to another.

Mammary Glands and Production of Milk

- The basic components of the mammary gland are the alveoli (hollow cavities, a few millimetres large) lined with milk-secreting cuboidal cells.
- These alveoli join up to form groups known as lobules, and each lobule has a lactiferous duct that drains into openings in the nipple.

Mammary Glands and Production of Milk



Mammary Glands and Production of Milk

- The lobules group together to form a lobe.
- There are 15 -20 lobes in each MG.

END

Animal Form & Function II

Module 146

Animal Form & Function II

Reproductive Cycles in Mammals

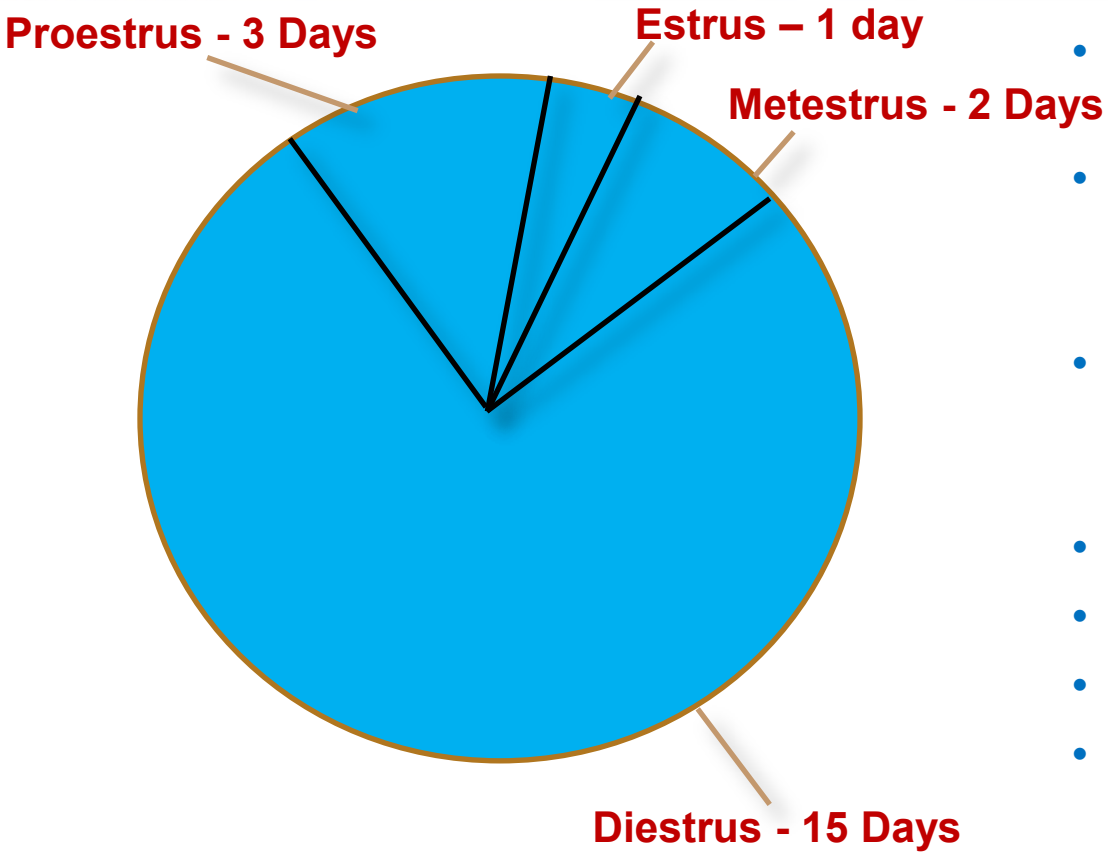
Reproductive Cycles in Mammals

- **Reproductive Cycle** is a set of physiologic changes that recur regularly in the reproductive organs of female mammals.
- There are two types of Reproductive Cycles found in mammals:
  - Estrous Cycle and
  - Menstrual Cycle

Reproductive Cycles in Mammals

- **Estrous Cycle:** In animals with estrous cycles, the female is sexually receptive ("in heat") only during the estrus phase of the cycle.
- In animals with estrous cycles, the uterus reabsorbs the thickened lining of the endometrium if conception does not occur.

Reproductive Cycles in Mammals



- The frequency and length of the estrus phase varies.
- For example, In cow the estrus phase lasts avg. 24 hours in 21 days cycle.
- In dogs, the estrus cycle typically occurs twice a year with:
- Proestrus lasts 4-20 days
- Estrus 4-13 days
- Metestrus 60-90 days
- Diestrus lasts 90-150 days

Reproductive Cycles in Mammals

- The frequency and length of the estrus phase varies.
- For example, In cow the estrus phase lasts avg. 24 hours in 21 days cycle.
- In dogs, the estrus cycle typically occurs twice a year with:
- Proestrus lasts 4-20 days
- Estrus 4-13 days
- Metestrus 60-90 days
- Diestrus lasts 90-150 days

Reproductive Cycles in Mammals

- In rats and mice the estrous cycle lasts four days and is characterized as: proestrus, estrus, metestrus and diestrus. the estrous period is about 12 hours.

Reproductive Cycles in Mammals

- **Menstrual Cycle**. Adult female mammals without estrous cycle have the MC.
- In these mammals uterus sheds the thickened lining of the endometrium along with blood and mucosal tissue through the vaginal opening if conception does not occur.
- In adult human females the MC is of 28 days and the menstrual period averages 5 days.

Reproductive Cycles in Mammals

- Menstruation in mammals other than human occur in primates (apes and monkeys), chimpanzee and gibbon.
- Ape species include humans, gorillas, chimpanzees, gibbons.
- Monkey species include baboons, macaques, marmosets.

END

Animal Form & Function II

Module 147

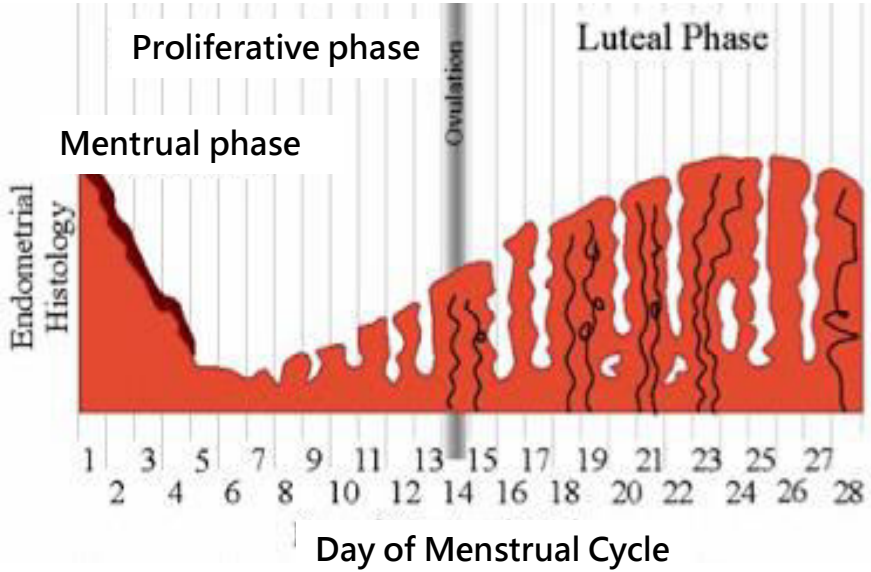
Animal Form & Function II

Human Female Reproductive Cycle

Human Female Reproductive Cycle

- **Human Female Reproductive Cycle** is a rhythmical series of physiological changes that occur in fertile woman.
- Necessary for reproduction.
- Average length of the cycle is 28 days.

Human Female Reproductive Cycle



Menstual cycle

Menstrual Cycle can be divided into three phases:

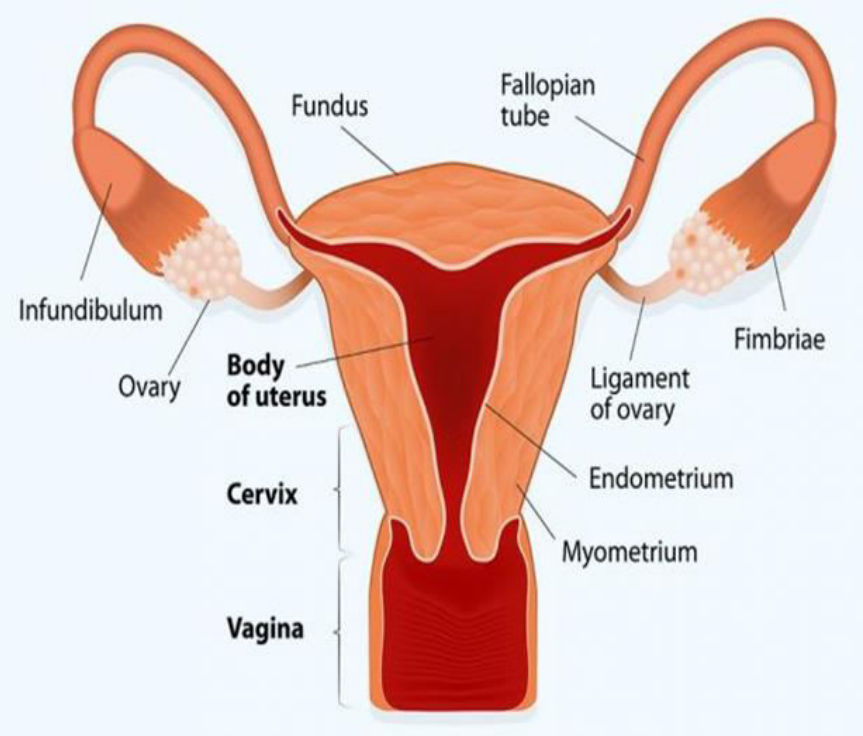
1. **Menstrual phase**
2. **Proliferative and**
3. **Luteal/Secretory phase**

Human Female Reproductive Cycle

The menstrual phase

- It is the first stage of the menstrual cycle that lasts for 4-5 days.
- This phase starts when an egg from the previous cycle is not fertilized, because pregnancy has not taken place.

Human Female Reproductive Cycle



- The thickened lining of the uterus, which supports pregnancy, is no longer needed, so it sheds through vagina.
- During this stage, a combination of blood, mucus, and tissue is discharged from the uterus through vaginal opening out.

Human Female Reproductive Cycle

The proliferative phase

- is the second phase of the cycle when the lining of the uterus grows, or proliferate.
- It lasts for 8 to 9 days.

The luteal phase

- begins after ovulation.
- It lasts about 14 days (unless fertilization occurs) and ends just before menstrual period.

END

Animal Form & Function II

Module 148

Animal Form & Function II

Hormonal Control of Menstrual Cycle Part I

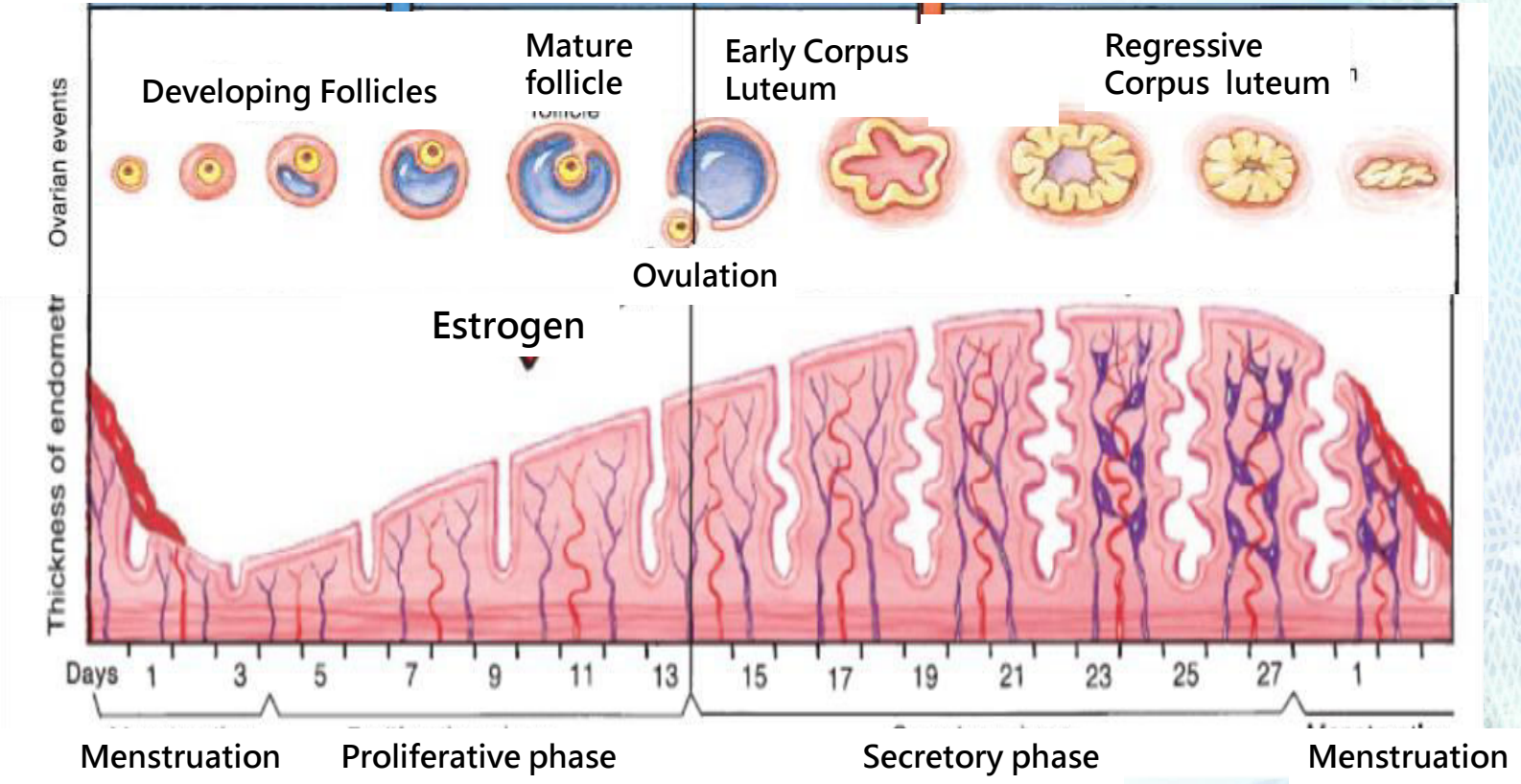
Hormonal Control of Menstrual Cycle Part I

- The male is continuously fertile from puberty to old age, and throughout that period, sex hormones are continuously secreted.
- The female, however, is fertile only during a few days each month.
- This pattern is related to the cyclical release of hormones from the ovary.

Hormonal Control of Menstrual Cycle Part I

- The cyclical production of ovarian hormones regulate the menstrual cycle, as well as the ovarian cycle, during which the oocytes mature and ovulation occurs.
- The development of the oocyte in one of the immature ovarian follicles produce estrogen, causing proliferation of the endometrium.

Hormonal Control of Menstrual Cycle Part I



Hormonal Control of Menstrual Cycle Part I

- The elevated estrogen level about midway in the cycle triggers the maturing follicle to enlarge rapidly and release the secondary oocyte (ovulation).
- The ruptured follicle becomes another endocrine tissue, the corpus luteum.

Hormonal Control of Menstrual Cycle Part I

- The CL secretes progesterone, which act to complete the development of the endometrium and maintain it for 10 to 14 days. (Luteal Phase)
- If the oocyte is not fertilized, the CL disintegrates into a corpus albicans, and the secretion of estrogen and progesterone cease.

Hormonal Control of Menstrual Cycle Part I

- **Without estrogen and progesterone, the endometrium breaks down, and menstruation occurs.**

END

Animal Form & Function II

Module 149

Animal Form & Function II

Hormonal Control of Menstrual Cycle Part II

Hormonal Control of Menstrual Cycle Part II

- The controlling center for ovulation and menstruation is the hypothalamus.
- The hypothalamus releases, on a regular cycle, GnRH, which stimulates the anterior lobe of the pituitary gland to secrete follicle stimulating hormone (FSH) and luteinizing hormone (LH).

Hormonal Control of Menstrual Cycle Part II

- The FSH brings about the maturation of oocyte up to its the Graafian follicle.
- The elevated estrogen level about midway in the cycle triggers the anterior pituitary to secrete another hormone LH, which causes the mature follicle to enlarge rapidly to form Graafian follicle.

Hormonal Control of Menstrual Cycle Part II

- The growing follicles produce estrogen, causing a buildup and proliferation of the endometrium, as well as the inhibition of FSH production.
- The increasing level of LH and the decreasing level of FSH bring about the release of oocyte from the Graafian follicle (ovulation).

Hormonal Control of Menstrual Cycle Part II

- LH also causes the ruptured follicle to become another endocrine tissue, the corpus luteum.
- The corpus luteum starts producing another hormone, progesteron.
- As long as the progesteron is released the endometrium is intact. (this is normally for 8 to 9 days).

Hormonal Control of Menstrual Cycle Part II

- The immediate breakdown of copus luteum stops the production of progesterone.
- This causes the breakdown of endometrium and the M phase starts.
- The decreasing level of progesteron renews active secretion of FSH, which stimulates the development of another follicle, and the monthly cycle begins again.

Hormonal Control of Menstrual Cycle Part II

- When a female reaches 45 to 55 years of age, the ovaries lose their sensitivity to FSH and LH,
- They stop making normal amounts of estrogen and progesterone and the monthly cycle ceases to occur permanantly
- This condition is called Menopause.

- end

Hormonal Control of Menstrual Cycle Part II

Animal Form & Function II

Module 150

Animal Form & Function II

Summary of Menstrual Cycle

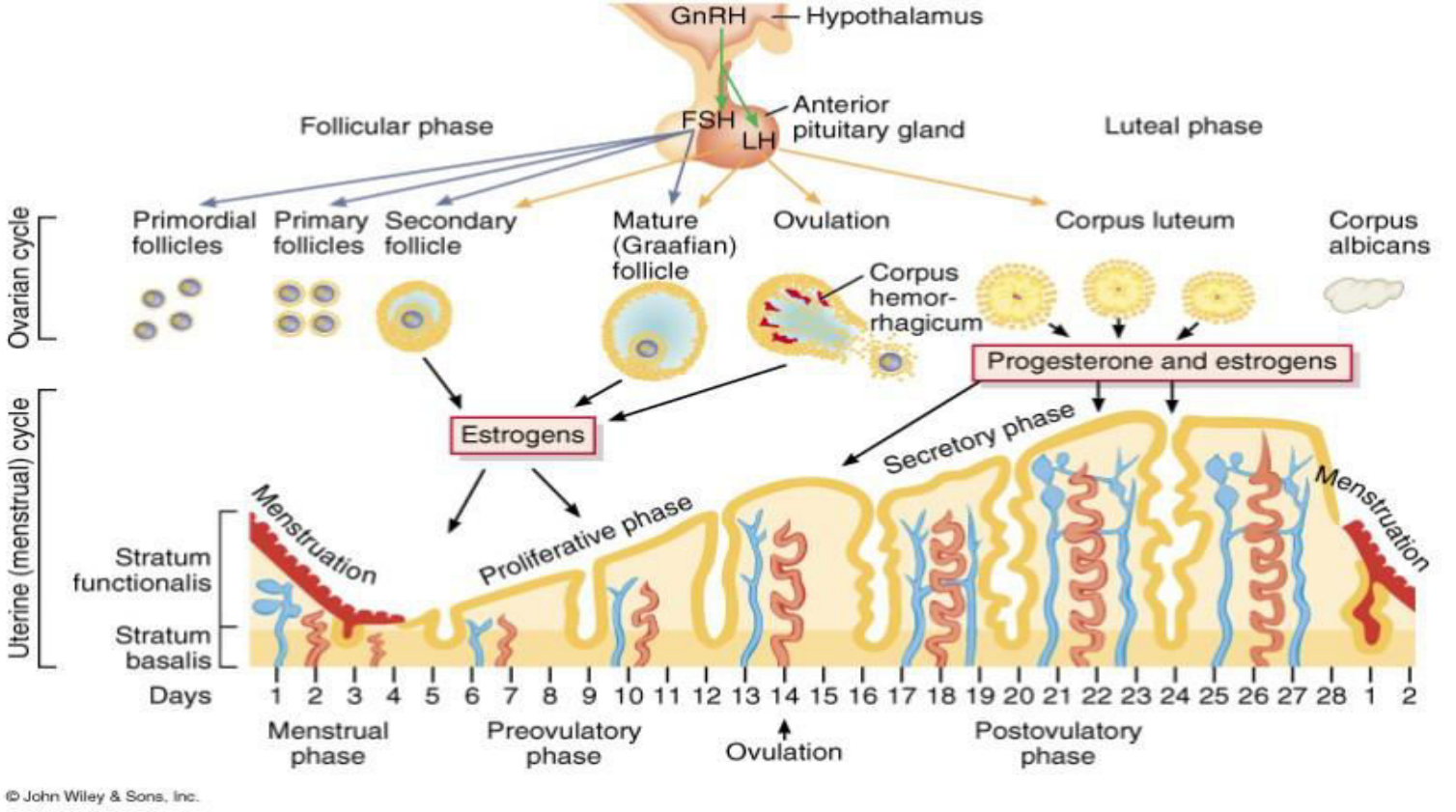
Summary of Menstrual Cycle

- The important events of the menstrual cycle, are ovulation and menstruation
- Endocrine regulation of female reproduction involves the hypothalamus, pituitary gland, and ovaries.
- The first day of the menstrual cycle is marked by the beginning of menstrual bleeding.

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Summary of Menstrual Cycle



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Summary of Menstrual Cycle

- Ovulation occurs at about day 14 in a typical 28-day menstrual cycle.
- During the preovulatory phase, gonadotropin-releasing hormone (GnRH) from the hypothalamus stimulates the anterior lobe of the pituitary to secrete follicle-stimulating hormone (FSH) and luteinizing hormone (LH).

Summary of Menstrual Cycle

- FSH stimulates follicle development and stimulates the granulosa cells to produce estrogen.
- LH stimulates theca cells to multiply and produce androgens, which are converted to estrogen.

Summary of Menstrual Cycle

- Estrogen is responsible for primary and secondary female sex characteristics and stimulates development of the endometrium.
- After the first week, only one follicle continues to develop.
- Although at relatively low concentration, estrogen inhibits FSH secretion by negative feedback.

Summary of Menstrual Cycle

- Granulosa cells produce inhibin, which also inhibits FSH secretion
- During the late preovulatory phase, estrogen concentration peaks and through positive feedback signals the anterior pituitary to secrete LH.

Summary of Menstrual Cycle

- LH stimulates final maturation of the follicle and stimulates ovulation.
- During the postovulatory phase, LH promotes development of the corpus luteum.
- The corpus luteum secretes rogesterone and estrogen, which stimulate final preparation of the uterus for pregnancy.

Summary of Menstrual Cycle

- During the postovulatory phase, progesterone, along with estrogen, inhibits secretion of GnRH, FSH, and LH.
- If fertilization does not occur, the corpus luteum degenerates, concentrations of estrogen and progesterone in the blood fall, and menstruation occurs.

END

Animal Form & Function II

Module 151

Animal Form & Function II

Hormonal Regulation in Pregnant Females

Hormonal Regulation in Pregnant Females

- **Pregnancy/Gestation**, is the time during which one or more offspring develops inside a woman.
- Pregnancy sets into motion a new series of physiological events.
- In pregnancy ovaries are directly affected.

Hormonal Regulation in Pregnant Females

- As the embryo develops, the placenta releases a hormone, human chorionic gonadotropin (hCG), which keeps the corpus luteum from disintegrating.

Hormonal Regulation in Pregnant Females

- The progesterone is necessary to maintain the uterine lining.
- After a time, the placenta takes over progesterone production, and the corpus luteum degenerates.

Hormonal Regulation in Pregnant Females

- By the end of two weeks following implantation, the concentration of hCG increases in the female's blood as well as in urine.
- hCG immunological test can check for pregnancy.

Hormonal Regulation in Pregnant Females

- As the embryo develops, other hormones such as prolactin and oxytocin are secreted.
- Prolactin induce the mammary glands to secrete and eject milk after childbirth.
- Oxytocin stimulate the uterine contractions during childbirth.
- It is also important for lactation, and certain aspects of human behavior.

end

Animal Form & Function II

Module 152

Animal Form & Function II

Embryonic Development in Human

Embryonic Development in Human

- This presentation covers the main event in reproduction—the nine month pregnancy/gestation period, during which time the woman carries, nourishes, and protects the embryo that grows to a full-term baby.

Embryonic Development in Human

- The development of a human being can be divided into: prenatal (before birth) and postnatal (after birth) development.
- During the prenatal period, the developing individual takes its start becomes a ball of cells called a morula, and eventually becomes a blastocyst that implants in the endometrium.

Embryonic Development in Human

- From two weeks after fertilization until the end of the eighth week of its existence, the individual is called an embryo.
- From nine weeks until birth, it is a fetus.
- During or after birth, it is called a newborn baby, or neonate.

Embryonic Development in Human

- Pregnancy is arbitrarily divided into three-month periods called trimesters.
- The first trimester begins at fertilization, and during this time most of the organs are formed.
- The next two trimesters are mainly periods of growth for the fetus.

end

Embryonic Development in Human

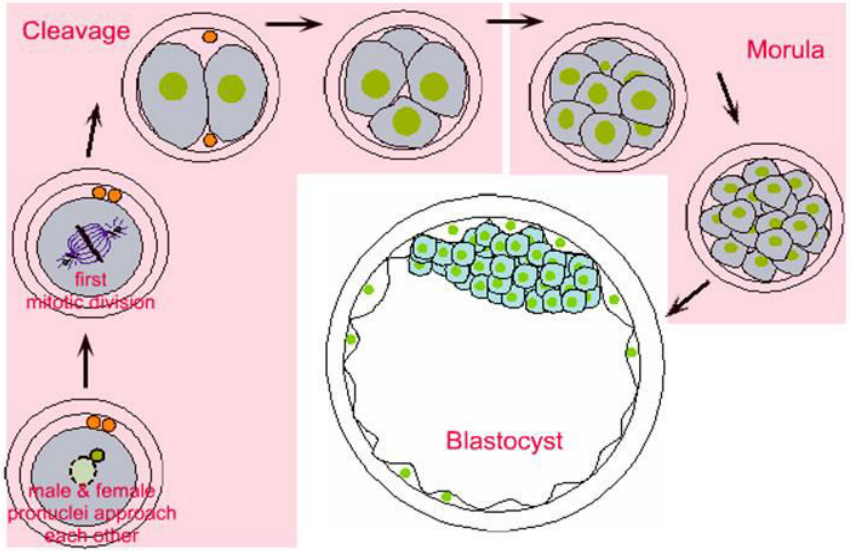
Animal Form & Function II

Module 153

Animal Form & Function II

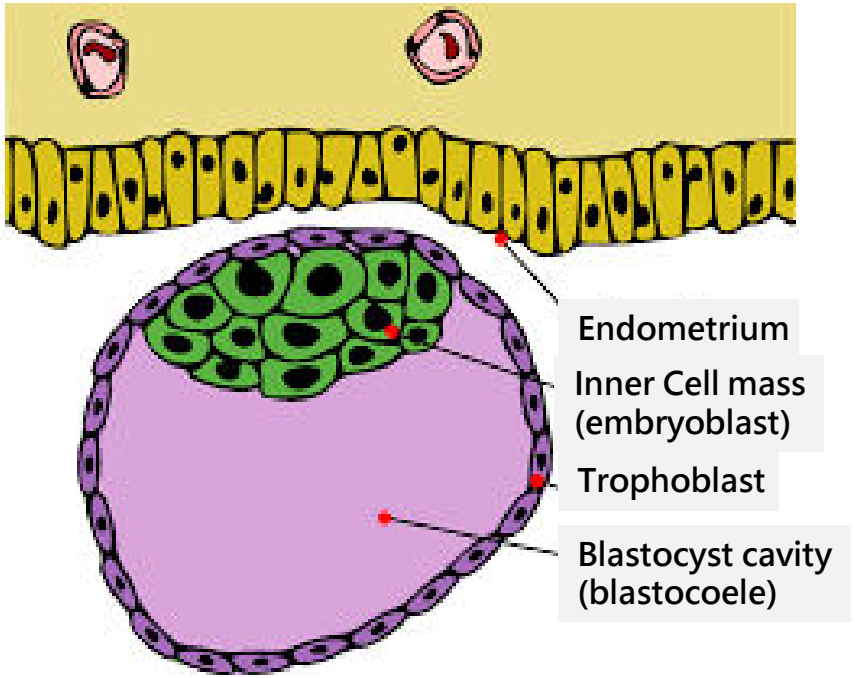
Prenatal Development (First Trimester)

Prenatal Development (First Trimester)



- **After fertilization, usually in the upper third of the uterine tube, he zygote goes through several cleavages as it moves down the tube.**
- **It eventually becomes a solid ball of cells called a morula, and by the fourth day, it develops into a 50 to120-cell blastula stage called a blastocyst.**

Prenatal Development (First Trimester)



- The next stage of development occurs when the blastocyst adheres to the uterine wall and implants.
- During implantation, the outer cells of the blastocyst, called the trophoblast, invade the endometrium.

Prenatal Development (First Trimester)

- Implantation is usually completed 11 to 12 days after fertilization; from then on, the female is considered to be pregnant.
- One of the unique features of mammalian development is that most of the cells of the early embryo make no contribution to the embryo's body, giving rise instead to supportive and protective membranes.

Prenatal Development (First Trimester)

- Only the inner cell mass gives rise to the embryonic body.
- Eventually, these cells arrange in a flat sheet that undergoes a gastrulation similar to that of reptiles and birds.
- Once gastrulation is completed, the rest of the first trimester is devoted to organogenesis and growth.

Prenatal Development (First Trimester)

- **Regulatory events and inductive-tissue interactions shape most of the organ systems.**
- **By the middle of the first trimester, all of the major body systems begin to develop.**

end

Animal Form & Function II

Module 154

Prenatal Development (Second Trimester)

Prenatal Development (Second Trimester)

The Second Trimester

- In the second trimester (4th to 6th month), fetal growth is spectacular.
- By now, the pregnant mother is aware of fetal movements.
- The heartbeat can be heard with a stethoscope.

Prenatal Development (Second Trimester)



- During the sixth month, the upper and lower eyelids separate, and the eyelashes appear.
- During this period, the bones begin to ossify.

Prenatal Development (Second Trimester)

- During the final weeks of second trimester, major organs — such as the liver and pancreas — become clear.
- By the end of 6th month of pregnancy, the fetus is about 10 to 11 inches long and weighs about 1 pound.

End

Animal Form & Function II

Module 155

Prenatal Development (Third Trimester)

Prenatal Development (Third Trimester)

- The third trimester extends from the seventh month until birth.
- During this time, the fetus has developed sufficiently (with respect to the circulatory and respiratory systems) to potentially survive if born prematurely.

Prenatal Development (Third Trimester)

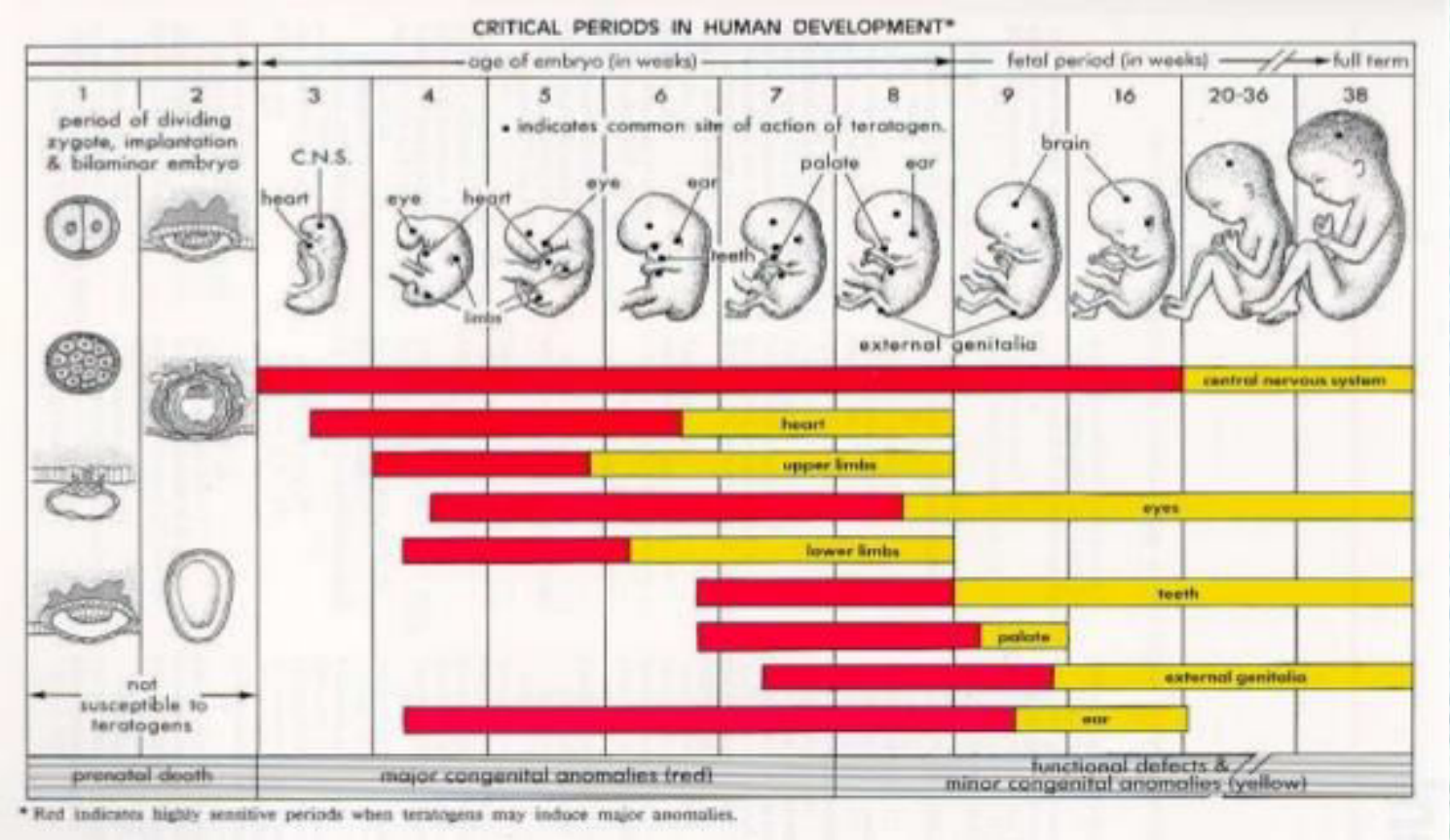
- Surfactant is a liquid made by the lungs that keeps the airways (alveoli) open.
- This liquid makes it possible for babies to breathe in air after delivery.
- An unborn baby starts to make surfactant at about 26 weeks of pregnancy.

Prenatal Development (Third Trimester)



- Fetus prepares for birth towards the end of 9th month
- The fetus is positioned head down with limbs curled up.
- During the last month, fetal weight doubles.

Prenatal Development (Third Trimester)



Animal Form & Function II

Module 156

Placenta: Structure and Function

Placenta: Structure and Function

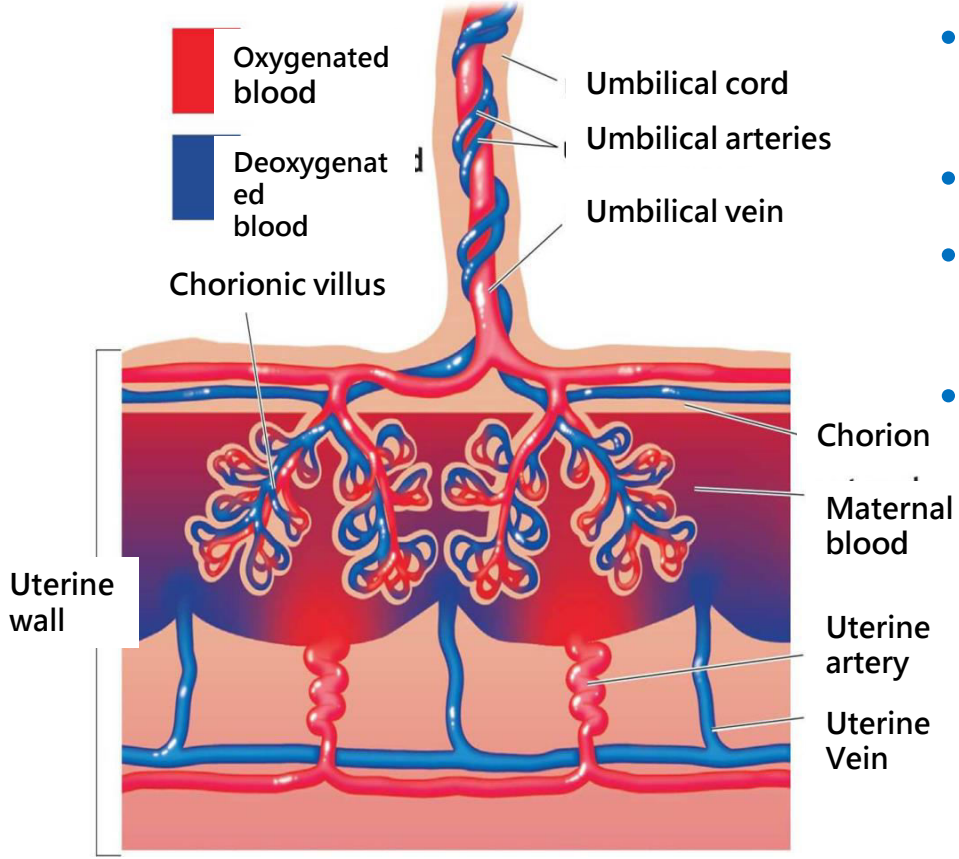


- **The placenta** is the organ that sustains the embryo and fetus throughout the pregnancy and through which gases are exchanged, nutrients are supplied and wastes are removed from the fetal systems to the maternal end.

Placenta: Structure and Function

- The tiny fingerlike projections are sent out from the blastocyst during implantation develop into numerous chorionic villi, which contain embryonic blood vessels.
- These blood vessels do not merge with those of the mother; the two bloodstreams remain separate throughout the pregnancy

Placenta: Structure and Function



- Placenta is a feto-maternal organ
- It has two components
- Fetal part, develops from the chorionic sac.
- Maternal part, derived from endometrium.

The placenta and the umbilical cord is the system for transporting substances between the mother and the fetus.

Placenta: Structure and Function

- Branches of the mother's arteries in the wall of her uterus open into pools near the chorionic villi.
- Oxygen and nutrients from the mother's blood diffuse into the fetal capillaries of the placenta.

Placenta: Structure and Function

Functions of the placenta

- Gas exchange, nutrition, hormone secretion, Excretion and fetal protection.
- Nutrient and drug transfer across the placenta are by passive diffusion, facilitated diffusion, active transport, and pinocytosis.

Placenta: Structure and Function

- The umbilical cord connects the placenta to the abdomen of the fetus.
- Two fetal umbilical arteries and one fetal umbilical vein spiral about each other in the umbilical cord.

Placenta: Structure and Function

- Oxygen and nutrients from the mother's blood diffuse into the fetal capillaries of the placenta.
- The fetal capillaries lead into the umbilical vein, which is enclosed within the umbilical cord.
- From here, the fresh blood circulates through the fetus's body.

end

Animal Form & Function II

Module 157

Animal Form & Function II

Childbirth

Childbirth

- About 266 days after fertilization, or 280 days from the beginning of the last menstrual period, the human baby is born.
- The birth process is called parturition.
- During parturition, the mother's uterine muscles begin to contract, and the cervix begins to dilate, or open.

Childbirth

- The hormone relaxin, produced by the ovaries and placenta, causes the mother's pelvic bones to slightly separate so that the baby can pass through the birth canal.
- Changing hormone levels initiate parturition.

Childbirth

- When it is time for the baby to be born, its pituitary gland secretes adrenocorticotropic hormone (ACTH), which stimulates the adrenal glands to secrete steroids.
- These steroids stimulate the placenta to produce prostaglandins and pituitary to produce oxytocin.

Childbirth

- These hormones cause the uterus to begin powerful muscular contractions.
- The contractions build in length and increase in frequency over a period that usually lasts from 2 to 18 hours.
- During that time, the cervix becomes fully dilated, and the amniotic sac ruptures.

Childbirth

- Usually within an hour of these events, the baby is expelled from the uterus .
- After the baby emerges, uterine contractions continue to expel the afterbirth (Placenta and umbilical cord).
- The umbilical cord is severed, and the newborn gets on the outside world.

Childbirth

- In mammals other than humans, the female bites through the cord to sever it.

end

Childbirth

Animal Form & Function II

Module 158

Milk Production and Lactation

Milk Production and Lactation

- **Lactation** includes both milk secretion (production) by the mammary glands and milk release from the breasts.
- During pregnancy, the breasts enlarge in response to increasing levels of the hormone prolactin.
- Before birth, placental secretions of estrogen and progesterone inhibit milk secretion from the breasts.

Milk Production and Lactation

- After the placenta has been expelled from the uterus, the concentrations of estrogen and progesterone drop, and the breasts begin to produce copious amounts of milk.
- The mother's breasts do not actually release milk until one to three days after the baby is born.

Milk Production and Lactation

- During these first days, the suckling baby receives colostrum, a high-protein fluid present in the breast at birth.
- Colostrum contains an abundance of maternal antibodies and thus helps strengthen the baby's immune system.

Milk Production and Lactation

- It also functions as a laxative, removing fetal wastes, called meconium, retained in the intestines.
- After about three days, the prolactin secreted from the pituitary stimulates milk production.
- The newborn's suckling stimulates the pituitary to release oxytocin as well as prolactin.

Milk Production and Lactation

- **Prolactin triggers milk release from the mammary glands.**

end

Animal Form & Function II

Module 159

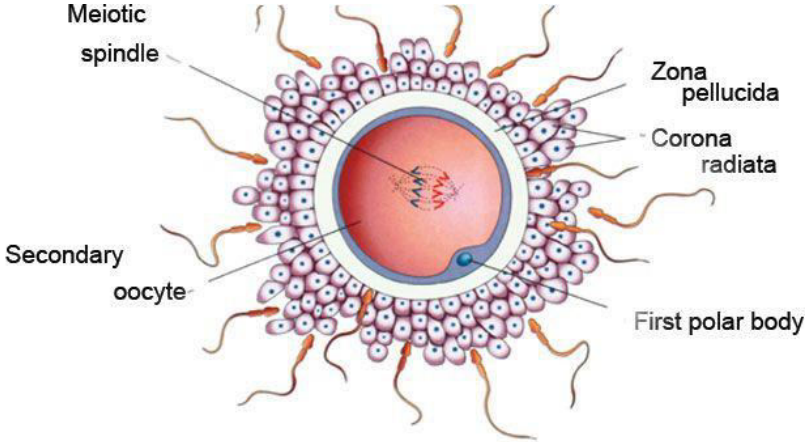
Animal Form & Function II

Process of Fertilization

Process of Fertilization

- **Fertilization** involves four events:
- i) the sperm contacts the egg and recognition occurs.
- ii) the sperm or sperm nucleus enters the egg.
- (iii) the egg becomes activated, and certain developmental changes begin.
- (iv) the sperm and egg nuclei fuse.

Process of Fertilization



Source: www.pinterest.com/pin/378302437433677592/

- An egg is surrounded by a plasma membrane and by one or more external coverings that are important in fertilization.
- For example, a mammalian egg is enclosed by a thick, noncellular, zona pellucida, which is surrounded by a layer of granulosa cells derived from the follicle in which the egg develops.

Process of Fertilization

- Before fertilization the mammalian sperm undergoes capacitation, a maturation process in the in the female reproductive tract.
- During capacitation, the sperms become increasingly motile and capable of undergoing an acrosome reaction when they encounter an egg.

Process of Fertilization

- **Sperm Entry**
- **The plasma membrane of the egg is covered with microvilli, which elongate to surround the head of the sperm.**
- **As they do so, the plasma membranes of sperm and egg fuse and a fertilization cone is formed that contracts to draw the sperm into the egg.**

Process of Fertilization

- **Fertilization activates the egg**
- **Release of calcium ions into the egg cytoplasm triggers the activation program, a series of metabolic changes within the egg.**
- **Aerobic respiration increases, this causes a burst of protein synthesis after sperm entry.**

Process of Fertilization

- In addition, the egg nucleus is stimulated to complete meiosis.
- The egg, which is actually a secondary oocyte, arrested early in the second meiotic division.

Process of Fertilization

- **Sperm and egg pronuclei fuse, restoring the diploid state.**
- **After the sperm nucleus enters the egg, it swells and forms the male pronucleus.**
- **The nucleus formed during completion of meiosis in the egg becomes the female pronucleus**

Process of Fertilization

- **The haploid male and female pronuclei then fuse to form the diploid nucleus of the zygote, and DNA synthesis occurs in preparation for the first cell division.**

end

Animal Form & Function II

Module 160

Animal Form & Function II

Fertilization and Implantation

Fertilization and Implantation

- **Fertilization** is the fusion of sperm and egg.
- Fertilization and the subsequent establishment of pregnancy together are referred to as conception.
- After ejaculation into the female reproductive tract, sperm remain alive and retain their ability to fertilize an ovum for an estimated 48 to 72 hours.

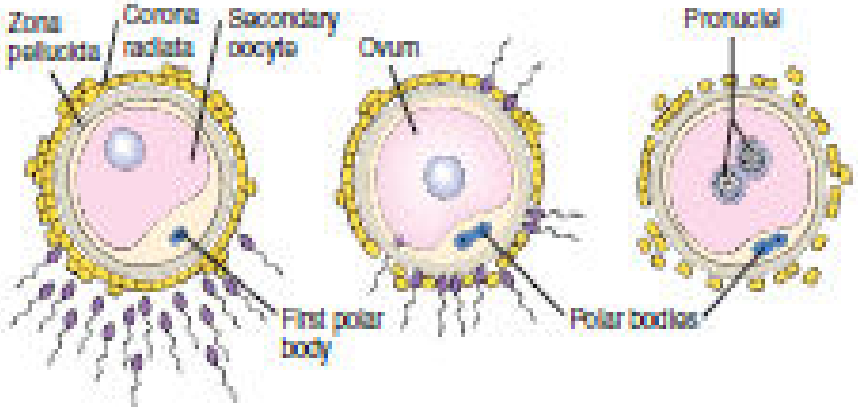
Fertilization and Implantation

- The ovum remains fertile for 12 to 24 hours after ovulation.
- Therefore, in a very regular 28-day menstrual cycle, sexual intercourse at mid cycle is most likely to result in fertilization.
- When conditions in the vagina and cervix are favorable, sperm arrive at the site of fertilization, the upper oviduct, within 5 minutes after ejaculation.

Fertilization and Implantation

- At the time of ovulation, when estrogen concentration is high, the cervical mucus has a thin consistency that permits passage of sperm from the vagina into the uterus.
- After ovulation, when progesterone concentration rises, the cervical mucus becomes thick and sticky,

Fertilization and Implantation



- This blocks the entrance of sperm.
- Once sperm enter the uterus, contractions of the uterine wall help transport them.
- These contractions are induced, in part, by prostaglandins in the semen.
- The sperm's own motility is important, especially in approaching and fertilizing the ovum.

Fertilization and Implantation

- When a sperm encounters an egg, openings develop in the sperm acrosome, exposing enzymes that digest a path through the zona pellucida surrounding the secondary oocyte.
- As soon as one sperm enters the secondary oocyte, changes occur that prevent the entrance of other sperm.

Fertilization and Implantation

- As the fertilizing sperm enters, it usually loses its flagellum.
- Sperm entry stimulates the secondary oocyte to complete its second meiotic division.
- The head of the haploid sperm then swells to form the male pronucleus and fuses with the female pronucleus to form the diploid nucleus of the zygote. The process of fertilization is described in m

Fertilization and Implantation

Process of Fertilization

Animal Form & Function II

Module 161

Animal Form & Function II

Cleavage and Egg Types

Cleavage and Egg Types

- The zygote formed as a result of fertilization is totipotent, i.e. it gives rise to all the cell types of the new individual.
- Because the ovum is very large compared with the sperm, the bulk of the zygote cytoplasm and organelles comes from the ovum.
- However, the sperm and ovum usually contribute equal numbers of chromosomes.

Cleavage and Egg Types

- Shortly after fertilization, the zygote undergoes cleavage, a series of rapid mitotic divisions with no period of growth during each cell cycle.
- For this reason, although the cell number increases, the embryo does not increase in size.
- The zygote initially divides to form a 2, to 4 , 4 to 8 celled embryo.

Cleavage and Egg Types

EggTypes

- On the basis of the amount of yolk, eggs are grouped into three types:
- 1. Alecithal Egg: When the egg contains no yolk. Example eutherian mammals.
- 2. Microlecithal Egg: When the egg contain small or negligible amount of yolk. Example, Amphioxus, Tunicates

Cleavage and Egg Types

- 3. Mesolecithal egg: When the amount of yolk present is moderate and is not high. Example, Dipnoi, Petromyzon, and amphibians.
- 4. Macrolecithal Egg: when the egg contains large amount of yolk. Such eggs are also called megalecithal or Polylecithal Example, Reptiles, Birds, Prototheria (Monotremata) Egg laying mammals.

Cleavage and Egg Types

- On the Basis of the distribution of yolk, Eggs may be:

Isolecithal or Homolecithal:

- Yolk is uniformly distributed throughout the ooplasm: Example: Echinoderms, Amphioxus, mammals.
- In this type of eggs, very little amount of yolk is present.

Cleavage and Egg Types

b. Telolecithal Eggs:

- **Eggs containing moderate or large quantity of yolk and the distribution is not uniform.**
- **Yolk is concentrated more towards the vegetal pole.**

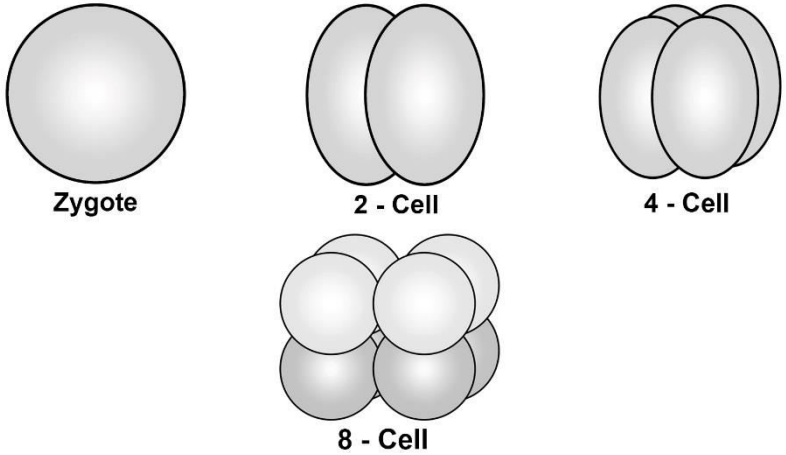
Animal Form & Function II

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Animal Form & Function II

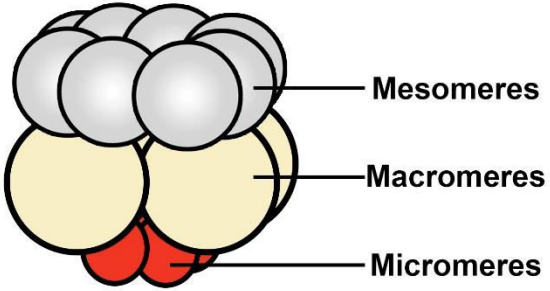
Echinoderm Embryology

Echinoderm Embryology

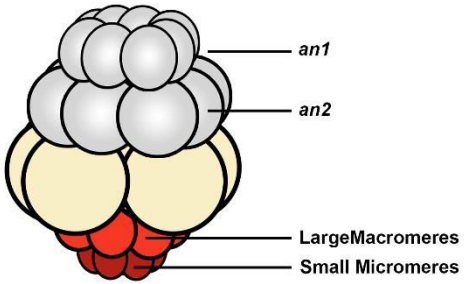


- **Echinoderms** (Sea Urchin) exhibit radial holoblastic cleavage.
- The first and second clevages are both meridional and at right angles to each other.
- It means the cleavage furrow pass through the animal and vegetal poles.
- The third cleavage is at right angles to the first two, dividing the embryo into 8 cells.

Echinoderm Embryology



16 - Cells



32 - Cells

- The vegetal tier, however, undergoes unequal equatorial cleavage to produce four large cells, the macromeres and micromeres at the vegetal pole.
- As the 16-cell embryo cleave, the 8 mesomeres divide to produce two 'animal tiers' an $_1$ and an$_2$, one above the other.
- The macromeres divide meridionally, forming a tier of 8 cells below an$_2$.

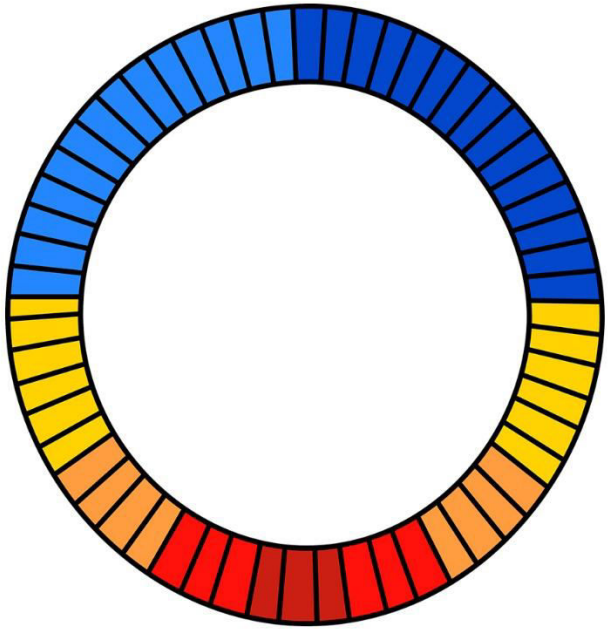
Echinoderm Embryology

- The 3rd cleavage also separates the four animal and vegetal tiers from each other.
- The 4th cleavage is very different from the first three. The four cells of the animal tier divide meridionally into 8 blastomeres. These cells are called mesomeres.

Echinoderm Embryology

- The micromeres also divide producing a small cluster beneath the larger tier.
- At the cleavage furrows of the 6th division are equatorial, and the 7th division is meridional, producing a 128- blatula.
- At this stage the cells form a hollow sphere surrounding the central cavity, blastocoel.
- All the cells at stage are of same size and tightly united.

Echinoderm Embryology



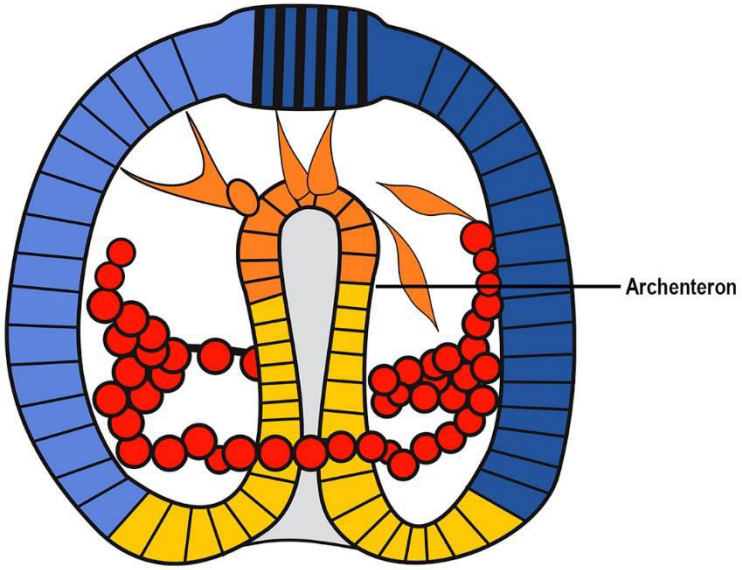
Hatched blastula

- The cells of blastula continue to divide forming one -cell-layer thick thinning out as it expands.
- After 9 or 10 cell division when the synchrony of cell division ends the non dividing cells develop cilia on their outer surfaces.
- The ciliated blastula then begins to rotate within the fertilization envelope.

Echinoderm Embryology

- Soon after the cells at the vegetal pole begins to thicken forming a vegetal plate.
- The cells of animal half synthesize and secrete a hatching enzyme that digests fertilization envelope
- The embryo is now a free swimming hatched larva.

Echinoderm Embryology



Late Gastrula

- Sea urchin gastrulation begins with the ingression of primary mesenchyme cells from the vegetal pole.
- Following ingression the vegetal plate invaginates to form a stout cylinder called archenteron.
- This stage is the gastrula which the develops into Pluteus larva.

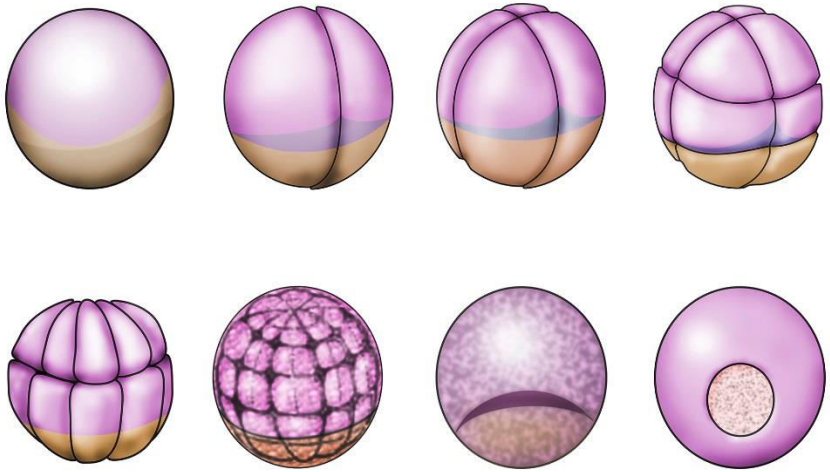
Animal Form & Function II

Module 163

Animal Form & Function II

Amphibian Embryology

Amphibian Embryology



- Cleavage in amphibians is radially symmetrical and holoblastic.
- The amphibian egg, contains much more yolk which is concentrated in the vegetal hemisphere.
- Here yolk is an obstruction to cleavage.
- The first division begins at the animal pole and slowly extends down into the vegetal region.

Amphibian Embryology

- The second cleavage is at right angles to the first one and the third cleavage is towards the animal hemisphere as expected, is equatorial.
- It divides the frog embryo into 8 cells, four small animal blastomeres (micromeres) and four large blastomeres (macromeres) in the vegetal region.

Amphibian Embryology

- As cleavage progresses, the animal region becomes packed with numerous small cells, while the vegetal region contains only a relatively small number of large, yolk-laden macromeres.
- An amphibian embryo containing 16 to 64 cells is commonly called a morula.

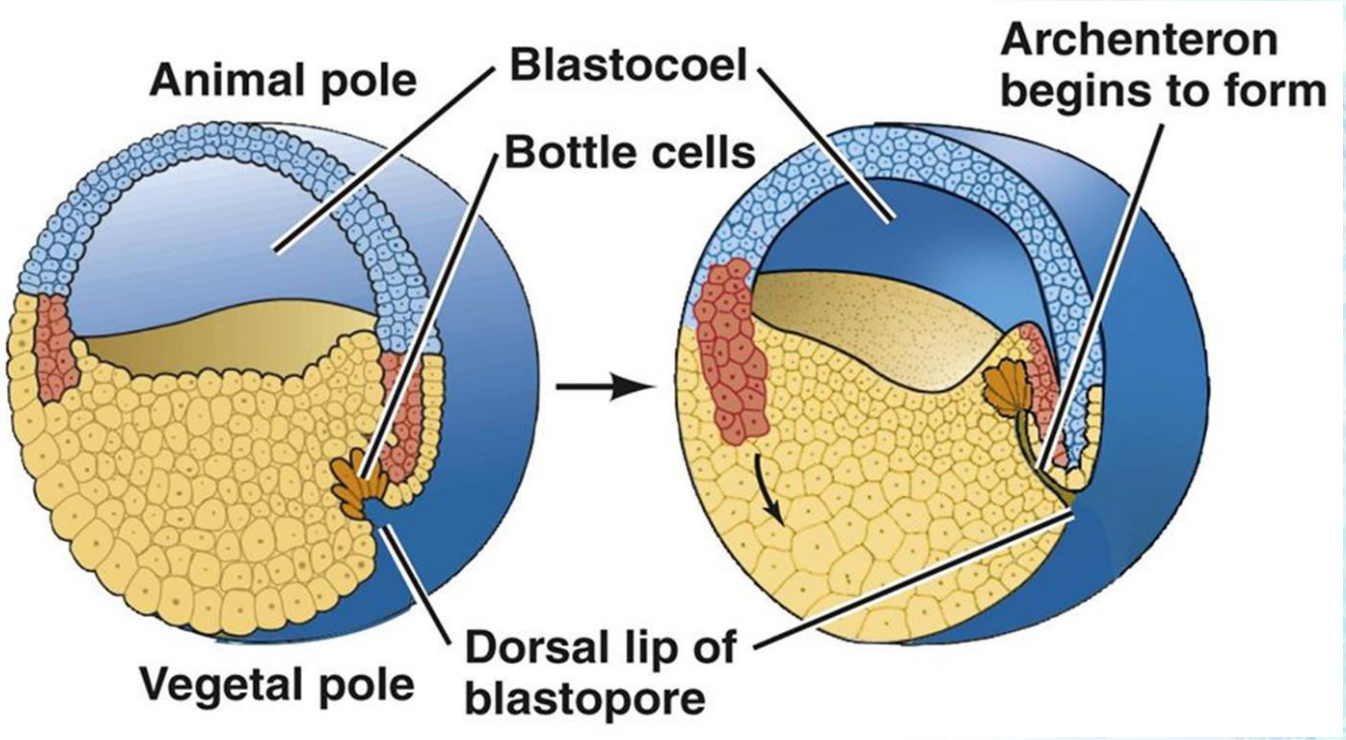
Amphibian Embryology

- At the 128-cell stage, the blastocoel becomes apparent, and the embryo is considered a blastula.
- The blastocoel probably serves two major functions in frog embryos:
- a. it permits cell migration during gastrulation, and
- b. it prevents the cells beneath it from interacting prematurely with the cells above it.

Amphibian Embryology

- **Gastrulation** is a phase early in the embryonic development of most animals, during which the blastula (a single-layered hollow sphere of cells) is reorganized into a multilayered structure known as the gastrula.

Amphibian Embryology



Amphibian Embryology

- 1. Gastrulation begins at the future dorsal side, below the equator when prospective endodermal cells (bottle cells) invaginate and form a slit (blastopore).
- The bottle cells line the early archenteron.

Amphibian Embryology

- 2. After the bottle cells, the next cells that enter the embryo form the prechordal plate, and the mesoderm of the notochord.
- 3. Simultaneously, the animal cells undergo epiboly, completely surrounding the surface of the embryo until a ventral lip of the blastopore forms.

Amphibian Embryology

- **4. The last endodermal cells on the surface appear as the yolk plug.**
- **5. The yolk plug is eventually covered by ectoderm.**

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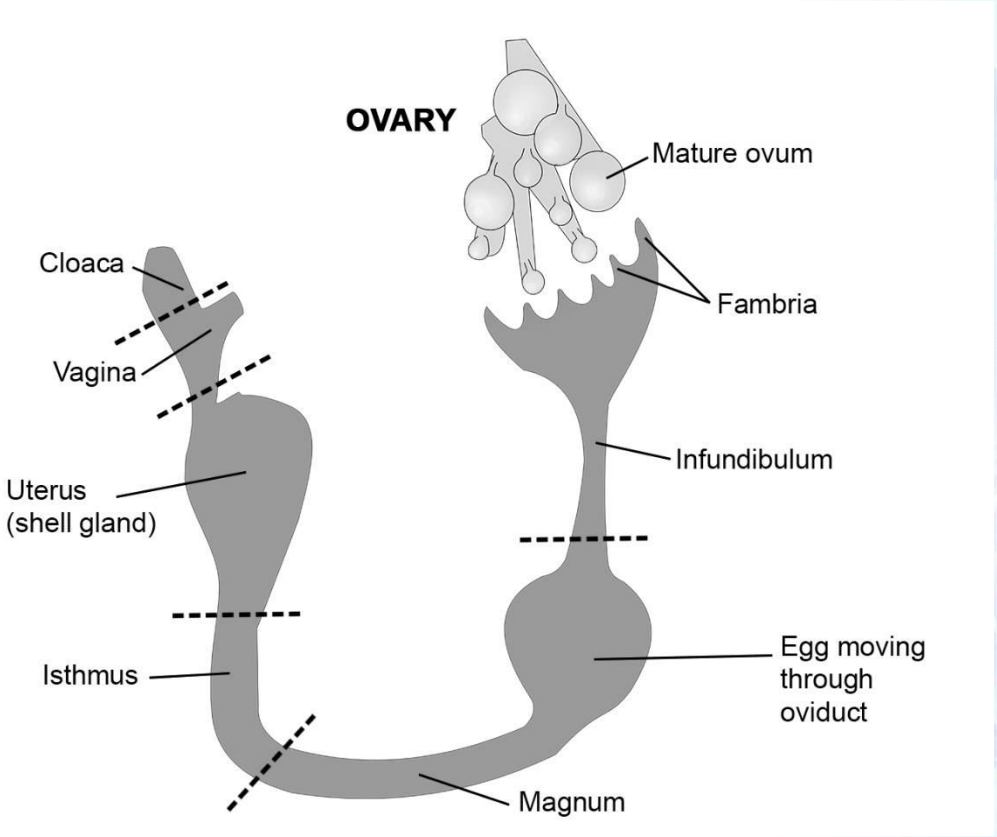
Animal Form & Function II

Module 164

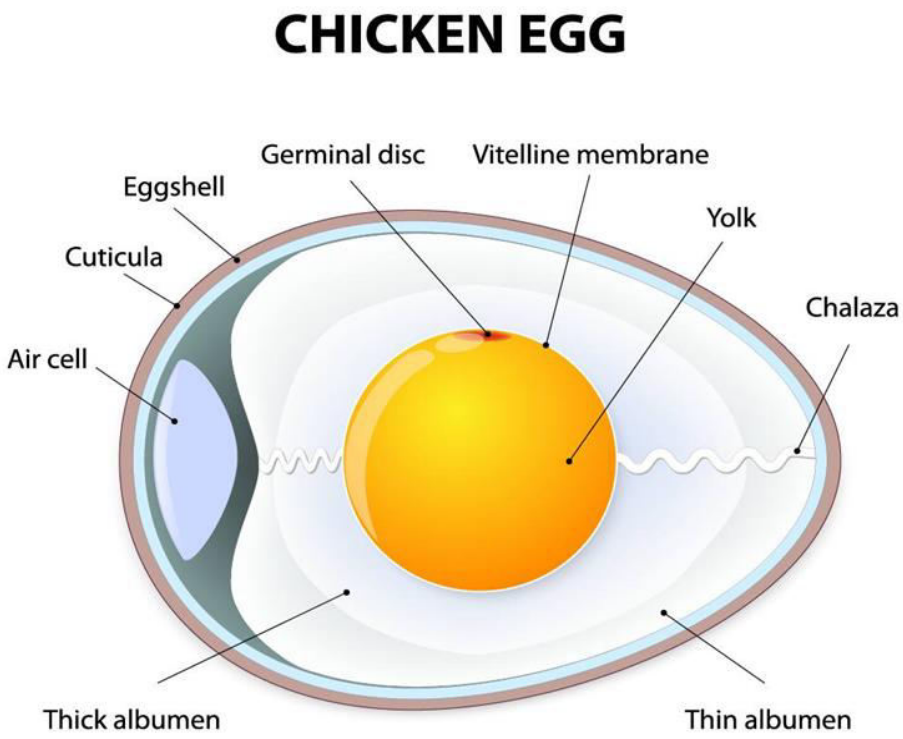
Animal Form & Function II

Avian Embryology

Avian Embryology Part I

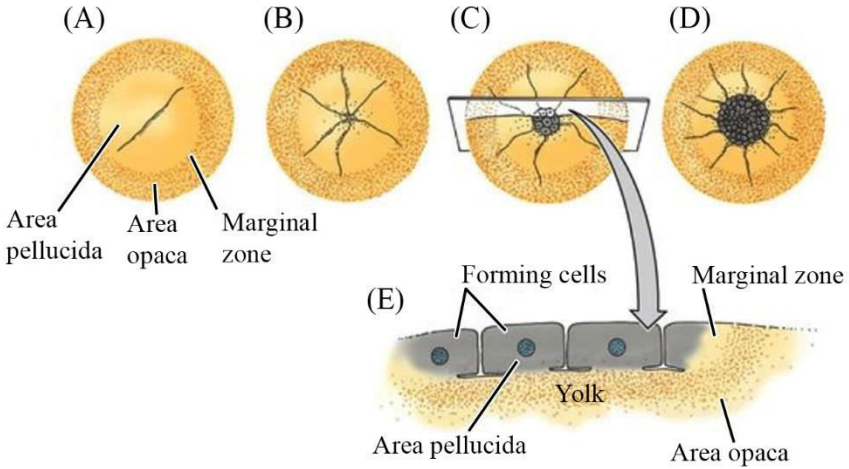


Avian Embryology Part I



- Fertilization of the chick egg occurs in the oviduct, before the albumen and the shell are secreted upon it.
- The egg is telolecithal with a small disc of cytoplasm sitting atop a large yolk.
- Cleavage occurs only in the blastodisc, a small disc of cytoplasm 2-3 mm in diameter at the animal pole of the egg.

Avian Embryology Part I

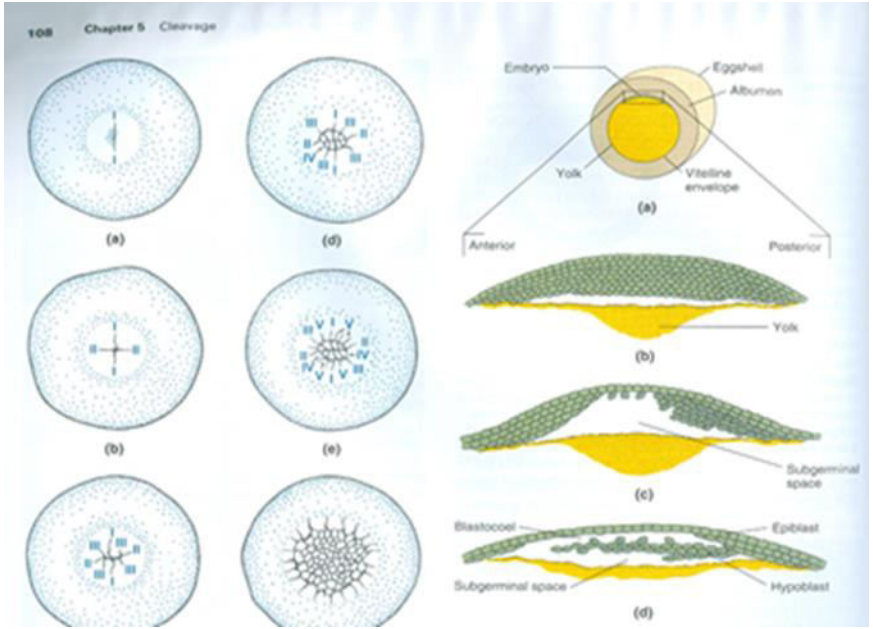


- The first cleavage furrow appears centrally in the blastodisc, and other cleavages follow to create a single-layered blastoderm.
- Thereafter, equatorial and vertical cleavages divide the blastoderm into a tissue five to six cell layers thick.

Avian Embryology Part I

- Between the blastoderm and the yolk there develops a space called the sub germinal cavity.
- This space is created when the blastoderm cells disintegrate leaving a single layer of cells called area pellucida/Epiblast.
- The peripheral ring of blastoderm cells constitutes the area opaca.

Avian Embryology Part I



- From the epiblast the ingressing cells unite with the sheet of cells migrating from the posterior margin of the blastoderm to form the hypoblast.
- The hypoblast thus divide the subgerminal space into compartments: blastocoel and subgerminal space.
- The blastocoel between epiblast and hypoblast layers.

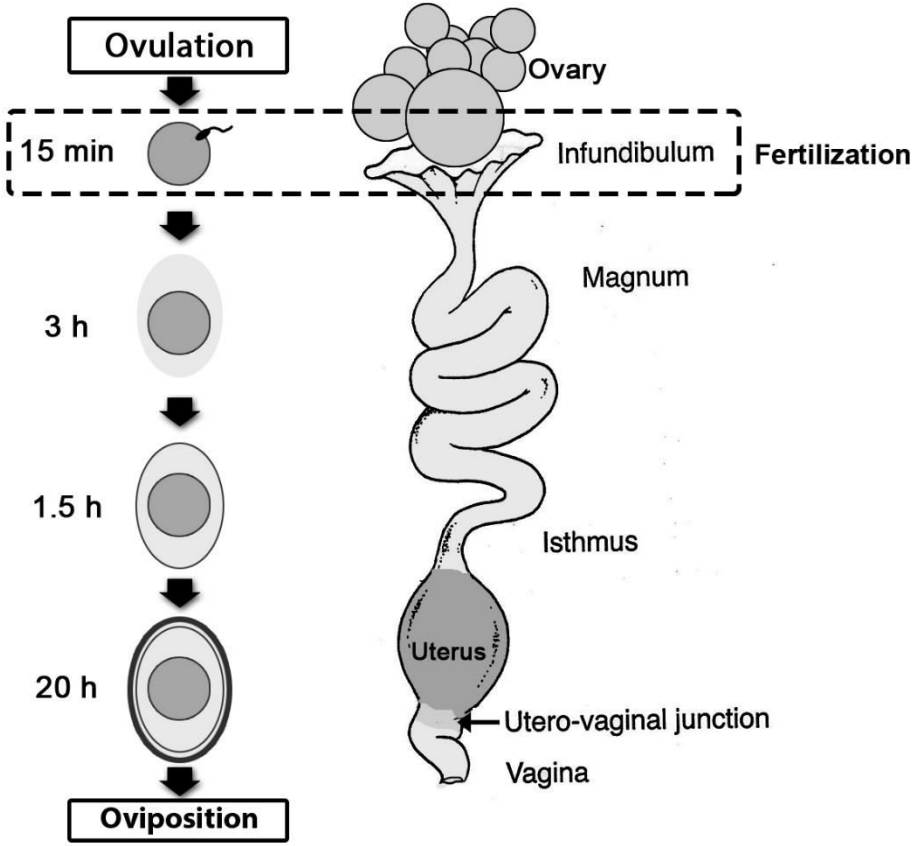
Avian Embryology Part I

- This two-layered blastoderm (epiblast and hypoblast) gaped apart by blastocoel and joined together at the margin of the area opaca, is called blastula.

Avian Embryology Part I

- The avian embryo come entirely from the epiblast.
- The hypoblast does not contribute any cells to the developing embryo.
- The hypoblast cells form portions of the external embryonic membranes (Amnion, Chorion and Allantois, especially the yolk sac and stalk)

Avian Embryology



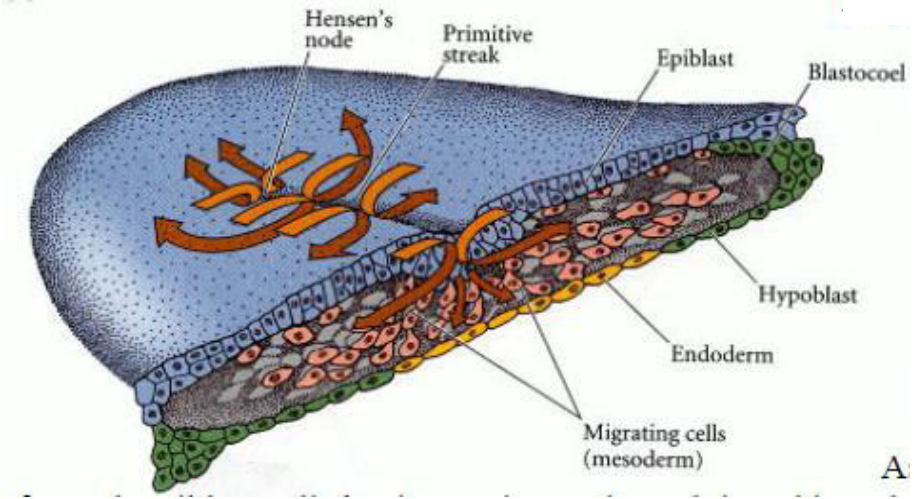
Animal Form & Function II

Module 165

Animal Form & Function II

Avian Embryology
Part II

Avian Embryology Part II



- **Primitive Streak** The major structural characteristic of avian, gastrulation is the primitive streak.
- This streak is first visible as a thickening of the epiblast at the posterior region of the embryo.
- As these cells migrate from the lateral region of the posterior epiblast toward the center the primitive streak, the elongates toward the future head region.

Avian Embryology Part II

- The streak eventually extends 60 75% of the length of the area pellucida.
- The primitive streak defines the axes of the embryo. It extends from posterior to anterior end.
- As cells converge to form the primitive streak, a depression forms within the streak. This depression is called the primitive groove.

Avian Embryology Part II

- This groove serves as an opening through which migrating cells pass into the blastocoel.
- At the anterior end of the primitive streak is a regional thickening of cells called Hensen's node.
- The center of this node contains a funnel-shaped depression, called the primitive pit.

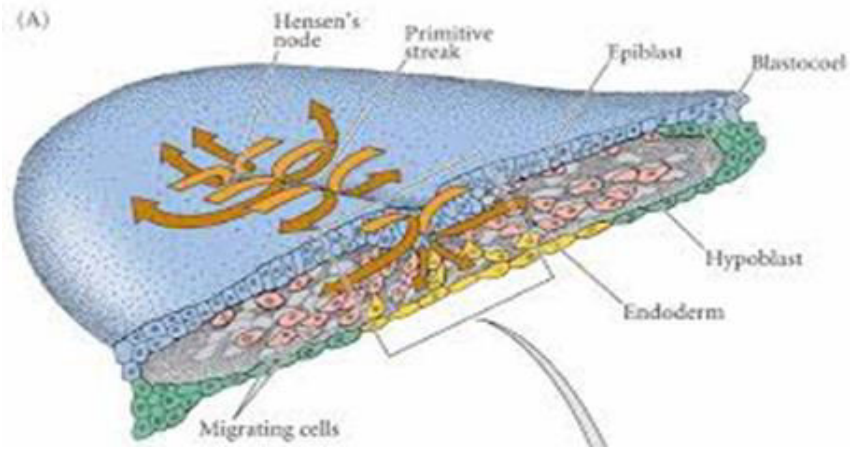
Avian Embryology Part II

- As soon as the primitive streak has formed, epiblast cells begin to migrate through it and into the blastocoel.
- The primitive streak has a continually changing cell population.
- Cells migrating through Hensen's node pass down into the blastocoel and migrate anteriorly, forming the notochord.

Avian Embryology Part II

- The cells passing through the lateral portions of the primitive streak give rise to endoderm and mesoderm layers.
- The epiblast remain out side and is transformed into the ectoderm.
- Now this three layered embryo is called gastrula which has formed undergoing a process, the Gastrulation.

Avian Embryology Part II



- **Gastrulation**
- **It is the process of highly integrated cell and tissue movements and their rearrangements so as to develop a three layered embryo composed of ectoderm, mesoderm and endoderm.**

Animal Form & Function II

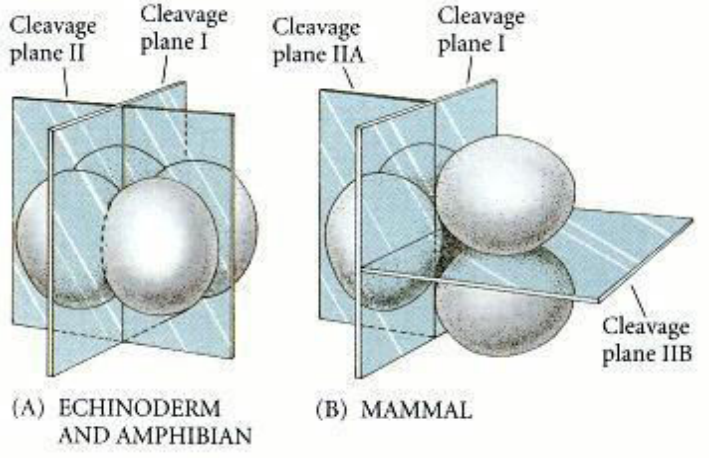
Module 166

Mammalian Embryology

Mammalian Embryology

- The mammalian oocyte after its release from the ovary is swept by the fimbriae into the oviduct
- Fertilization occurs in the ampulla of the oviduct, a region close to the ovary.
- Meiosis is completed at this time, and first cleavage begins about a day later.

Mammalian Embryology

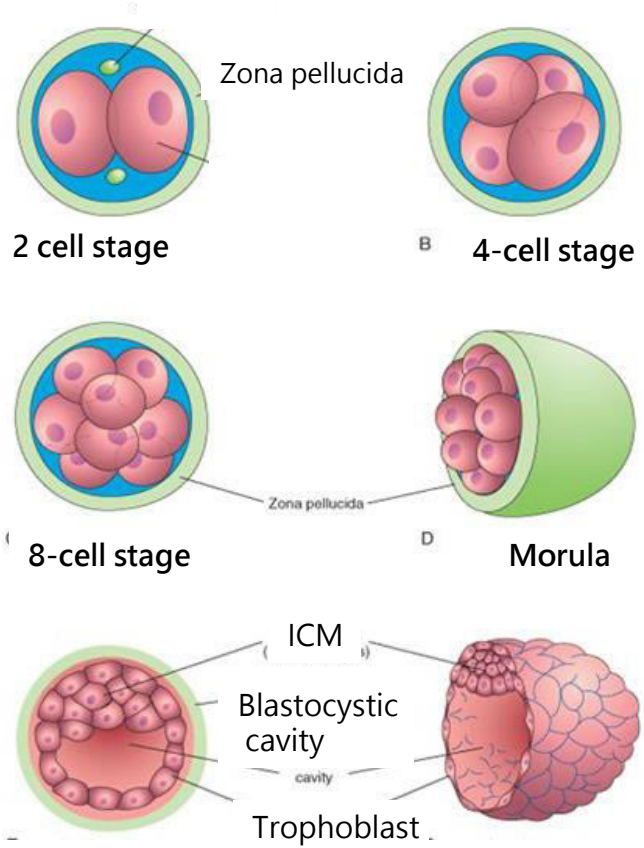


- The cilia in the oviduct push the embryo toward the uterus; the early cleavages occur along this journey.
- In mammals the cleavage is rotational cleavage.
- Following the third cleavage,, the 8 blastomeres undergo a spectacular change in their behavior.

Mammalian Embryology

- The 8-cell embryo divide to produce a 16-cell morula consisting of a small group of internal cells (Inner cell mass) surrounded by a larger group of external cells.
- Most of the descendants of the external cells become the trophoblast cells.
- This group of cells produces no embryonic structures.

Mammalian Embryology



- It forms the tissue of the chorion, the embryonic portion of the placenta.
- The chorion enables the embryo to get oxygen and nourishment from the mother.
- In mammals the embryo proper is derived from the descendants of the ICM.
- The development of the embryo proper can wait until after that attachment occurs.

Mammalian Embryology

- The ICM is positioned on one side of the ring of trophoblast cells.
- The resulting structure, called the blastocyst.
- While the embryo is moving through the oviduct en route to the uterus, the blastocyst expands within the zona pellucida and ultimately hatches out.

END

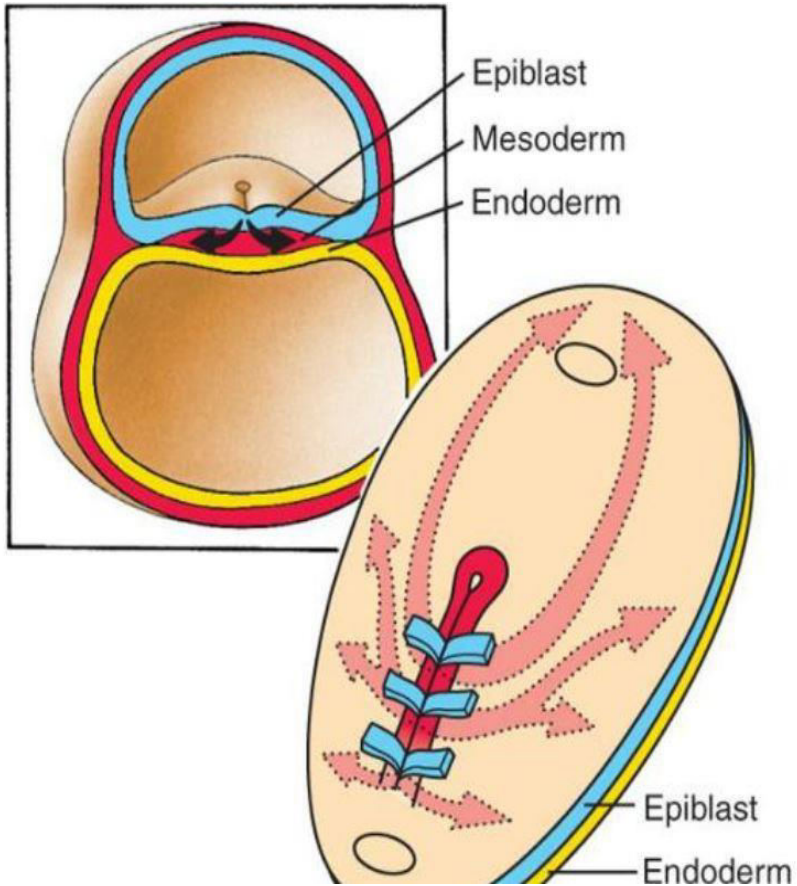
Animal Form & Function II

Module 167

Animal Form & Function II

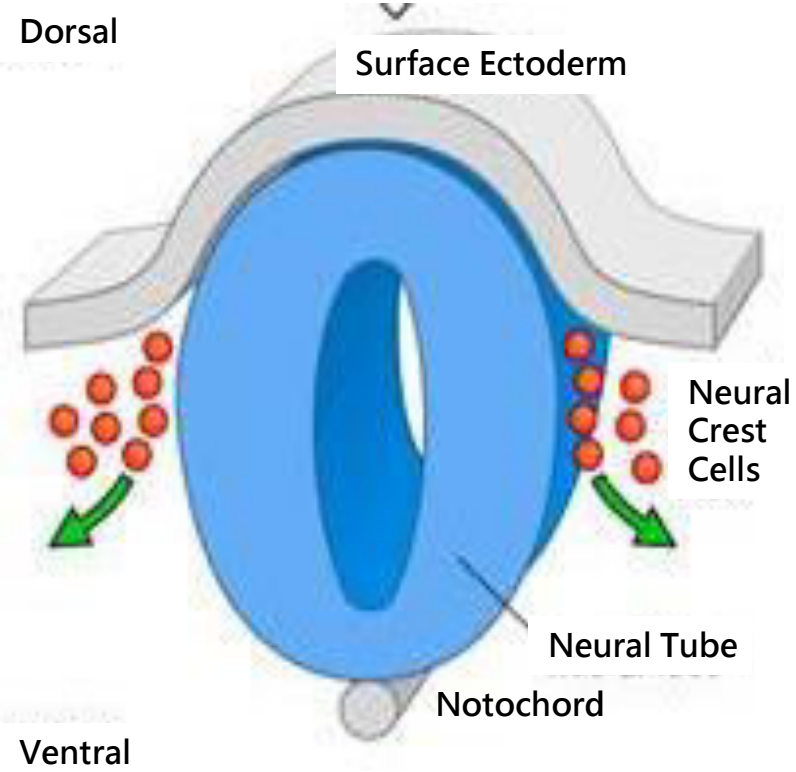
Primary Germ Layers

Primary Germ Layers



- **The germ layers** develop early in embryonic life, through the process of gastrulation.
- During gastrulation, the embryo develops/differentiate into three primary germ layers:
- an inner layer, endoderm,
- an outer layer, ectoderm and
- a middle layer, mesoderm.

Primary Germ Layers



- **Ectoderm**
- **The ectoderm develops from the embryo's epiblast.**
- **The ectoderm develops into the surface ectoderm, neural crest, and the neural tube.**
- **The surface ectoderm develops into: epidermis, hair, nails, lens of the eye, sebaceous glands, cornea, tooth enamel, the epithelium of the mouth and nose and anterior pituitary.**

Primary Germ Layers

- **The neural crest of the ectoderm develops into: peripheral nervous system, adrenal medulla, melanocytes, facial cartilage (ear and nose)**
- **The neural tube of the ectoderm develops into: brain, spinal cord, posterior pituitary, motor neurons, retina.**

end

Animal Form & Function II

Module 168

Animal Form & Function II

Primary Germ Layers
Part II

Primary Germ Layers Part II

- **The endoderm** is one of the germ layers formed during animal embryonic development.
- Cells migrating inward along the archenteron form the inner layer of the gastrula, which develops into th eendoderm.

Primary Germ Layers Part II

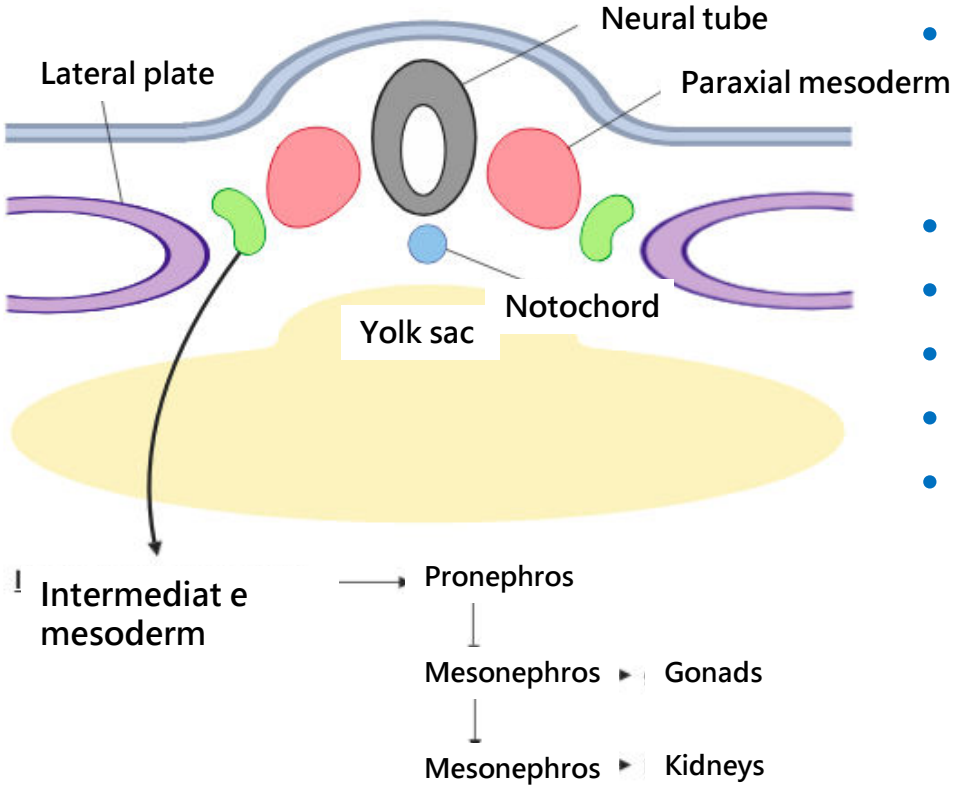
The endoderm gives rise to:

- **the epithelial lining of the gastrointestinal and respiratory tracts;**
- **the liver, the thymus, the thyroid, the parathyroids, and the pancreas;**
- **the epithelial lining of the urinary bladder and urethra;and**
- **the epithelial lining of the tympanic cavity, tympanic antrum, and auditory tube.**

Primary Germ Layers Part II

- **The mesoderm** forms in the embryos of triploblastic animals.
- During gastrulation, some of the cells migrating inward contribute to the mesoderm, an additional layer between the endoderm and the ectoderm.
- The formation of a mesoderm leads to the development of a coelom.

Primary Germ Layers Part II



- The mesoderm has several components which develop into tissues:
- 1. intermediate mesoderm,
- 2. paraxial mesoderm,
- 3. lateral plate mesoderm and
- 4. chorda-mesoderm.
- human body.

Primary Germ Layers Part II

- **The chorda-mesoderm** develops into the notochord.
- **The intermediate mesoderm** develops into kidneys and gonads.
- **The paraxial mesoderm** develops into cartilage, skeletal muscle, and dermis.
- **The lateral plate mesoderm** develops into the circulatory system (including the heart and spleen) and the wall of the gut.

END